MATERIAL SCIENCE

FIRST EDITION

SUNIL RISAL
KHEM GYANWALI

Material Science

First Edition

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FOREWORD

It is my pleasure to go through the manuscript of "Material Science", written by Sunil Risal and Khem Gyanwali. The authors have written this book using their extensive teaching experience in the field of engineering Material Science in Institute of Engineering, Tribhuvan University to meet the need of a text book for engineering undergraduates and a professional reference book for practicing engineers.

Material Science is an interdisciplinary field applying the synthesis of essential features and properties of material to various technologies. The technologies needed for the development activities should be affordable, energy efficient and less carbon consuming. Basic knowledge of material science is necessary to understand the working of various devices, machines, equipment. It is very important to engineers and technologists to learn the properties of materials in order to design, fabricate and carry out maintenance of equipment and machines.

The book not only covers all the contents of materials science but also provides many examples in each chapter. Each chapter is presented with well-designed simplicity requiring no special pre-requisite knowledge of supporting subjects. Self-explanatory figures and diagrams have been generously used to curtail long and wordy explanations.

I congratulate the authors, Sunil Risal and Khem Gyanwali on bringing out this book for the benefit of students in engineering discipline. I hope the students will be motivated to understand the fundamentals of Material Science. With great enthusiasm, I recommend this book to students and practicing engineers.

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Kathmandu, Nepal

PREFACE

This book titled 'Material Science' is designed to provide fundamental concept of metallurgical phenomenon in concise form to the students of mechanical and industrial engineering

It is an attempt to simplify some complex systems of material science and guide the students towards informative knowledge.

The contents are arranged based on the revised curriculum of Institute of Engineering, Tribhuvan University and this book will broadly fulfill the requirements of a complete text book to the students as well as faculty members involved in teaching practices.

Years of teaching experiences and difficulties encountered has strengthened the eagerness to publish a book.

Suggestions regarding corrections and improvement of this book are highly welcomed.

April, 2014

Sunil Risal

Khem Gyanwali

ACKNOWLEDGEMENT

We are indebted to several persons who have helped us in various ways in preparing this book.

We would like to express our deep sense of gratitude to Professor Dr. Bhakta Bahadur Ale for his thorough review and valuable guidance and feedbacks.

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Mr. Shyam Krishna Khadka, Prakash Man Shakya deserves special thanks for their support in typing and printing this book.

Last but not least, we are very grateful to our family members for their precious support.

Sunil Risal
Khem Gyanwali

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CHAPTER 1
INTRODUCTION TO MATERIALS

1.1 Introduction

Material science is one of the oldest forms of engineering and applied science and is the study of the properties of materials. It focuses on the factors that make one material different from another. There are many such factors, which include elemental composition, arrangement, bonding, impurities, surface structure, length, scale and so on. The ability to understand the relationships between these factors and the properties of a material has been crucial to the study of material science.

Materials science is an interdisciplinary field involving the properties of matter and its applications to various areas of science and engineering. Modern material science evolved directly from metallurgy, which itself evolved from mining.

A material is any substance or mixture of substances that occupy a volume, and has a mass. Material science encompasses various classes of materials, each of which may constitute a separate field. It includes elements of applied physics and chemistry, as well as chemical, mechanical, civil and electrical engineering.

Material science refers to *the branch of applied science which is concerned with investigating the relationship existing between the structures of materials, their properties and their inter-disciplinary study for practical purposes.*

In material science, one aims to understand materials fundamentally so that new material with the desired properties can be created rather than haphazardly looking for and discovering materials and exploiting their properties.

The basis of all material science involves relating the desired properties and relative performance of a material in a certain application to the structure of the atoms and phases in that material through characterization.

1.2 Types of materials

In material science, we confine our attention to solid materials only. So the subject is related to solid-state physics and solid-solid chemistry. The engineering materials may be classified as follows:

- Metals
- Polymers
- Ceramics
- Glasses
- Composites
- Semi-conductors

Metals

Metals are elements that generally have good electrical and thermal conductivity. Many metals have high strength, high stiffness, and have good ductility. Some metals, such as iron, cobalt and nickel are magnetic. At extremely low temperatures, some metals and inter-metallic compounds become superconductors.

Pure metals

Pure metals are elements which come from a particular area of the periodic table. Most pure metals are too soft, brittle and/or chemically reactive. Examples of pure metals include copper in electrical wires and aluminum in cooking foil and beverage cans.

Metal alloys

Metal alloys contain more than one metallic element. Their properties can be changed by changing the elements present in the *alloy. Examples of metal alloys include stainless steel which is an alloy of iron, nickel, and chromium; and gold jewelry which usually contains an alloy of gold and nickel.*

The most important properties of metals include density, fracture toughness, strength and plastic deformation. The atomic bonding of metals also affects their properties. In metals, the outer valence electrons are shared among all atoms, and are free to travel everywhere. Since electrons conduct heat and electricity, metals make good cooking pans and electrical wires.

Polymers

A polymer has a repeating structure, usually based on a carbon backbone. The repeating structure results in large chain-like molecules. Polymers are useful because they are lightweight, corrosion resistant, easy to process at low temperatures and are generally inexpensive. Some important characteristics of polymers include their size (or molecular weight), softening and melting points, crystallinity and structure. The mechanical properties of polymers generally include low strength and high toughness. Their strength is often improved using reinforced composite structures. One of the distinct properties of polymers is that they are poor conductors of electricity and heat, which makes them good insulators.

Ceramics

A ceramic is often broadly defined as any inorganic non-metallic material. Examples of such materials can be anything from NaCl (table salt) to clay (a complex silicate). Some of the useful properties of ceramics and glasses include high melting temperature, low density, high strength, stiffness, hardness, wear resistance and corrosion resistance. Many ceramics are good electrical and thermal insulators. Some ceramics have special properties; some ceramics are magnetic materials; some are piezoelectric materials; and a few special ceramics are superconductors at very low temperatures. Ceramics and glasses have one major drawback: they are brittle.

Glasses

A glass is an inorganic non-metallic material that does not have a crystalline structure. Such materials are said to be amorphous. Examples of glasses range from the soda-lime silicate glass in soda bottles to the extremely high purity silica glass in optical fibers.

Composites

Composites are formed from two or more types of materials. Examples include polymer/ceramic and metal/ceramic composites. Composites are used because their overall properties are superior to those of the individual components. For example: polymer/ceramic composites have a greater modulus than the polymer component, but aren't as brittle as ceramics.

Semiconductors

A semiconductor is a substance, usually a solid chemical element or compound that can conduct electricity under some conditions but not others, making it a good medium for the control of electrical current. Its conductance varies depending on the current or voltage applied to a control electrode, or on the intensity of irradiation by infrared (IR), visible light, ultraviolet (UV) or X-rays. Elemental semiconductors include antimony, arsenic, boron, carbon, germanium, selenium, silicon, sulfur, and tellurium. Silicon is the best-known of these, forming the basis of most integrated circuits (ICs).

1.3 Material properties

The presence, absence or variation of minute quantities of secondary elements and compounds in a material will have a great impact on the final properties of the materials produced. The important properties of materials are:

Physical Properties

Physical properties are characteristics of materials that are determined by nature. There are so many physical properties that it is not possible to even briefly describe all of them. However, important physical properties include color, density, melting point, electrical conductivity, thermal expansion and magnetic properties.

Chemical Properties

Chemical properties are material characteristics that relate to the structure of a material and its formation from the elements. The important chemical properties are composition, structure and corrosion resistance. Composition of material can be determined by analytical chemistry in metals, composition usually means the percentage of the various elements that make up the metal. Studies of the structure of the materials is one of the most useful tools of the metallurgist. Structure of a material generally refers to its micro-structure, i.e. the components seen when the metal is examined under a microscope.

Mechanical Properties

Mechanical properties are the characteristics of a material that are exhibited when a force is applied to the material. They usually relate to the elastic and plastic behavior of the material. Important mechanical properties are hardness, modulus of elasticity, yield strength, tensile strength, percent elongation, reduction in area, impact, fatigue strength, creep strength and wear resistance. Mechanical properties are of foremost importance in selecting materials for structural machine components.

Electrical properties

Electrical properties are the physical conditions that allow an electrical charge to move from atom to atom in a specific material. These properties differ greatly between the three major states of materials: solids, liquids and gases. Solids and electricity often are a perfect combination for conductivity. The electrical properties of copper, steel, and other metals provide the optimum opportunity because of the physical closeness of atoms. When electrons can pass easily between atoms, it promotes electrical conductivity. Solids like silver, copper and aluminum are popular in electrical work because very little energy is lost when electricity travels through these metals.

Magnetic properties

Materials may be classified by their response to externally applied magnetic fields as diamagnetic, paramagnetic or ferromagnetic. These magnetic responses differ greatly in strength. Diamagnetism is a property of all materials and opposes applied magnetic field but is very weak. Paramagnetism, when present, is stronger than diamagnetism and produces magnetization in the direction of the applied field and proportional to the applied field. Ferromagnetic effects are very large, producing magnetizations sometimes orders of magnitude greater than the applied field and as such are much larger than either diamagnetic or paramagnetic effects.

Dimensional Properties

The available size, shape and tolerances on materials are important in selecting a material. Other dimensional properties include surface roughness, waviness and micro-topography. The microscopic characteristics of a surface is called micro-topography and it includes various surface characteristics such as surface layout, roughness height, roughness width, waviness height, waviness width, etc.

1.4 Material structure

All solid substances are either amorphous solids or crystalline solids. The differences between crystalline and amorphous solids are given below:

Characteristic Geometry

In crystalline solids, the particles (atoms, ions, or molecules) are definitely and orderly arranged; thus these have characteristic geometry while amorphous solids do not have characteristic geometry.

Melting Point

A crystalline solid has a sharp melting point, i.e. it changes into liquid state at a definite temperature. On the contrary an amorphous solid does not have a sharp melting point. For example, when glass is heated, it softens and then starts flowing without undergoing any abrupt or sharp change from solid to liquid state. Therefore, amorphous solids are regarded as “liquids at all temperatures”

Cutting

Crystalline solids give clean cleavage while amorphous solids give irregular cut, due to conchoidal fracture on cutting with a sharp edged tool.

Cooling curve

Amorphous solids show smooth cooling curve while crystalline solids show two breaks in cooling curve. In the case of crystalline solids, two break points ‘a’ and ‘b’ appear. These points indicate the beginning and the end of the process of crystallization. In this time interval, temperature remains constant. This is due to the fact that during crystallization process energy is liberated which compensates for the loss of heat, thus the temperature remains constant.

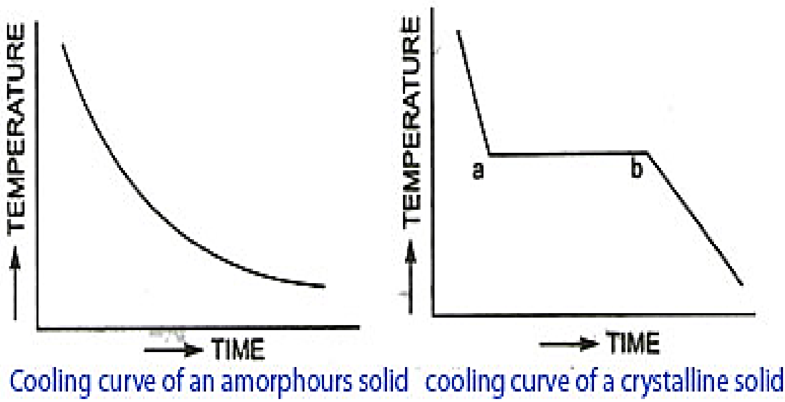


Figure 1.1: Cooling curve of amorphous and crystalline solid

1.2 Materials processing

Material processing is the series of operations that transforms industrial materials from a raw state into finished parts or products. The processes used to convert raw materials into finished products perform one or both of two major functions: first, they form the material into the desired shape; second, they alter or improve the properties of the material. Some of the material processing methods are casting, rolling, extrusion, compaction, etc.

1.5 Relationship among structures, processing and properties

The properties of all materials arise from their structure, i.e. from the manner in which their atoms aggregate into hierarchies of molecular or crystalline order or into disordered amorphous structures. Some of the properties, structures and processing of different types of materials are tabulated below.

Table 1: Relationship among structures, processing and properties

	Metals and alloys	**Ceramics and glasses**	**Polymers**	**Composites**
Structure	Crystal structures, Point defects, Dislocations	Crystal structures, Defect reactions, The glassy state	Configuration, conformation, molecular weight	Matrices, Reinforcements
Thermo - dynamics	Phase equilibrium, Gibbs rule, Lever rule	Ternary systems, Surface energy, Sintering	Phase separation, Polymer solutions, Polymer bends	Adhesion, Cohesion, Spreading
Mechanical properties	Stress-strain, Elasticity, Ductility	Fatigue, Fracture, Creep	Viscoelasticity, Elastomers	Laminates
Electrical, Magnetic, Optical properties	Resistivity, magnetism, Reflectance	Dielectrics, Ferrites, Absorbance	Ion conductors, Molecular magnets, LCDs	Dielectrics, Storage media
Processing	Casting, Rolling, Compaction	Pressing, CVD/CVI, Sol-Gel	Extrusion, Injection moulding, Blow moulding	Pultrusion, RTM, CVD/CVI

The properties of bulk matter of all kinds depend strongly on the nature and distribution of imperfections, either chemical or architectural, in the main array. Creating a material with the desired properties can be obtained by altering the defects in crystalline material such as precipitates, grain boundaries, interstitial atoms, vacancies or substitutional atoms.

Most of the properties observed and exploited in materials are co-operative properties of the aggregate rather than of the constituent atoms and all materials do not have a regular crystal structure. Polymers display crystallinity. Glasses, some ceramics, and many natural materials are amorphous, not possessing any long-range order in their atomic arrangements.

Industrial applications of materials science include materials design, cost-benefit trade-offs in industrial production of materials, processing techniques as: casting, rolling, welding, crystal growth, glass-blowing, extraction of materials and their conversion into useful forms.

1.6 Material selection for design

Material selection is a step in the process of designing any object (physical). In the context of product design, the main goal of material selection is to minimize cost while meeting product performance goals. Systematic selection of the best material for a given application begins with properties and costs of candidate materials.

While selecting materials for manufacturing in the initial evaluation of material alternatives, many can be rejected on the basis of absolute product parameters such as strength requirements achievable by the mechanical properties of the selected materials, conductivity by the thermal properties, density, melting point and specifications. Besides product specifications, materials should also meet performance specifications. Performance specifications delineate the basic functional requirements of the product and set out the basic parameters from which the design can be developed. They are based on the need the product is intended to satisfy and an evaluation of the likely risk and consequences of failure.

Performance requirements of materials

- **Function**: This is based on the thorough understanding of the product application, the functions the product must perform in terms of mechanical, physical, electrical and thermal properties of the material.
- **Appearance**: The material makes a major contribution to the aesthetics of product. Compact, nicely shaped, attractive color products jump ahead in the market.
- **Reliability**: Reliability is gaining greater importance as a criterion for material selection due to increasing consumer demand for trouble free products.
- **Service Life**: Larger the length of service life over which the material maintains its desirable characteristics is an important consideration.
- **Environment**: Environment effects are to be minimum during the product life of the material.
- **Compatibility**: When more than one type of material is used in a product or assembly, materials should work harmoniously with each other.
- **Producibility**: The selection of material in relation to the ease of producibility of an item is an important consideration.
- **Cost**: Because cost is so important in selecting materials, it is logical to consider cost at the outset of the materials selection process. A detailed cost comparison is not possible at the concept of formulation stage, but it is often possible to set a target cost &

eliminate the materials that obviously are too expensive (e.g. Silver wire for transmission wire).

Engineering requirements of materials

It means that a material must be expected to be used successfully for making engineering components such as crankshaft, spanners etc. The main engineering requirements are as follows:

- **Fabrication Requirements**: Material should be able to get shaped (e.g. cast, rolled, forged) and joined (e.g. welded, brazed etc.) easily.
- **Service Requirements**: Selected materials must stand up to service demand e.g. proper strength, wear and corrosion resistant.
- **Economic Requirements**: The engineering parts should be made with minimum overall cost.

Steps for selection of materials

The steps generally followed during selection of materials for material design are:

Analysis of the material requirements

Firstly, the fundamental properties required are analyzed. Suppose, to select the material for conduction wire, the fundamental properties required is good electrical conductivity. So electrical conductivity is the primary requirement.

Screening the candidate materials

Materials possessing the basic requirements are listed down. For conduction wire, materials like copper, silver etc. are taken into consideration.

Selection of candidate material

Screened materials are evaluated and rated among many criteria's and the material with optimum performance is selected. Copper may be selected for conduction wire since it is less costly.

Development of design data

Finally selected material is used for the required purpose.

REVIEW QUESTIONS

1. What is material science? Briefly explain different types of materials.
2. Describe briefly the important properties of materials.
3. How structures, processing and properties of materials are inter-related?
4. What are the factors to be considered while selecting material for design?

CHAPTER 2

ATOMIC STRUCTURE AND ARRANGEMENTS

2.1 Atomic structure

2.1.1 Atom

All matter is considered to be composed of unit substances known as chemical elements. These are the smallest units that are distinguishable on the basis of their chemical activity and physical properties. The elements are composed of atoms which have distinct structure characteristic of each element. The smallest particle of an element which retains the distinct structure characteristic of an element is called *atom*.

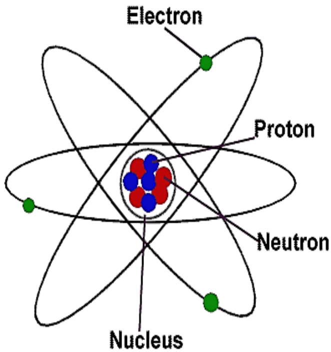


Figure 2.1: Structure of atom

Each atom consists of a nucleus and electrons. Figure 2.1 shows the structure of an atom. Central nucleus is surrounded by orbital electrons which move in concentric spherical shells. The weight of the atom and its radioactive properties are associated with the nucleus. The chemical properties and spectrum on the other hand depend on the planetary electrons.

The following fundamental particles are important from the subject point of view:

1. Electrons: These are negatively charged particles. The mass of an electron is 9.1×10^{-31} kg. It is equal to 1/1836th of the mass of a hydrogen atom. Each electron possesses a unit negative charge of electricity, equal to 1.602×10^{-19} coulomb.
2. Protons: These are positively charged particles. The mass of a proton is 1.672×10^{-27} kg. Each proton possesses a unit positive charge of electricity, this charge is also equal to 1.602×10^{-19} coulomb.
3. Neutrons: These are electrically neutral particles. The mass of each neutron is 1.675×10^{-27} kg which is approximately equal to the mass of a proton. Each neutron is composed of one proton and one electron.

2.1.2 Atomic number

The atomic number of an element is numerically equal to the value of the positive charge on the nucleus of an atom or the number of protons present in the nucleus. Since an atom contains equal number of protons and electrons, therefore atomic number of an element may also be defined as the number of electrons present outside the nucleus of its atom. It may be noted that all the atoms of the same element possess the same atomic number which identifies the element. Thus atomic number is a fundamental property of the atom and is generally denoted by the letter 'Z'.

Atomic number (Z) = number of electrons = number of protons

$$Z = e = p$$

2.1.3 Atomic weight

It is also called atomic mass or mass number. The atomic weight is numerically equal to the sum of protons and neutrons present in the nucleus.

Mathematically,

If we let A = atomic weight, then

$$A = e + n$$

$$A = p + n$$

Where p = number of protons, n = number of neutrons and e = number of electrons

For example, the atomic number of iron is 26 (Z=26) and the atomic mass number is 56 (A = 56).

2.1.4 Isotopes

Isotopes are the atoms of the same element that have different atomic mass numbers. Isotopes of an element have the same atomic number but differ in their atomic mass number because, even though they have the same number of electrons and protons (electrically neutral), they differ in the number of neutrons.

For example, atomic number of hydrogen is 1. Hydrogen exists in 3 isotopic forms as given below:

1. Ordinary hydrogen with atomic weight equal to 1.
2. Deuterium with atomic weight equal to 2.
3. Tritium with atomic weight equal to 3.

All the three isotopes of hydrogen contain 1 proton and 1 electron but different number of neutrons. Thus an ordinary hydrogen atom does not have a neutron, the deuterium atom has 1 neutron in its nucleus, while tritium has 2 neutrons in its nucleus.

2.1.5 Periodic table

A periodic table is a tabular display of the chemical elements. The rows of the table are called periods; the columns are called groups.

Mendeleev’s periodic table

It was proposed by Mendeleev, a Russian scientist in 1869, when he discovered that if the elements are arranged in the order of their increasing atomic weights, the elements with similar properties occur at regular intervals. Based on this observation, he gave a law which states that the properties of the elements are periodic functions of their atomic weights.

Mendeleev arranged all the elements (known at that time) in the order of their increasing atomic weights in horizontal rows in such a way that the elements having similar properties come directly under one another in the same vertical columns or groups. In this table, there were seven horizontal rows (or periods) and eight vertical columns (or groups).

Though the Mendeleev’s Periodic Table was of great significance in the study of elements, yet it could not explain the position of isotopes. The isotopes were not given a separate place in the periodic table. In certain cases, it was found that the elements with higher atomic weight came first and the elements with lower atomic weight came later. As a result of this, it was felt that the atomic weight cannot be taken as the basis for the classification of elements.

Modern periodic table

In 1913, Mosley discovered that the properties of elements are periodic functions of their atomic numbers and not atomic weights. The atomic number of an element is fixed and no two elements can have the same atomic number. It is thus obvious that the atomic number is more a fundamental property than the atomic weight for the classification of elements. After this discovery, Mosley proposed a periodic table, in which the elements were arranged in the order of their increasing atomic numbers.

Later on, the periodic table proposed by Mosley was modified by Bohr, which is now popularly known as Modern Periodic Table. In this table, the elements are also arranged in the order of their increasing atomic numbers as shown in figure 2.2 (refer appendix).

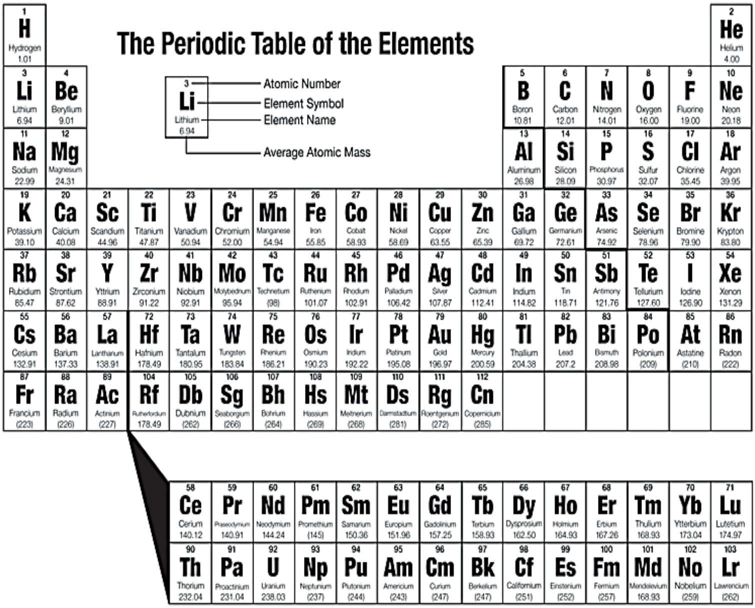


Figure 2.2: Modern Periodic Table

2.1.6 Binding energy

Binding energy is the mechanical energy required to disassemble a whole into separate parts. A bound system typically has a lower potential energy than its constituent parts; this is what keeps the system together, often this means that energy is released upon the creation of a bound state. The usual convention is that this corresponds to a *negative* binding energy.

2.1.7 Atomic bonds

Primary bonds:

Primary bonds are the strongest bonds which hold atoms together. The three types of primary bonds are:

a) Metallic bonds

This type of bond results when each atom of the metal contributes its valence electrons to the formation of an electron cloud that spreads throughout the solid metal. A characteristic of the metallic bond is that the conduction of electricity and heat are produced by the free movement of valence electrons through the metal. All metal conductors show this type of bond. The resistance to the free movement of the electron occurs if it collides with other electrons. This collision of the electrons interferes with the flow of the electrons and accounts for the resistivity in metals. Metallic crystals are malleable and have variable hardness and melting point.

b) Covalent bonds

This bond is formed by sharing of electrons between adjacent atoms. An excellent example of covalent bonding is found in the chlorine molecule. Here, the outer shell of each atom possesses seven electrons. Each chlorine atom would like to gain an electron, and thus form a stable octet. This can be done by sharing of two electrons between pairs of chlorine atoms. Each atom contributes one electron for the sharing process. The diamond from of carbon is an example of this bond where four valence electrons are shared between four neighboring atoms. Other examples of this type of bond are hydrogen, nitrogen etc. Covalent crystals are characterized by poor electrical conductivity and high hardness.

c) Ionic bonds

This bond exists between two unlike atoms. If an electron is transferred from a metallic atom to a non-metallic atom, the two resulting ions are held together by electrostatic attraction. Examples are the sodium and chlorine atoms. Sodium atom gives away its valence electron and becomes a positive ion, while chlorine atom takes the electron to fill its last orbit and becomes a negative ion. Ionic crystals have poor electrical conductivity, high hardness and high melting point.

Secondary bonds: Secondary bonds are much weaker than primary bonds. They often provide a "weak link" for deformation or fracture. The examples for secondary bonds are:

a) Hydrogen bond

Hydrogen bonding is the attraction between hydrogen in a highly polar molecule and the electronegative atom in another polar molecule. In the water molecule, oxygen draws much of the electron density around it, creating positively charged centers at the two hydrogen atoms. These positively charged hydrogen atoms can interact with the negative center around the oxygen in adjacent water molecules. This bond can have a profound influence on the properties of a material, such as boiling and melting points. In addition to having important chemical and physical implications, hydrogen bonding plays an important role in many biological and environmental phenomena. It is responsible for causing ice to be less dense than water, an occurrence that allows fish to survive at the bottom of frozen lakes.

b) Van der Waal's bond

Inert gases and molecules like methane, which have no valence electrons available for crystalline binding, obtain a weak attractive force as a result of polarization of electrical charges. Polarization is displacement of the centers of positive and negative charges in an electrically neutral atom or molecule when as it is brought close to its neighboring atom. Its neighbors also become polarized. The resulting weak electrical attraction between neighboring atoms or molecules is the Van der Waals force. This force can be overcome by the disrupting effect of thermal motion of atoms and molecules at higher temperatures. The additional forces which hold the molecules together are of the Van der Waals type. Elements possessing Van der Waals bond are soft, have poor electrical conductivity and low melting point.

2.2 Atomic arrangements

2.2.1 Crystal

The term ‘crystal’ of a material may be defined as a small body having a regular polyhedral form, bounded by smooth surfaces, which are acquired under the action of its intermolecular forces. The crystals are also known as grains. The boundary, separating the two adjacent grains, is called grain boundary.

2.2.2 Unit cell

A crystal is composed of unit cells. A unit cell contains the smaller number of atoms, which when taken together have all the properties of the crystals of the particular metal. The unit cells are arranged like building blocks in a crystal i.e. they have same orientation and their similar faces are parallel.

A unit cell may also be defined as the smallest parallelepiped which could be transposed in three co-ordinate directions to build the space lattice. The space lattices of various substances differ in size and shape of their unit cells.

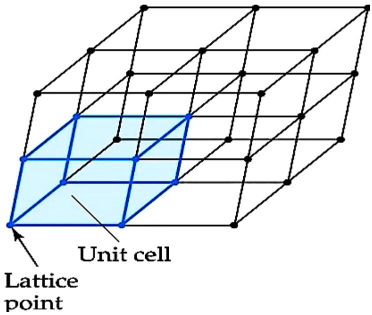


Figure 2.3: Unit cell

2.2.3 Crystalline and amorphous substances

All solid substances are either amorphous solids or crystalline solids. In amorphous solids, the atoms are arranged chaotically i.e., the atoms are not arranged in a systematic order. The common amorphous solids are wood, plastics, glass, paper, rubber etc. In crystalline solids, the atoms making up the crystals arrange themselves in a definite and orderly manner & form. All solid metals such as iron, copper, aluminum etc. are crystalline solids. The definite and orderly manner & forms of atoms producing a geometrical shape in the aggregate is called space lattice or crystal lattice.

2.2.4 Lattice

The three dimensional array formed by the unit cells of a crystal is called lattice.

2.2.5 Lattice point and Array

If each atom in a lattice is replaced by a point; then each point is called a lattice point and the arrangement of the points is referred to as the (three dimensional) lattice array.

2.2.6 Space lattice

A space lattice is defined as an array of points in three dimensions in which every point has surroundings identical to that every other point in the array. The distance between the atom- points is called interatomic or lattice spacing. A space lattice is a conventional geometrical basis by which crystal structures can be destroyed. Each point of a space lattice has identical surroundings.

2.2.7 Lattice parameters of a unit cell

Figure 2.4 shows a unit cell of a three dimensional crystal lattice. It is formed by primitives or intercepts a, b and c along the three axes respectively. The three angles (α, β and γ) are called interfacial angles. A little consideration will show, that both the intercepts and interfacial angles constitute the lattice parameters of the unit cell. It is thus obvious, that if the values of intercepts and interfacial angles are known, actual size and form of unit cell can be determined.

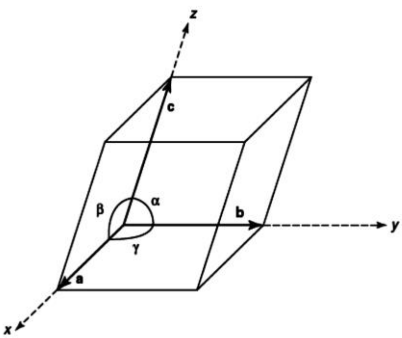


Figure 2.4: Crystal lattice of a unit cell

2.2.8 Types of crystal systems

There are 32 classes of crystal systems based on the geometrical consideration (i.e., symmetry and internal structure). But it is a common practice to divide all the crystal systems into seven groups or basic systems. These seven basic crystal systems are distinguished from one another by the angles between the three axes and the intercepts of the faces along them. These seven basic crystal systems are:

1. Cubic Crystal system

This system includes all those crystals, which have three equal axes and at right angles to each other. The most common examples of cubic crystal system are cube and octahedron.

2. Tetragonal crystal system

This system includes all those crystals, which have three axes at right angles to each other. Two of these axes (say horizontal) are equal, while the third (say vertical) is different (i.e., either longer or shorter than the other two). The most common examples of tetragonal crystal system are regular tetragonal prisms and pyramids.

3. Hexagonal crystal system

This system includes all those crystals, which have four axes. Three of these axes (say horizontal) are equal and meet each other at an angle of 60^0. The fourth axis (say vertical) is different (i.e., either longer or shorter than the other three axes). The most common examples of hexagonal crystal system are regular hexagonal prisms and hexagonal pyramids.

4. Orthorhombic crystal system

This system includes all those crystals, which have three axes at right angles to each other. But all the three axes are essentially of unequal lengths. The most common examples of orthorhombic crystal system are orthorhombic prisms and pyramids.

5. Rhombohedral crystal system

This system includes all those crystals, which have three equal axes. All the three axes are inclined at angles other than right angles with each other. The most common examples of rhombohedral crystal systems are rhombohedral prisms and pyramids.

6. Monoclinic crystal system

This system includes all those crystals, which have three essentially unequal axes. One of these axes is at right angles with the other two. But the other two axes are not at right angles to each other.

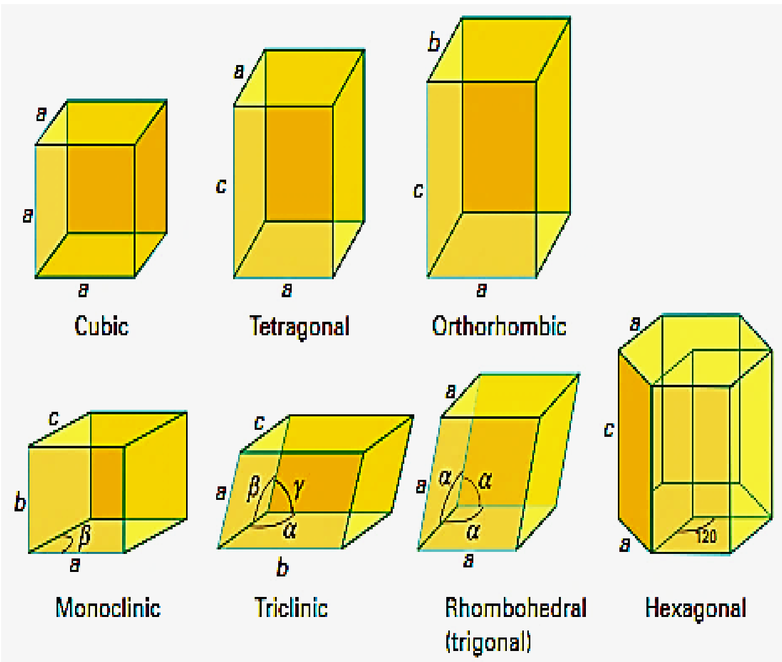


Figure 2.5: Seven crystal systems

7. Triclinic crystal system

This system includes all those crystals, which have three unequal axes and none of them is at right angles to the other two axes. A little consideration will show that all the irregular crystals belong to this crystal system.

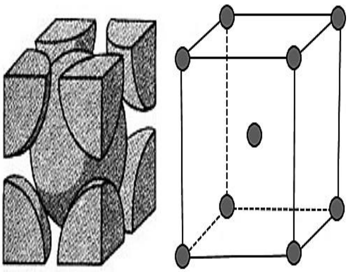
The following table 2 shows the seven basic crystal systems along with their characteristics:

Table 2: Seven crystal systems

S.no.	Crystal system	Axial lengths (a, b and c)	Inter-axial angles (α,β,γ)
1.	Cubic	$a = b = c$	$\alpha = \beta = \gamma = 90^0$
2.	Tetragonal	$a = b \neq c$	$\alpha = \beta = \gamma = 90^0$
3.	Hexagonal	$a = b \neq c$	$\alpha = \beta = 90^0; \gamma = 120^0$
4.	Orthorhombic	$a \neq b \neq c$	$\alpha = \beta = \gamma = 90^0$
5.	Rhombohedral	$a = b = c$	$\alpha = \beta = \gamma \neq 90^0$
6.	Monoclinic	$a \neq b \neq c$	$\alpha = \beta = 90^0; \gamma \neq 90^0$
7.	Triclinic	$a \neq b \neq c$	$\alpha \neq \beta \neq \gamma \neq 90^0$

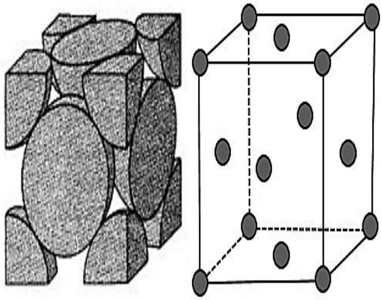
2.2.9 Crystal structure

According to Bravais, there are 14 possible types of space lattices in these 7 crystal systems. The majority of metals have one of three well-packed crystal structures:

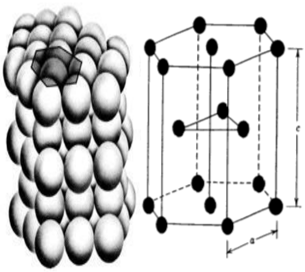


Body-centered cubic (B.C.C): In this type of unit cell of crystal structure the atoms are located at the corners of the cube and one atom at its center. This type of unit cell is found in metals like lithium, sodium, potassium, barium, vanadium, molybdenum, alpha-iron, tungsten, chromium, manganese, tantalum etc.

Face-centered cubic (F.C.C): In this type of unit cell of crystal structure the atoms are located at the corners of the cube and one atom at the center of each face. This type of unit cell is found in metals like gamma-iron, aluminum, copper, lead, silver, nickel, gold, platinum, calcium etc.



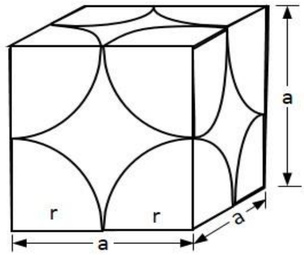
Hexagonal close packed (H.C.P): In this type of crystal structure, the unit cell has an atom at each of the twelve corners of the hexagonal prism, one atom at the center of each of the two hexagonal faces and three atoms in the body of the cell. This type of unit cell is found in metals like zinc, magnesium, cobalt, cadmium, antimony, bismuth, beryllium, titanium, zirconium etc.



2.2.10 Atomic radius

Atomic radius is defined as half the distance between nearest neighbors in a crystal of a pure element. It is possible to calculate the atomic radius by assuming that atoms are spheres in contact in a crystal if the structure and the lattice parameters are known.

(i) Simple cubic (S.C.) structure

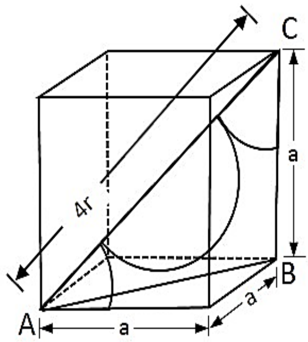


In this structure, atoms touch each other along the lattice.

Therefore, $a = 2r$

$\therefore Atomic\ radius, r = \dfrac{a}{2}$

(ii) Body centered cubic (B.C.C.) structure



In this structure, the atoms touch each other along the diagonal of the cube. Therefore, diagonal of the cube (AC) in this case is 4r.

Also, $AC = \sqrt{AB^2 + BC^2}$

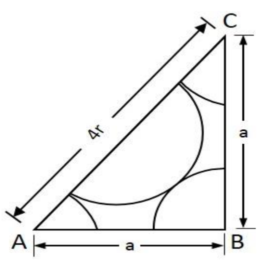
But, $AB = \sqrt{a^2 + a^2} = a\sqrt{2}$

$$AC = 4r = \sqrt{\left(a\sqrt{2}\right)^2 + a^2}$$

$\therefore Atomic\ radius, r = \dfrac{a\sqrt{3}}{4}$

(iii) Face centered cubic (F.C.C.) structure

In this structure, the atoms touch each other along the diagonal of any face of the cube. The diagonal of the cube (AC) is 4r.



Also, $AC^2 = AB^2 + BC^2$

$or, (4r)^2 = a^2 + a^2$

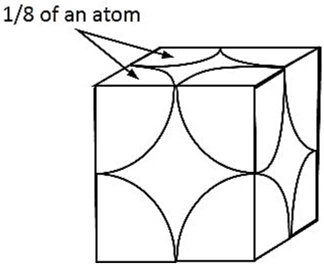
$or, 16r^2 = 2a^2$

$\therefore Atomic\ radius, r = \dfrac{a\sqrt{2}}{4}$

2.2.11 Number of atoms per unit cell

The number of atoms possessed in a unit cell can be calculated if the arrangement of atoms inside the unit cell is known. For three types of cube crystal, the number of atoms can be calculated as follows:

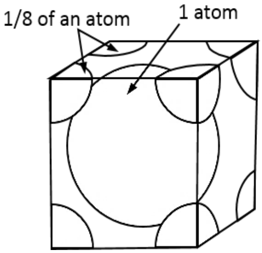
(i) Simple cubic (S.C.) structure



In this case, there are 8 atoms, one at each corner of the cube and all of them are shared by adjoining or surrounding cubes. Hence, the share of cube = 1/8th of each corner atom.

$$\therefore Total\, no.of atoms = \frac{1}{8} \times\ 8 = 1$$

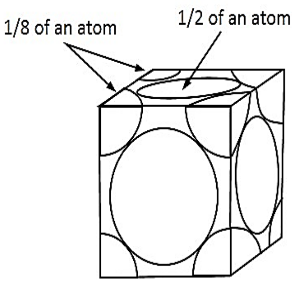
(ii)Body centered cubic (B.C.C.) structure



In this case, there are 8 atoms, one at each corner and are shared by 8 surrounding cubes, plus one center atom at the center of the cube.

$$\therefore Total\, no.of atoms = \frac{1}{8} \times\ 8 + 1\ \ = 2$$

(iii) Face centered cubic (F.C.C.) structure



In this case, there are 8 atoms, one at each corner of the cube plus 6 face centered atoms at 6 planes of the cube. Each corner atom is shared by 8 surrounding cubes, and each face centered atom is shared by two surrounding cubes.

$$\therefore Total\ no.\ of atoms = \frac{1}{8} \times\ 8 + \frac{1}{2} \times 6$$
$$= 1 + 3 = 4$$

2.2.12 Co-ordination number

The number of atoms touching a particular atom or the number of nearest neighbors, is the co-ordination number and is one indication of how closely (tightly) and efficiently atoms are packed together. The co-ordination number for BCC structure is 8 while it is 12 for both FCC and HCP structure.

2.2.13 Atomic packing factor (APF)

Atomic packing factor (also known as density of facing) is defined as the ratio of the volume of atoms per unit cell to the total volume occupied by unit cell.

$$A.P.F. = \frac{Volume\ of\ atoms\ per\ unit\ cell}{Volume\ of\ unit\ cell} = \frac{v}{V}$$

(i) Simple cubic (S.C.) structure

Atoms per unit cell = 1

Volume of one atom, $v = \frac{4\pi r^3}{3}$ where r is atomic radius.

Putting $r = a/2$, we get: $v = \frac{\pi a^3}{6}$

$\therefore$ Atomic packing factor $= \frac{v}{V} = \frac{\frac{\pi a^3}{6}}{a^3} = 0.5238 \approx$ **0.52**

(ii) Body centered cubic (B.C.C.) structure

Atoms per unit cell = 2

Volume of 2 atoms, $v = 2 \times \frac{4\pi r^3}{3}$,where r is atomic radius.

Putting $r = a\sqrt{3}/4$, we get: $v = \frac{\pi a^3\sqrt{3}}{8}$

$\therefore$ Atomic packing factor $= \frac{v}{V} = \frac{\frac{\pi a^3\sqrt{3}}{8}}{a^3} = \mathbf{0.68}$

(iii) Face centered cubic (F.C.C.) structure

Atoms per unit cell = 4

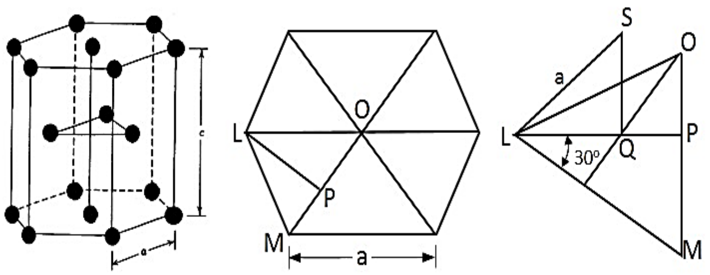
Volume of 4 atoms, $v = 4 \times \frac{4\pi r^3}{3}$ where r is atomic radius.

Putting $r = a\sqrt{2}/4$, we get: $v = \frac{\pi a^3\sqrt{2}}{6}$

$\therefore$ Atomic packing factor = $\frac{v}{V} = \frac{\frac{\pi a^3\sqrt{2}}{6}}{a^3} = \mathbf{0.74}$

(iv) Hexagonal close packed (H.C.P.) structure

The unit cell contains one atom at each corner, one atom each at the center of the two hexagonal faces and three atoms within the body of the cell. The atoms touch each other along the edge of the hexagon. Thus, $a = 2r$.



The top layer contains seven atoms. Each corner atom is shared by six surrounding hexagon cells and the center atom is shared by two surrounding cells. The three atoms within the body of the cell are fully contributing to the cell.

∴ Total no. of atoms in a unit cell = 3/2 + 3/2 + 3 = 6

Calculation of c/a ratio for an ideal hexagonal close packed structure

Let *c* be the height of the unit cell and *a* be the edge of the unit cell.

In $\Delta LMP,\ LP = LM\cos 30° = a\cos 30° = \frac{a\sqrt{3}}{2}$

In $\Delta LQS,\ LS^2 = LQ^2 + QS^2$ ..... *(i)*

Here, $LQ = \frac{2}{3}LP = \frac{2}{3} \times \frac{a\sqrt{3}}{2} = \frac{a}{\sqrt{3}}$

Substituting the appropriate values in equation (i), we get:

$$a^2 = \left(\frac{a}{\sqrt{3}}\right)^2 + \left(\frac{c}{2}\right)^2 = \frac{a^2}{3} + \frac{c^2}{4}$$

$$or, \frac{c^2}{4} = a^2 - \frac{a^2}{3} = \frac{2a^2}{3}$$

$$or, \frac{c^2}{a^2} = \frac{8}{3}$$

$$\therefore \frac{c}{a} = \sqrt{\frac{8}{3}}$$

Volume of unit cell, $V = Area\ of\ the\ base \times Height\ of\ cell$

$$= Six\ times\ the\ area\ of\ \Delta LOM \times c$$

$$= 6\left(\frac{1}{2} \times OM \times LP\right) \times c$$

$$= 6\left(\frac{1}{2} \times a \times \frac{a\sqrt{3}}{2}\right) \times c$$

$$\therefore V = \frac{3\sqrt{3}}{2} a^2 c$$

Volume of all the atoms in a unit cell, $v = 6 \times \frac{4}{3}\pi r^3$

$$or, v = 6 \times \frac{4}{3}\pi\left(\frac{a}{2}\right)^3 = \pi a^3$$

$\therefore$ Atomic packing factor, A.P.F. $= \frac{v}{V}$

$$= \frac{\pi a^3}{\frac{3\sqrt{3}}{2} a^2 c}$$

$$= \frac{2\pi}{3\sqrt{3}} \times \frac{a}{c}$$

$$= \frac{2\pi}{3\sqrt{3}} \times \sqrt{\frac{8}{3}}$$

$$= 0.74$$

Magnesium, Zinc and Cadmium crystallize in this structure.

The atomic packing factor of F.C.C and H.C.P is 74% and hence they are known as *closest packing structures.*

2.2.14 Miller indices

Miller Indices is a system of notation of planes within a crystal of space lattice. They are based on the intercepts of plane with the three crystal axes i.e. edges of the unit cell. The intercepts are measured in terms of the edge lengths or dimensions of the unit cell which are unit distances from the origin along three axes.

The Miller indices of a given crystal plane is determined as follows:

Step 1: Find the intercepts of the plane along the axes a, b, c (The intercepts are measured as multiples of the fundamental vector). i.e. 3, 2, 2

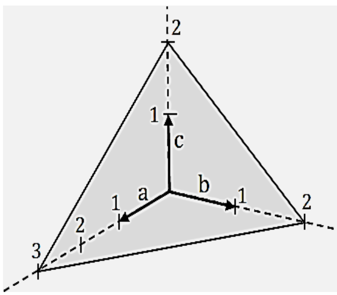


Figure 2.6: Crystal plane

Step 2: Take reciprocals of the intercepts i.e. 1/3, 1/2, 1/2.

Step 3: Convert into smallest integers in the same ratio i.e. 2, 3, 3.

Step 4: Enclose in parentheses, i.e. (2 3 3).

The factor that results in converting the reciprocals of integers may be indicated outside the brackets, but it is usually omitted.

In particular, a family of lattice planes is determined by three integers *h*, *k*, and *ℓ*, the *Miller indices*. They are written as (hkℓ), and each index denotes a plane orthogonal to a direction (h, k, ℓ) in the basis of the reciprocal lattice vectors. By convention, negative integers are written with a bar, as in 3 for −3. The integers are usually written in lowest terms, i.e. their greatest common divisor should be 1. Miller index 100 represents a plane orthogonal to direction *h*; index 010 represents a plane orthogonal to direction *k*, and index 001 represents a plane orthogonal to *ℓ*.

There are also several related notations:

- The notation {hkℓ} denotes the set of all planes that are equivalent to (hkℓ) by the symmetry of the lattice.

In the context of crystal directions (not planes), the corresponding notations are:

- [hkℓ], with square instead of round brackets, denotes a direction in the basis of the *direct* lattice vectors instead of the reciprocal lattice.
- Similarly, the notation ⟨hkℓ⟩ denotes the set of all directions that are equivalent to [hkℓ] by symmetry.

Miller indices were introduced in 1839 by the British mineralogist William Hallowes Miller. The method was also historically known as the Millerian system, and the indices as Millerian, although this is now rare.

2.2.15 Allotropic and polymorphic transformation

Polymorphism is a physical phenomenon where a material may have more than one crystal structure. A material that shows polymorphism exists in more than one type of space lattice in the solid state. If the change in structure is reversible, then the polymorphic change is known as allotropy. The prevailing crystal structure depends on both the temperature and the external pressure. One familiar example is found in carbon: Graphite is the stable polymorph at ambient conditions whereas diamond is formed at extremely high pressures.

The best known example for allotropy is iron. When iron crystallizes at 2800 F it is B.C.C. (delta-iron), at 2554 F the structure changes to F.C.C. (gamma-iron) and at 1670 F it again becomes B.C.C. (alpha-iron).

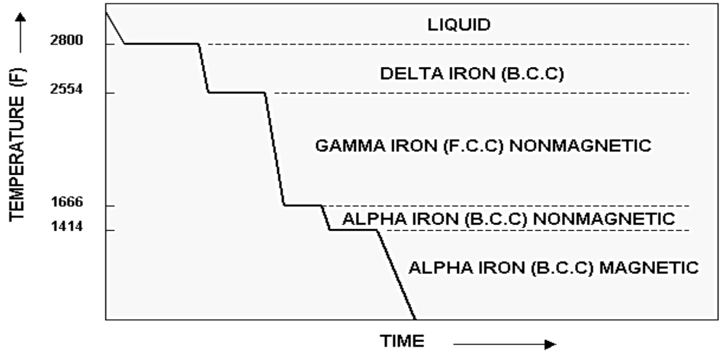


Figure 2.7: Cooling curve for pure iron (Allotropic behavior of pure iron)

2.3 Imperfections in the atomic arrangement

2.3.1 Imperfection

Imperfections are the results of deformation which can be classified into two types.

Elastic deformation

The term elastic deformation may be defined as the process of deformation which appears and disappears simultaneously with the application and removal of stress.

It has been observed that whenever a stress of low magnitude is applied to a piece of metal, it causes displacement of atoms from their original positions. But on the removal of stress, the atoms spring back and occupy their original position. In elastic deformation, the tensile strain is due to a slight elongation of the unit cell in the direction of the tensile load.

Plastic deformation

The term plastic deformation may be defined as the process of permanent deformation which exists in a metal even after the removal of the stress. It is due to this property, that the metals may be subjected to various operations like rolling, forging, drawing, spinning etc. The plastic deformation may occur under the tensile, compressive or torsional stresses. It is a function of stress, temperature and rate of straining. There are two basic modes of plastic deformation, namely slip or gliding and twinning.

The perfectly regular crystal structures that have been considered up to now are called ideal crystals in which atoms are arranged in a regular way. In actual crystals, however, imperfections or defects are always present and their nature and effects are very important in understanding the properties of crystals. These imperfections affect the properties of crystals such as mechanical strength, chemical reactions, electrical properties etc. to a great extent. By controlling lattice imperfections we create stronger metals and alloys, more powerful magnets, improved transistors and solar cells, glassware of striking colors and many other materials of practical importance.

The crystallographic defects are classified as follows:

1. Point Imperfections
2. Line Imperfections
3. Surface Imperfections

2.3.2 Point imperfections

These defects are localized disruptions of the lattice involving one or possibly several atoms. Point defects disturb the perfect arrangement of the surroundings atoms. The typical size of a point defect is one or two atomic diameters. Point imperfections are always present in crystals and their presence results in a decrease in the free energy.

The point defects may be created by thermal fluctuations, quenching from higher temperature, severe deformation of the crystal lattice e.g. hammering or rolling, external bombardment by atoms or high-energy particle etc. The various types of point defects are discussed below:

i) Vacancy: Vacancy is produced when an atom is missing from a normal lattice point. These defects may come up as a result of imperfect packing during the original crystallization. They may also arise from thermal vibrations of the atoms at high temperatures.

ii) Interstitial defect: An Interstitial defect is formed when an extra atom within a crystal structure is inserted into the lattice structure at a site which is not a normal lattice point. The addition of an extra atom is particularly due to low atomic packing factor. The vacancy and interstitial defect are therefore, inverse phenomena.

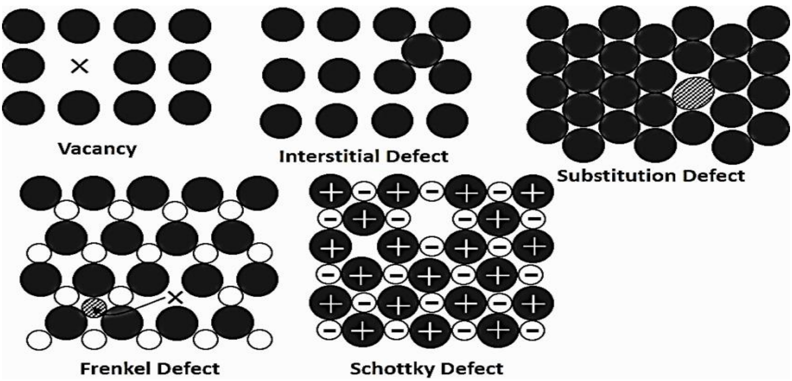


Figure 2.8: Point defects in a crystal

iii) Substitution defect: A substitution defect is introduced when a foreign atom substitutes for a parent atom in the lattice as shown. The substitutional atom remains at the original normal lattice point.

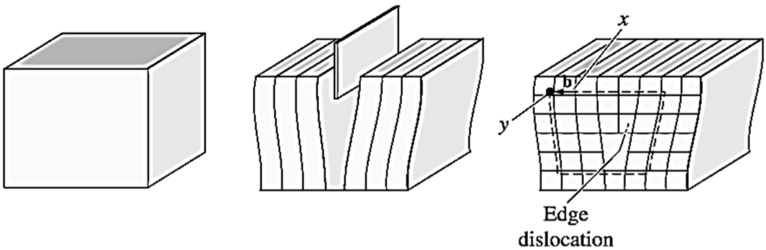
iv) Frenkel defect: An ion dislodged from the lattice into an interstitial site is called Frenkel defect. The interstitial and Frenkel defects are less in number than vacancies because additional energy is required to force the atom into the new position.

v) Schottky Defect: Whenever a pair of positive and negative ions is missing from a crystal, the defect caused is called Schottky defect. In such defect, the crystal is electrically neutral.

2.3.3 Line defects (dislocations)

A linear disturbance of the atomic arrangement, which can move very easily on the slip plane through the crystal is known as dislocation. The dislocation may be caused during growth of crystals from a melt or from a vapor or they may occur during a slip. It is important to note that dislocations cannot end inside a crystal. They must end at a crystal edge or other dislocations, or they must close back on themselves. The two basic types of dislocations are:

i) Edge dislocation: Whenever a half plane of atom is inserted between the planes of atoms in a perfect crystal, the defect so produced is known as edge dislocation. The imperfection may extend in a straight line all the way through the crystal or it may follow an irregular path. The symbol ⊥ represents a positive edge dislocation and the symbol **T** represents a negative edge dislocation.



(a) (b) (c)

Figure 2.9: The perfect crystal in (a) is cut and an extra half plane of atoms is inserted (b) the bottom edge of the extra half plane is an edge dislocation (c) positive edge dislocation

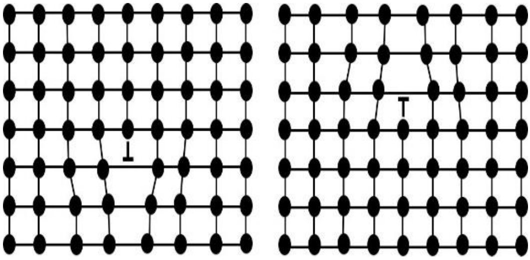


Figure 2.10: Positive and negative edge dislocation

ii) Screw dislocation: Whenever the atoms are displaced in two separate planes perpendicular to each other, the defect so produced is known as screw dislocation. Screw dislocations can be produced by tearing of the crystal parallel to the slip direction. If a screw dislocation is followed all the way around a complete circuit, it would show a slip pattern similar to that of screw thread.

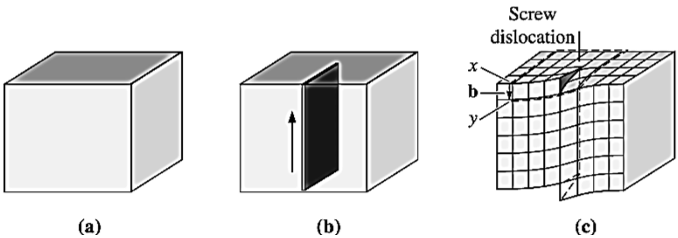


Figure 2.11: The perfect crystal (a) is cut and sheared one atom spacing, (b) and (c) the line along which shearing occurs is a screw dislocation

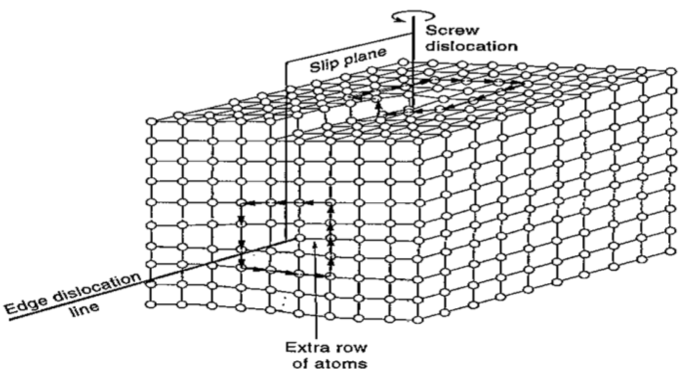


Figure 2.12: Geometry of edge and screw dislocation

2.3.4 Surface imperfections

Surface defects are the two dimensional regions in a crystal. They arise from a change in the stacking of atomic planes on or across a boundary.

i) External defects: The external surface of the material is an imperfection itself, because the atomic bonds do not extend beyond it. The surface atoms have neighbors on one side only, while atoms inside the crystal have neighbors on either side of them. Since these surface atoms are not entirely surrounded by others, they possess higher energy than those of internal atoms.

ii) Internal defects

a) Grain boundaries: Grain boundaries are those imperfections which separate crystals or grains of different orientation in polycrystalline aggregation during nucleation or crystallization. The mismatch of the orientation of adjacent grain produces a less efficient packing of atoms at the boundary. Thus, the atoms along the boundary have energy than those within the grains.

b) Twin boundaries: A twin boundary is a special type of grain boundary across which there is specific mirror lattice symmetry; i.e. atoms on one side of the boundary are located in mirror-image positions of the atoms on the other side. Twin results from atomic displacements that are produced from applied mechanical shear forces (mechanical turns) and also during annealing heat treatments following deformation (annealing twins). Annealing twins are typically found in crystals that have the F.C.C crystal structure, while mechanical twins are observed in B.C.C and H.C.P metals.

c) Tilt boundary: Tilt boundary in reality is a series of aligned dislocations which tend to anchor dislocation movements normally contributing to plastic deformation. Little energy is associated with this type of boundary.

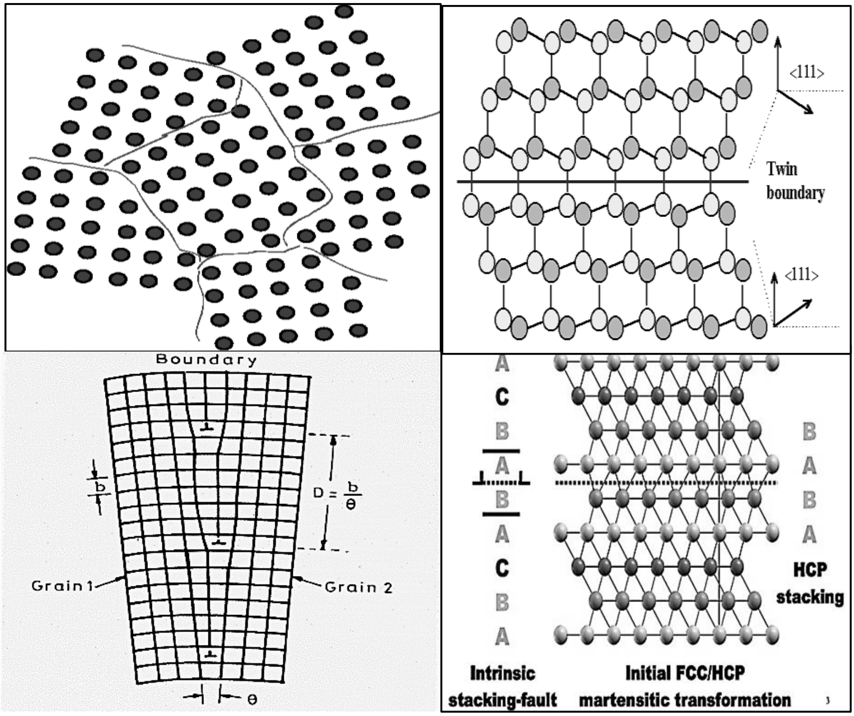


Figure 2.13: a) Grain Boundaries b) twin Boundary c) Tilt boundary d) Stacking faults

d) Stacking faults: It is a surface imperfection that arises from the stacking of one atomic plane out of sequence on another while the lattice on either side of the fault is perfect. Consider the stacking arrangement of plane in a F.C.C crystal: ….ACBA**C**BACB…. If a **C** plane is missing, the stacking sequence becomes ….ACBABACB…. The stacking in the missing region is ….BABA…. which is H.C.P staking. This thin region is a surface imperfection and is called stacking fault.

2.3.5 Volume defects

Volume defects such as cracks may arise in crystals when there is only small electrostatic dissimilarity between the stacking sequences of close packed plane in metals. Presence of a large vacancy or void space, when cluster of atom are missed is also considered as a volume imperfection.

Foreign particles inclusion and non-crystalline regions which have dimensions of the order of 0.20 nm are also called as volume imperfection.

2.3.6 Deformation by slip and twinning

Slip: Slip is defined as a shear deformation that moves atoms by many interatomic distance in one crystal plane over the atoms of another crystal plane. By an external force, parts of the crystal lattice glide along each other, resulting in a changed geometry of the material. Depending on the type of lattice, different slip systems are present in the material. The slip planes are normally the planes with the highest density of atoms, i.e. those which are most widely spaced, and the direction of the slip is the direction in the slip plane corresponding to one of the shortest lattice translation vectors. Often this is the direction in which the atoms are most closely spaced.

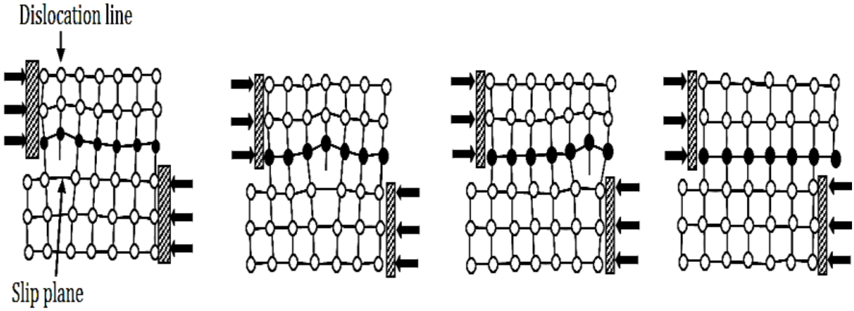


Figure 2.14: Movement of a dislocation through a crystal

A slip plane and a slip direction constitute a slip system. A characteristic shear stress is required to initiate a slip. Slip is an important mode of deformation mechanism in crystals. Slip induces plasticity in most of the metals

Twinning: Twinning is that process by which a portion of the crystal takes up an orientation which makes that portion a mirror image of the parent crystal. Deformation by twinning is most common in H.C.P metals and its effects on others are to move parts of the grains such that they acquire favorable orientation for the slip to occur. Twinning is produced suddenly and is accompanied with sound. The well-known cry of twin is the result of twinning. There are three modes of formation of twinned crystals. *Growth twins* are the result of an interruption or change in the lattice during formation or growth due to a possible deformation from a larger substituting ion. *Annealing* or *transformation twins* are the result of a change in crystal system during cooling as one *form* becomes unstable and the crystal structure must re-organize or *transform* into another more stable form.

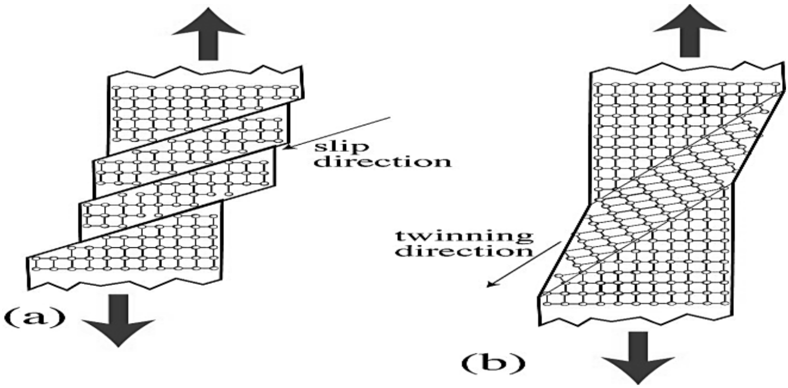


Figure 2.15: (a) Slip Direction (b) Twinning Direction

Deformation or *gliding twins* are the result of stress on the crystal after the crystal has formed. Deformation twinning is a common result of regional metamorphism.

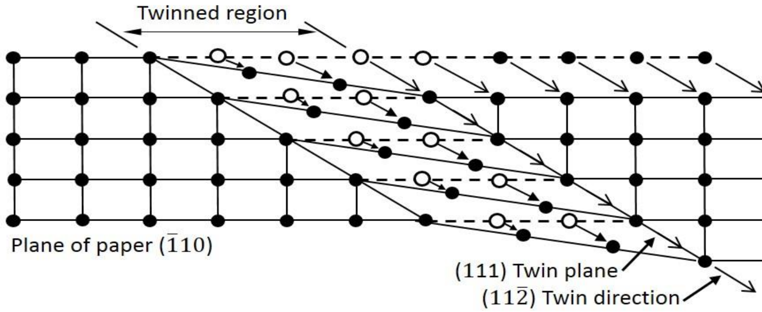


Figure 2.16: Schematic diagram of Twinning in a F.C.C Lattice

Comparison between Slip and Twinning

Slip	Twinning
1. It occurs in discrete multiples of atomic spacing.	1. Atom movements are much less than the atomic spacing.
2. The orientation of the crystal above and below the slip plane is the same after deformation as before.	2. Orientation difference takes place across the twin plane.
3. It occurs over wide planes.	3. Every atomic plane is involved.
4. Slip begins when shearing stress on the slip plane in the slip direction reaches a threshold value called the critical resolved shear stress.	4. There is no critical resolved shear stress for twinning.
5. It takes place in several milliseconds.	5. It takes place in a few micro-seconds.

2.3.7 Significance of imperfections in heat treatment

The role of imperfections in heat treatment is very important. Imperfections account for crystal growth, diffusion mechanism, annealing and precipitation. Besides this, other metallurgical phenomena, such as oxidation, corrosion, yield strength, creep, fatigue and fracture, are governed by imperfections. Imperfections are not always harmful to metals. Sometimes they are generated to obtain the desired properties. For example, carbon is added in steel as interstitial impurity to improve the mechanical properties, and these properties are further improved by different heat treatment processes.

2.3.4 Schmid's law

Schmid's law defines the relationship between shear stress, the applied stress, and the orientation of the slip system. It is an equation for finding the stress in the slip plane given an axial force and the angle of the slip plane. Schmid's law can help to explain the differences in behavior of different metals when subjected to a unidirectional force.

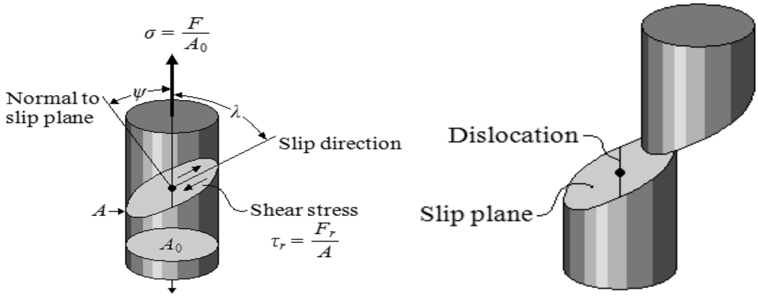


Figure 2.17: Force applied in a cylinder

Mathematically, Schmid's law is given by:

$$\boldsymbol{\tau_r = \sigma \cos\psi \cos\lambda}$$

Where, F = Unidirectional force,

λ = Angle defining slip direction relative to the force

F_r =Shear force

ψ = Angle defining the normal to the slip plane

A =Area of slip plane

τ_r =Resolved shear stress in the slip direction

σ = Unidirectional stress applied to the cylinder

The differences in behavior of metals that have different crystal structures can be understood by examining the force required to initiate the slip process.

Suppose a unidirectional force F is applied to a cylinder of metal that is a single crystal as shown in figure 2.17.

The slip plane and slip direction to the applied force can be oriented by defining the angles λ and ψ. λ is the angle between the slip direction and the applied force, and ψ is the angle between the normal to the slip plane and the applied force.

In order for the dislocation to move in its slip system, a shear force acting in the slip direction must be produced by the applied force. This resolved shear force F_r is given by:

$$\boldsymbol{F_r = F cos\lambda}$$

If the equation is divided by the area of the slip plane,

$A = A_o/cos\psi$,

Schmid's law is obtained:

$$\boldsymbol{\tau_r = \sigma \cos\psi \cos\lambda}$$

Where

$\tau_r = F_r A$ and $\sigma = F A_o$

2.4 Movement of atoms in materials

2.4.1 Fick's first law

Fick's laws of diffusion describe diffusion and can be used to solve for the diffusion coefficient, D. They were derived by Adolf Fick in the year 1855.

Fick's first law relates the diffusive flux to the concentration under the assumption of steady state. It postulates that the flux goes from regions of high concentration to regions of low concentration, with a magnitude that is proportional to the concentration gradient (spatial derivative). In one dimension, the law is mathematically represented as:

$$\boldsymbol{J = -D\frac{\partial \emptyset}{\partial x}}$$

Where

J is the "diffusion flux" [(amount of substance) per unit area per unit time], example (mol./m^2.s). It measures the amount of substance that will flow through a small area during a small time interval.

D is the diffusion coefficient or diffusivity in dimensions of [$length^2$ $time^{-1}$], example (m^2/s)

$\emptyset$ (for ideal mixtures) is the concentration in dimensions of [amount of substance per unit volume], example (mol/m^3)

x is the position [length] (m)

2.4.2 Fick's second law

Fick's second law predicts how diffusion causes the concentration to change with time.

Mathematically,

$$\boldsymbol{\frac{\partial \emptyset}{\partial x} = D\frac{\partial^2 \emptyset}{\partial x^2}}$$

Where

$\emptyset$ is the concentration in dimensions of [(amount of substance) length^{-3}], (mol./m^3)

t is time (s)

D is the diffusion coefficient in dimensions of [length^2 time^{-1}], (m^2/s)

x is the position [length], (m)

Equations based on Fick's law have been commonly used to model transport processes in foods, neurons, biopolymers, pharmaceuticals, porous soils, population dynamics, nuclear materials, semiconductor doping process, etc. Theory of all volumetric methods is based on solutions of Fick's equation.

REVIEW QUESTIONS

1. Briefly introduce the terms atom, atomic number, atomic weight.
2. What are isotopes? Why hydrogen is considered as an isotope?
3. What is periodic table? What are the differences in Mendeleev's and Modern periodic table?
4. Illustrate different types of atomic bonds.
5. What is crystal? Differentiate between crystalline and amorphous substance.
6. Explain the terms unit cell, lattice, space lattice.
7. Explain the seven different types of crystal systems?
8. What is crystal structure? Explain three types of Bravais crystal structure?
9. What are imperfections? Explain different types of imperfection found in materials.
10. Differentiate between slip and twinning?
11. Explain Schmid's law.
12. Explain Fick's first and second law.

CHAPTER 3

MECHANICAL PROPERTIES AND THEIR TESTS

The mechanical properties of metals are strength, elasticity, plasticity, ductility, malleability, toughness, brittleness, hardness, fatigue, fracture and creep.

The factors affecting mechanical properties of a metal are grain size, temperature, heat treatment and atmospheric exposure.

The materials are tested for one or more of the following purposes:

- To assess numerically the fundamental mechanical properties of ductility, malleability, toughness etc.
- To check chemical composition.
- To determine the suitability of a material for a particular application.
- To determine data i.e. force deformation (or stress) values to draw up sets of specifications upon which the engineer can base his design.
- To determine the surface or surface defects in raw materials or processed parts.

The tests on materials may be classified as:

1. **Non-destructive tests**: In this type of test, the component does not break and so even after being tested, it can be used for the purpose for which it was made. E.g. Radiography, ultrasonic, inspection etc.
2. **Destructive tests**: In this type of test, the component or specimen either breaks or remains no longer useful for further use. E.g. tensile test, impact test, torsion test etc.

3.1 Tensile test

Tensile testing, also known as tension testing, is a fundamental material science test in which a sample is subjected to uniaxial tension until failure. The results from the test are commonly used to select a material for an application, for quality control, and to predict how a material will react under other types of forces.

3.1.1 Stress-strain curve

During tensile testing of a material sample, the stress-strain curve is a graphical representation of the relationship between stress derived from measuring the deformation of the sample, i.e. elongation, compression or distortion.

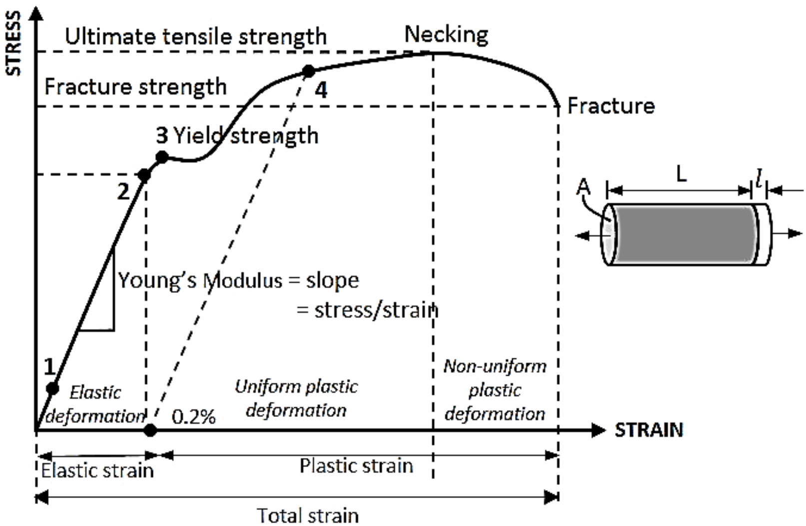


Figure 3.1: Stress-strain curve

1: True elastic limit — 2: Proportionality limit

3: Elastic limit — 4: Offset yield strength, usually defined at e = 0.2%

σ: Engineering stress — ε: Engineering strain

L: Undeformed length l: Elongation

A: Undeformed cross-sectional area F: Uniaxial load

The various properties connected with tensile test are given more elaborately in the following parts.

1. Tangent modulus: The slope of stress-strain curve at any point is called the tangent modulus

2. Modulus of elasticity: Modulus of elasticity or Young's modulus is the measure of material stiffness (given by the slope of stress-strain curve). Modulus of elasticity is determined by the binding forces between atoms (structure intensive property). It is slightly affected by alloying addition, heat treatment or cold work. Young's modulus decreases when there is increase in temperature. The slope of stress-strain curve at any point is called the tangent modulus

3. True elastic limit: The lowest stress at which dislocations move. This definition is rarely used, since dislocations move at very low stresses, and detecting such movement is very difficult.

4. Proportionality limit: It is the maximum stress at which stress remains directly proportional to strain. The proportional limit is determined from the stress-strain curve by drawing straight line tangent at the origin and noting the first deviation of the plot from the line.

5. Elastic limit: The elastic limit is the maximum stress which the material can withstand without causing permanent deformation which remains after removal of stress.

6. Yield strength: The yield strength is the stress at which a material exhibits a specified limiting permanent set. The tensile yield strength indicates resistance to permanent deformation by tensile loads. Because of this and the ease of its measurement the tensile yield strength is used widely as a factor of design.

7. Yield point: The yield point is the stress at which there first occurs a marked increase in strain without an increase in stress.

8. Offset yield strength: The offset yield strength can be determined by the stress corresponding to the intersection of the stress-strain curve and a line parallel to elastic line offset by a strain of 0.2 or 0.1% (e = 0.002 or 0.001). In Great Britain, the offset stress is referred to proof stress either at 0.1 or 0.5% strain. It is used for design and specification purposes to avoid the practical difficulties of measuring the elastic or proportional limit.

9. Tensile strength (ultimate or maximum strength): It is calculated by dividing the maximum load carried by the specimen during a tension test by the original cross-sectional area of the specimen. Tensile strength is widely used design factor, although there is more justification for yield strength.

10. Rupture strength: It is determined by dividing the load at the time of fracture by the original cross-sectional area. If the rupture load is divided by the actual cross-section at the time of fracture, the time rupture strength is obtained.

11. Resilience: Resilience is an ability of a material to absorb energy when elastically deformed and to return it when unloaded. It is usually measured by modulus of resilience (strain energy per unit volume required to stress the material from zero to yield stress. Coil springs are examples of resilient materials.

12. Toughness: The toughness of a material refers to that material's ability to absorb large amounts of stress, deforming plastically before breaking. Polythene and steel are examples of tough materials. On the other hand, brittle material, such as glass is elastic over a very limited range of stresses and undergoes fracture without plastic deformation.

13. Ductility: It measures the amount of deformation that a material can withstand without breaking. There are two ways of expressing ductility. First, we could measure the distance the gage marks on our specimen before and after the test. The % elongation describes the amount that the specimen stretches before fracture.

$$\% \text{ Elongation} = (L_f - L_o)/L_o * 100$$

Where,

L_f is the distance between the gage marks after the specimen breaks.

A second approach is to measure the percent change in cross-sectional area at the point of fracture before and after the test. This % reduction in area describes the amount of thinning that the specimen undergoes during the test.

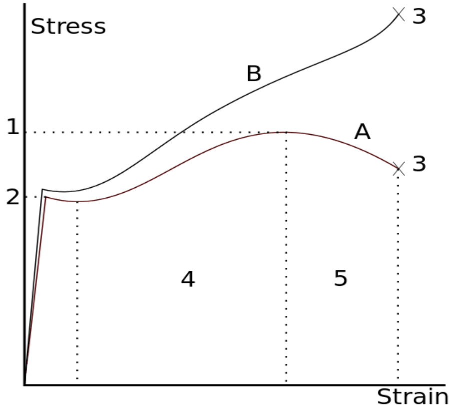
$$\% \text{ Reduction in area} = (A_o - A_f)/A_o * 100$$

Where, A_f is the first cross-sectional area at the fracture surface.

3.1.2 True stress-strain diagram

True stress-strain curve gives the true indication of deformation characteristics because it is based on the instantaneous dimension of the specimen. The true stress-stain curve is also known as *flow curve.*

In engineering stress-strain curve, stress drops down after necking since it is based on the original area. In true stress-strain curve, the stress however increases after necking since the cross sectional area of the specimen decreases rapidly after necking which can be seen in the figure 3.2.



1– Ultimate Strength

2 – Yield strength

3 – Rupture

4– Strain hardening region

5– Necking region

A– Apparent stress (F/A_o)

B– Actual stress (F/A)

Figure 3.2: True stress-strain curve vs. engineering stress-strain curve

3.1.3 Engineering stress-strain diagram

3.1.3.1 Ductile materials

Ductile materials generally exhibits a very linear stress–strain relationship up to a well-defined yield point. After the yield point, the curve typically decreases slightly. As deformation continues, the stress increases on account of strain hardening until it reaches the ultimate strength. Until this point, the cross-sectional area decreases uniformly because of Poisson contractions. The actual rupture point is in the same vertical line as the visual rupture point.

However, beyond this point a neck forms where the local cross-sectional area decreases more quickly than the rest of the sample resulting in an increase in the true stress as shown in figure 3.1. On an engineering stress–strain curve this is seen as a decrease in the apparent stress. However, if the curve is plotted in terms of true stress and true strain the stress will continue to rise until failure. Eventually the neck becomes unstable and the specimen ruptures (fractures).

3.1.3.2 Brittle materials

Brittle materials such as concrete or carbon fiber do not have a yield point, and do not strain-harden. Therefore the ultimate strength and breaking strength are the same. A typical stress-strain curve is shown in figure 3.2. Typical brittle materials like glass do not show any plastic deformation but fail while the deformation is elastic. One of the characteristics of a brittle failure is that the two broken parts can be reassembled to produce the same shape as the original component as there will not be a neck formation like in the case of ductile materials. A typical stress-strain curve for a brittle material will be linear. Testing of several identical specimen, cast iron, or soil, tensile strength is negligible compared to the compressive strength and it is assumed zero for many engineering applications.

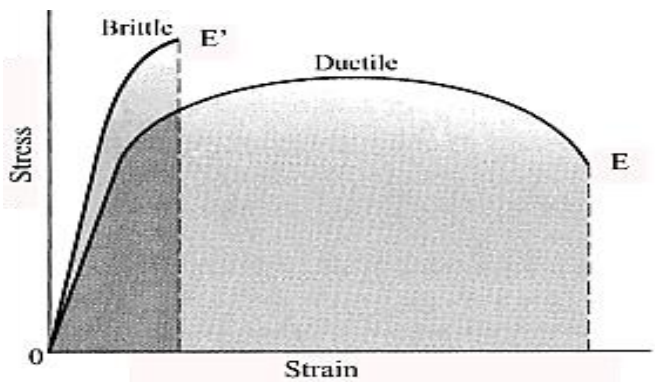


Figure 3.3: Ductile vs. brittle materials

3.1.4 Effect of temperature on stress-strain curve

Temperature strongly affects the stress-strain curve and the flow and fracture properties. With the increase in temperature, strength of material decreases but the ductility increases and vice-versa. Thermally activated processes assist deformation (dislocation motion) and reduce strength at elevated temperatures. Structural changes can occur at certain temperature ranges (high temp/ long term exposure) to alter the general behavior.

3.1.5 Brittle behavior and notch effects

The behavior of materials can be broadly classified into two categories; brittle and ductile. Glass and cast iron fall in the class of brittle materials. Brittle materials often have relatively large Young's moduli and ultimate stresses in comparison to ductile materials. These differences are a major consideration for design.

The sensitivity of metals and alloys to notch effects is termed as *notch sensitivity*. This sensitivity is quantified through the ratio of notch strength to smooth bar tensile strength. Metals or alloys that are notch sensitive have the ratio less than one. Smooth bar tensile data for these materials are not satisfactory predictors of material behavior under service conditions. Tough, ductile metals or alloys frequently are notch-strengthened and have notch sensitivity ratios greater than one, thus the standard tensile test is a conservative predictor of performance for these materials.

3.2 Hardness test

Hardness is the property of a material that enables it to resist plastic deformation, usually by penetration. However, the term hardness may also refer to resistance to bending, scratching, abrasion or cutting. Hardness is not an intrinsic material property dictated by precise definitions in terms of fundamental units of mass, length and time. A hardness property value is the result of a defined measurement procedure. Hardness tests can be used for many engineering applications to achieve basic requirement of mechanical property. For examples: surface treatments where surface hardness has much been improved, powder metallurgy, fabricated parts like forgings, rolled plates, extrusions, machined parts.

3.2.1 Main hardness testing methods

There are three general types of hardness measurements.

1. **Scratch hardness**: A. Martens in 1890 defined scratch hardness or the load in grams under which 90° diamond will produce a scratch 0.01 mm wide. According to Mohr's test, iron and steel range in hardness from 4 to 9 depending upon the heat treatment and the alloy content. It is important to mineralogist and not suited for metals.

2. **Indentation hardness**: This type of test is carried out in major engineering metals of interest. This includes Brinell, Rockwell, Vickers, Meyer hardness test.

3. **Rebound or dynamic hardness**: The indenter is dropped onto the metal surface and hardness is expressed as energy of impact. This test was carried out by A.F. shore in 1910. In this test a steel cylinder hammer is dropped from a height of 25 cm. through a glass tube on the surface to be tested. The height of the rebound is used as a measure of hardness of surface. The surface to be tested should be smooth, free from oil and tube should be truly vertical.

For practical and calibration reasons, each of these methods is divided into a range of scales, defined by a combination of applied load and indenter geometry.

3.2.2 Brinell hardness test

J.A. Brinell introduced the first standardized indentation hardness test in 1900. The Brinell hardness test consists in indenting the metal surface with a 10mm diameter steel ball at a load range of 500-3000 kg, depending of hardness of particular materials.

The load is applied for a standard time (30 seconds), and the diameter of the indentation is measured giving an average value of two readings of the diameter of the indentation at right angle**.**

The Brinell hardness number (**BHN** or $\mathbf{H_B}$) is expressed as the load **P** divided by the surface area of the indentation.

$$\boldsymbol{BHN} = \frac{\mathrm{P}}{\left(\frac{\pi D}{2}\right)\left(D - \sqrt{D^2 - d^2}\right)} = \frac{\mathrm{P}}{\pi \mathrm{Dt}}$$

Where, P is applied load (kg),

D is diameter of ball (mm),

d is diameter of indentation (mm) and

t is depth of impression (mm).

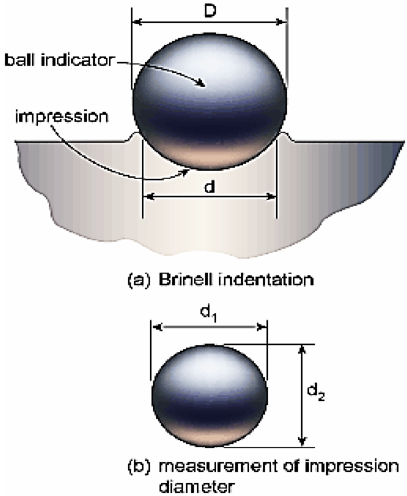


Figure 3.4: Brinell hardness test

Advantages and disadvantages of Brinell hardness Test

- Large indentation averages out local heterogeneities of microstructure.
- Different loads are used to cover a wide range of hardness of commercial metals.
- Brinell hardness test is less influenced by surface scratches and roughness than other hardness tests.
- The test has limitation on small specimens or in critically stressed parts where indentation could be a possible site of failure.

3.2.3 Rockwell hardness test

Rockwell hardness test is the most widely used hardness test and generally accepted due to its speed, freedom from personal error, ability to distinguish small hardness difference and small size indentation. The hardness is measured according to the depth of indentation under a constant load.

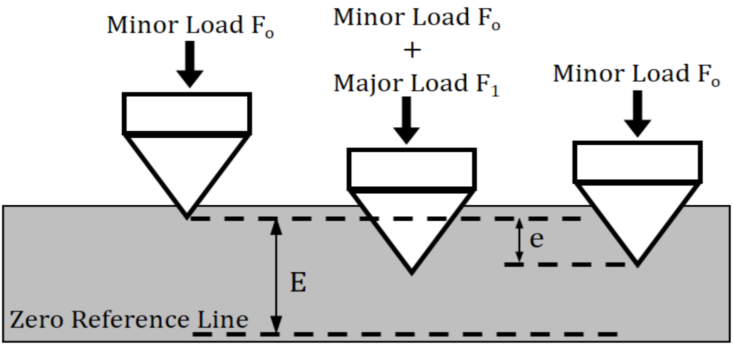


Figure 3.5: Rockwell hardness test

To carry out this test, firstly the surface area to be measured is positioned close to the indenter and the minor load is applied and a zero reference position is established. The major load is applied for a specified time period (*dwell time*) beyond zero and the major load is released leaving behind the minor load applied.

The dial contains 100 divisions, each division representing a penetration of 0.002 mm. The Rockwell number represents the difference in depth from the zero reference position as a result of the applied major load and deeper the indentation, softer is the material.

Rockwell hardness number (**RHN**) can be represented in different scale A, B, C… depending on types of indenters and major loads used. The scale is usable for materials from annealed brass to cemented carbides. Other scales are available for special purposes.

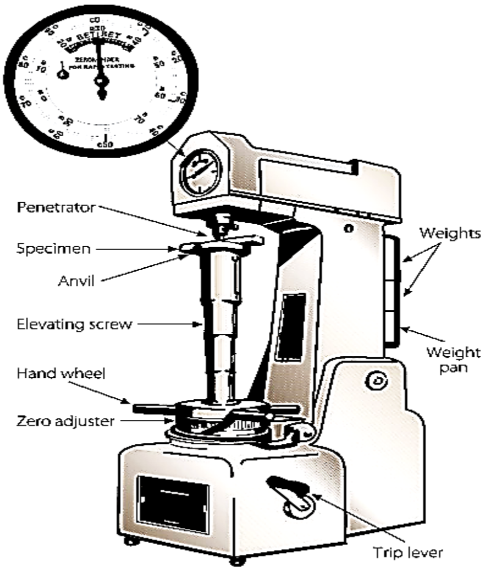


Figure 3.6: Rockwell hardness tester

The instructions for Rockwell hardness test are listed below:

- Cleaned and well seated indenter and anvil.
- Surface which is clean and dry, smooth and free from oxide.
- Flat surface, which is perpendicular to the indenter.

- Cylindrical surface gives low readings, depending on the curvature.
- Thickness should be 10 times higher than the depth of the indenter.
- Loading speed should be standardized.
- The spacing between the indentations should be 3 or 5 times the diameter of the indentation.

3.2.4 Vickers hardness test

Vickers hardness test uses a square-base diamond pyramid as the indenter with the included angle between opposite faces of the pyramid of 136°. The Vickers hardness number (VHN) is defined as the load divided by the surface area of the indentation.

$$\boldsymbol{VHN} = \frac{2P\sin\left(\frac{\theta}{2}\right)}{L^2} = \frac{1.854P}{L^2}$$

Where,

P is applied load, (kg)

L is the average length of diagonals, (mm)

θ is the angle between the opposite faces of diamond = 136°

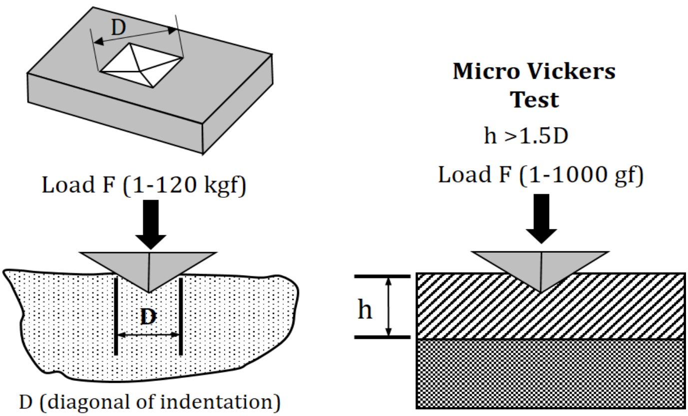


Figure 3.7: Vickers hardness test specifications

Vickers hardness test uses the loads ranging from 1-120kgf, applied for between 10 and 15 seconds. A fairly wide acceptance for research work should be provided because it provides a continuous scale of hardness, for a given load. Since, VHN = 5-1,500 can be obtained at the same load level that makes ease for comparison.

3.2.5 Knoop test

The Knoop indenter (diamond-shape) is used for measuring in a small area, such as at the cross section of the heat-treated metal surface. The Knoop Hardness Number (**KHN**) is the applied load divided by the uncovered projected area of the indentation.

$$\mathbf{KHN} = \frac{\mathrm{P}}{A_P} = \frac{P}{L^2C}$$

Where

P = applied load, kg

A_P = uncovered projected area of the indentation, mm^2

L = length of long diagonal, mm

C = a constant for each indenter supplied by manufacturer

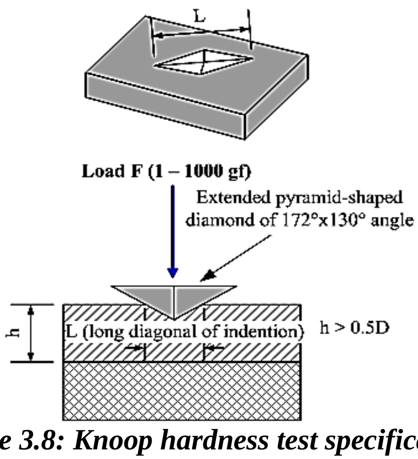


Figure 3.8: Knoop hardness test specifications

3.2.6 Microhardness test

Microhardness Testing of metals,composites and ceramics are employed where a 'macro' hardness test is not usable. Microhardness tests can be used to provide necessary data when measuring individual microstructures within a larger matrix, or testing very thin foil like materials, or when determining the hardness gradient of a specimen along a cross section.

Figure 3.9: Micro-hardness tester

The term Microhardness testingusually refers to static indentations made by loads of 1kgf or less. The Baby Brinell Hardness Test uses a 1mm carbide ball, while the Vickers Hardness Test employs a diamond with an apical angle of 136°, and the Knoop Hardness Test uses a narrow rhombus shaped diamond indenter. The test surface usually must be highly polished. The smaller the force applied, the higher the metallographic finish required.

Microscopes with a magnification of around 500x are required to accurately measure the indentationsproduced.

3.2.7 Hardness conversion table

Hardness conversions are empirical relationships for Brinell, Rockwell and Vickers hardness values. These hardness conversions are applicable to heat-treated carbon and alloy steels in many heat treatment conditions (or alloys with similar elastic modulus). For soft metals, indentation of hardness depends on the strain hardening behavior of the materials.

ROCKWELL				SUPERFICIAL ROCKWELL			BRINELL		VICKERS OR FIRTH DIAMOND HARDNESS NUMBER	SCLERO-SCOPE	TENSILE STRENGTH
							10 m/m Ball				
Diamond Brale			1/16" Bull	"N" Brale Penetrater			3000 kgm Load				
150 kgm C Scale	60 kgm A Scale	100 kgm D Scale	100 kgm B Scale	15 kg Load 15N	30 kg Load 30N	45 kg Load 45N	Diam. Of Ball Impression in m/m	Hardness Number			Equivalent 1000 lb. Sq. In.
80	92	87		97	92	87			1865		
79	92	86			92	87			1787		
78	91	85		96	91	86			1710		
77	91	84			91	85			1633		
76	90	83		96	90	84			1556		
75	90	83			89	83			1478		
74	89	82		95	89	82			1400		
73	89	81			88	81			1323		
72	88	80		95	87	80			1245		
71	87	80			87	79			1160	99	
70	87	79		94	86	78			1076	98	
69	86	78		94	85	77			1004	97	
68	86	77			85	79			942	96	
67	85	76		93	84	75			894	95	
66	85	76		93	83	73			854	93	
65	84	75		92	82	72	2.25	745	820	91	
64	84	74			81	74	2.3	710	789	88	
63	83	73		92	80	70	2.3	710	763	87	
62	83	73		91	79	69	2.35	682	746	85	
61	82	72		91	79	68	2.35	682	720	83	
60	81	71		90	78	67	2.4	653	697	82	
59	81	70		90	77	66	2.45	627	674	80	326
58	80	69		89	76	65	2.55	578	653	78	315
57	80	69		89	75	63	2.55	578	633	77	304
56	79	68		88	74	62	2.6	555	613	75	294
55	79	67		88	73	61	2.6	555	595	74	287
54	78	66		87	72	60	2.65	534	577	72	279
53	77	65		87	71	59	2.7	514	560	71	269
52	77	65		86	70	57	2.75	495	544	69	261
51	76	64		86	69	56	2.75	495	528	68	254

Figure 3.10: Hardness conversion table (refer appendix)

Hardness at elevated temperatures gives a good indication of potential usefulness of an alloy for high-temperature strength applications. Hot hardness tester uses a Vickers indenter made of sapphire and with provisions for testing in either vacuum or an inert atmosphere. The temperature dependence of hardness could be expressed as follows;

$$H = Ae^{-BT}$$

Where

H = Hardness, kgf.mm^2

T = Test temperature, K

A, B = constants

3.3 Impact test

Impact test is a method for determining behavior of material subjected to shock loading in bending tension, or torsion. An impact test signifies toughness of material that is ability of material, to absorb energy during plastic deformation. The test measures the energy necessary to fracture a standard notch bar by applying an impulse load i.e. notch toughness of material under shock loading.

The pendulum type impact testing machine is generally used for conducting notched bar impact tests. The following type of impact tests are performed on this machine.

1. Izod Test
2. Charpy Test

3.3.1 Toughness

The toughness of a material refers to that material's ability to absorb large amounts of stress, deforming plastically before breaking or the ability to withstand occasional stresses above the yield stress without fracture. It can be simply defined by the area under the stress-strain curve (amount of work per unit volume that the material can withstand without failure). Polythene and steel are examples of tough materials. The structural steel although has a lower yield point but more ductile than high carbon spring steel. So, structural steel is therefore tougher.

$$\text{Toughness} = \text{Strength} + \text{Ductility}$$

3.3.2 Izod test

The test uses a cantilever test piece of 10mm x 10mm section specimen having standard 45° notch 2 mm deep. This is broken by means of a swinging pendulum which is allowed to fall from a certain height to cause an impact load on the specimen. This angle rise of the pendulum after rupture of the specimen or energy to rupture the specimen is indicated on the graduated scale by a pointer. The energy required to rupture the specimen is the function of the angle of rise.

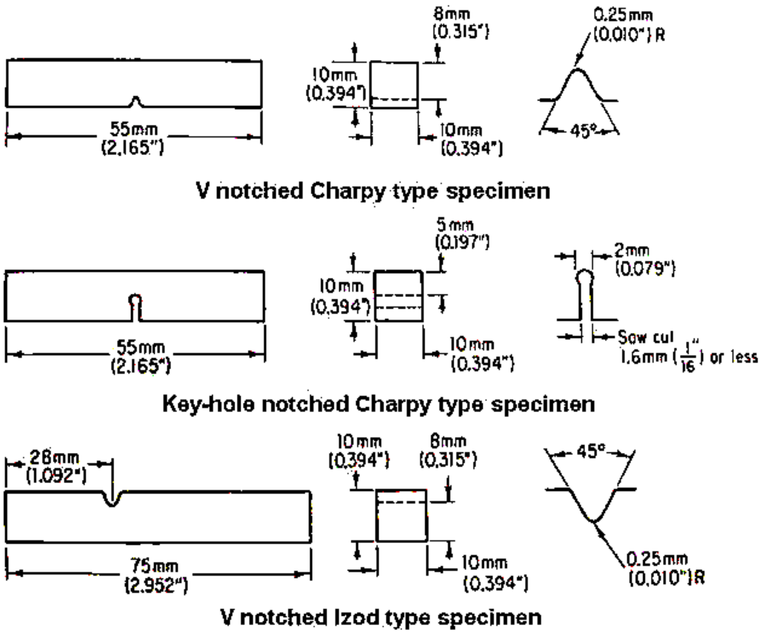


Figure 3.11: Charpy and Izod specimens

3.3.3 Charpy test

This test is more common than Izod test and it uses simply supported test piece of 10mm x 10mm section. The specimen is placed on supports or anvil so that the blow of striker is opposite to the notch.

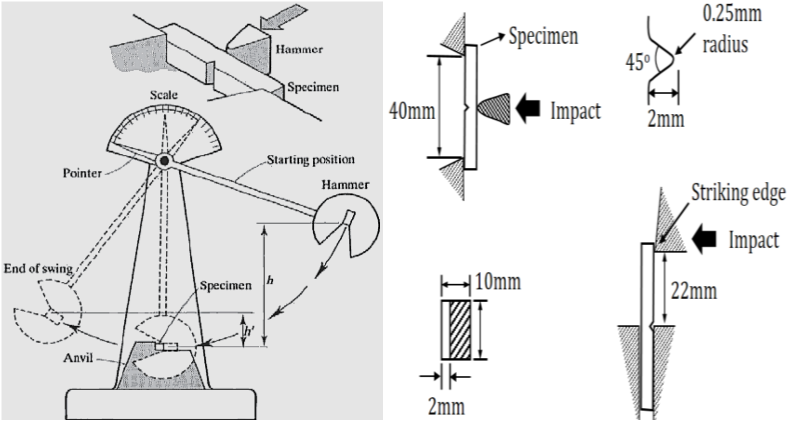


Figure 3.12: Experimental set up

The energy used in rupturing the specimen in both Charpy and Izod tests is calculated as follows:

Initial energy

$$= Wh = W(R - R\cos\alpha) = WR\,(1 - \cos\alpha)$$

Energy after rupture

$$= Wh' = W\,(R - R\cos\beta) = WR\,(1 - \cos\beta)$$

Energy used to rupture specimen

$$= Wh - Wh' = WR\,[(1 - \cos\alpha) - (1 - \cos\beta)]$$
$$= WR\,(\cos\beta - \cos\alpha)$$

Where,

h = Height of fall of center of gravity of pendulum/striker

α = Angle of fall

h' = Height of rise of center of gravity of pendulum/striker

β = Angle of rise

R = Distance from C.G. of pendulum/striker to axis of rotation O.

W = Weight of pendulum/striker.

The principal advantage of the Charpy V-notch impact test is that it is a relatively simple test that utilizes a relatively cheap, small test specimen. Tests can readily be carried out over a range of sub ambient temperatures. Moreover, the design of the test specimen is well suited for measuring differences in notch toughness in low-strength materials such as structural steels. The test is used for comparing the influence of alloy studies and heat treatment on notch toughness. It frequently is used for quality control and material acceptance purposes.

3.3.4 Notch sensitivity

It is a measure of the reduction in strength of a metal caused by the presence of a notch or the extent to which the sensitivity of a material to fracture is increased by the presence of a surface in homogeneity such as a notch, a sudden change in section, a crack, or a scratch. Low notch sensitivity is usually associated with ductile materials, and high notch sensitivity with brittle materials.

3.3.5 Significance of transition-temperature curve

The chief engineering use of the Charpy test is in selecting materials which are resistant to brittle fracture by means of transition-temperature curves. The design philosophy is to select a material which has sufficient notch toughness when subjected to severe service conditions so that the load-carrying ability of the structural member can be calculated by standard strength of materials methods without considering the fracture properties of the material or stress concentration effects of cracks or flaws.

The transition-temperature behavior of a wide spectrum of materials falls into the three categories. Medium- and low-strength F.C.C metals and most H.C.P metals have such high notch toughness that brittle fracture is not a problem unless there is some special reactive chemical environment. High-strength materials (S_0> E/150) have such low notch toughness that brittle fracture can occur at nominal stresses in the elastic range at all temperatures and strain rates when flaws ace present. High-strength steel, aluminum and titanium alloys fall into this category. At low temperature fracture occurs by brittle cleavage, while at higher temperatures fracture occurs by low-energy rupture. It is under these conditions that fracture mechanics analysis is useful and appropriate. The notch toughness of low- and medium-strength B.C.C metals, as well as Beryllium (Be), Zinc (Zn), and ceramic materials is strongly dependent on temperature. At low temperature the fracture occurs by cleavage while at high temperature the fracture occurs by ductile rupture.

Thus, there is a transition from notch brittle to notch tough behavior with increasing temperature. In metals this transition occurs at 0.1 to 0.2 of the absolute melting temperature T_m, while in ceramics the transition occurs at about 0.5 to 0.7 T_m. A well-defined criterion is to base the transition temperature on the temperature at which the fracture becomes 100 percent cleavage. This point is known as nil ductility temperature (NDT). The NDT is the temperature at which fracture initiates with essentially no prior plastic deformation. Below the NDT the probability of ductile fracture is negligible.

3.4 Fatigue test

3.4.1 Fatigue failure

In materials science, fatigue is the progressive and localized structural damage that occurs when a material is subjected to cyclic loading. The nominal maximum stress values are less than the ultimate tensile stress limit, and may be below the yield stress limit of the material. Fatigue occurs, when a material is subjected to repeated loading and unloading.

If the loads are above a certain threshold, microscopic cracks will begin to form at the stress concentrators such as the surface, persistent slip bands (PSBs), and grain interfaces. Eventually a crack will reach a critical size, and the structure will suddenly fracture. The shape of the structure will significantly affect the fatigue life; square holes or sharp corners will lead to elevated local stresses where fatigue cracks can initiate. Round holes and smooth transitions or fillets are therefore important to increase the fatigue strength of the structure.

Types of fatigue loading

Alternating /reversing stress

In determining fatigue stress levels using standard test equipment the test specimens are subject to alternative/reversed stress levels. The cyclic stress varies from σ_a tensile to $\sigma_a$compressive. The mean stress is equal to zero. The fatigue strength values for metals are based on this loading condition with the published fatigue strength values S'$_n$, representing the material fatigue strength property, being the σ_{amp}value at failure.

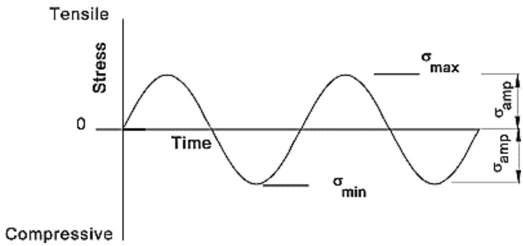


Figure 3.13: Alternating/reversing stress

Repeated stress

In the repeated stress loading condition the stress varies between zero and a maximum tensile (or compressive) stress in a cyclic manner.

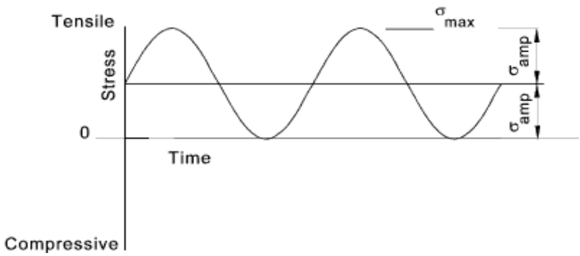


Figure 3.14: Repeated stress

Combined Steady and Cyclic Stress

A stress condition normally experienced in practice is a cyclic stress imposed on a steady load stress.

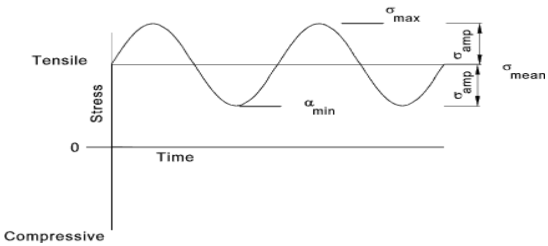


Figure 3.15: Combined steady and cyclic stress

3.4.2 S-N curve

A common method to measure the resistance to fatigue is the rotating cantilever beam test. One end of a machined cylindrical specimen is mounted in a motor-driven chuck. A weight is suspended from the other end.

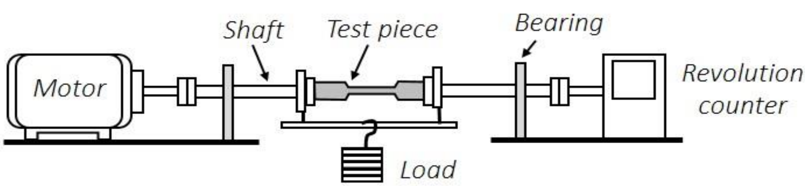


Figure 3.16: Rotating beam fatigue testing machine

The specimen initially has tensile force acting on the top surface, while the bottom surface is compressed. After the specimen turns 90°, the locations that were originally in tension are now in compression. Thus, the stress at any one point goes through a complete cycle from zero stress to maximum tensile stress to zero stress to maximum compressive stress.

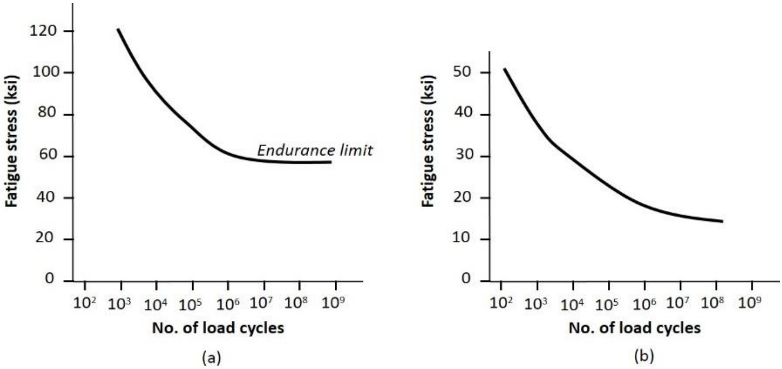


Figure 3.17: (a) S-N curve, steel and stainless steel (b) S-N curve, aluminum

After sufficient number of cycles, the specimen may fail. Generally, a series of specimens are tested at different applied stresses and the stress plotted versus the number of cycles at a particular stress and the endurance limit for the material. The fatigue life tells us how long a component survives when a stress is repeatedly applied to the material.

3.4.3 Endurance limit

If we are designing a tool steel part that must undergo 100,000 cycles during its lifetime, then the part must be designed so that the applied stress is lower than 90,000 psi. The endurance limit is stress below which failure by fatigue never occurs. To prevent the tool steep part from failing, we must be sure that the applied stress is below 60,000 psi.

3.4.4 Fatigue strength versus fatigue limit

The American Society for Testing and Materials (ASTM) defines fatigue strength (S_{Nf}), as the value of stress at which failure occurs after N_f cycles, and fatigue limit (S_f), as the limiting value of stress at which failure occurs as N_f becomes very large. ASTM does not define endurance limit, the stress value below which the material will withstand many load cycles, but implies that it is similar to fatigue limit.

3.4.5 Preventions

The most effective method of improving fatigue performance is improvements in design:

- Eliminate or reduce stress raisers by streamlining the part.
- Avoid sharp surface tears resulting from punching, stamping, shearing, or other processes.
- Prevent the development of surface discontinuities during processing.
- Reduce or eliminate tensile residual stresses caused by manufacturing.
- Improve the details of fabrication and fastening procedures.

3.5 Creep test

Creep is the slow plastic deformation of metals under constant stresses (the applied stress is less than the yield strength at that temperature) or under prolonged loading usually at high temperature. It can take place and lead to fracture at static stresses much smaller than those which will break the specimen by loading it quickly. Creep is especially taken care of while designing I.C.

engines, boilers and turbines. Creep is more severe in materials that are subjected to heat for long periods, and near melting point. Creep always increases with temperature.

3.5.1 Creep failure

Creep is a deformation mechanism that may or may not constitute a failure mode. Moderate creep in concrete is sometimes welcomed because it relieves tensile stresses that might otherwise lead to cracking. The temperature range in which creep deformation may occur differs in various materials. For example, tungsten requires a temperature in the thousands of degrees before creep deformation can occur while ice will creep near 0 °C (32 °F).

The rate of this deformation is a function of the material properties, exposure time, exposure temperature and the applied structural load. Depending on the magnitude of the applied stress and its duration, the deformation may become so large that a component can no longer perform its function - for example creep of a turbine blade will cause the blade to contact the casing, resulting in the failure of the blade. Creep is usually of concern to engineers and metallurgists when evaluating components that operate under high stresses or high temperatures.

The installation for creep test is shown in figure 3.18. The specimen to be tested is placed in the electric furnace where it is heated to a given temperature and is constantly subjected to a load. The strain variations are measured with strain gauges.

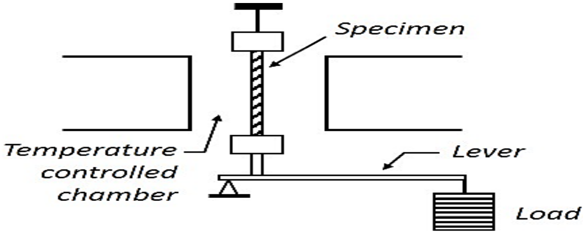


Figure 3.18: Schematic of creep test

3.5.2 Creep and stress rupture curve

As soon as stress is applied, the specimen elastically elongates a small amount E_0 depending on the applied stress and the modulus of elasticity of a material at the high temperature. The plot of strain with time, i.e. the creep curve will be as shown in figure 3.19.

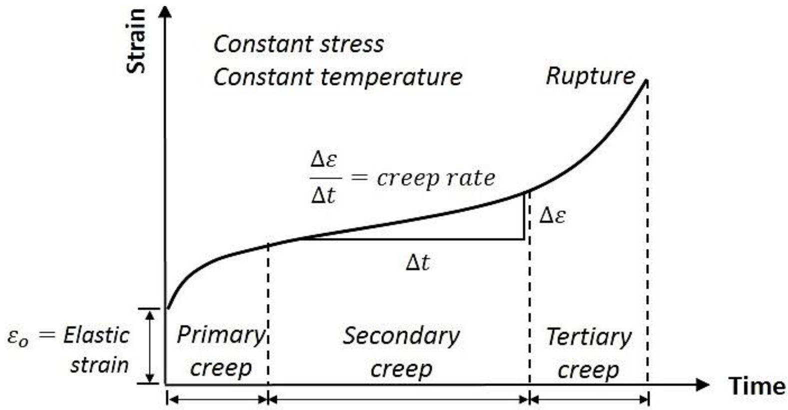


Figure 3.15: Creep Rupture Curve

The creep curve can be studied in the following three stages:

(i) Primary Creep: This is the initial part of the curve. In this part, the curve indicates rapid extension of the specimen. This part of the curve helps us in providing proper clearances in the various components of machines. Primary creep represents a transient stage in which the resistance of the metal increases due to its own deformation. It forms part of the total extension reached in a given time affecting clearance.

(ii)Secondary Creep: This state of the curve indicates the period of extension which the creep occurs at a more or less constant rate having its minimum creep rate. Secondary creep is the result of balance between processes of strain hardening and recovery. It is the important part of the curve and covers the whole of estimated service life of the alloy.

(iii) Tertiary Creep: In this stage, the rate of extension or strain rate accelerates, rapidly leading to rupture finally. In this stage, the uses of alloys should be avoided. Tertiary creep is considered to be due to necking.

3.5.3 Effect of temperature and stress level

As the temperature or the applied stress increases, the creep curve shifts upwards and the time duration of the various stages reduces significantly as shown in figure 3.11. This indicates that the value of creep increases with the increase in temperature as well as stress. Therefore, the materials to be used at a high temperature must have high melting point.

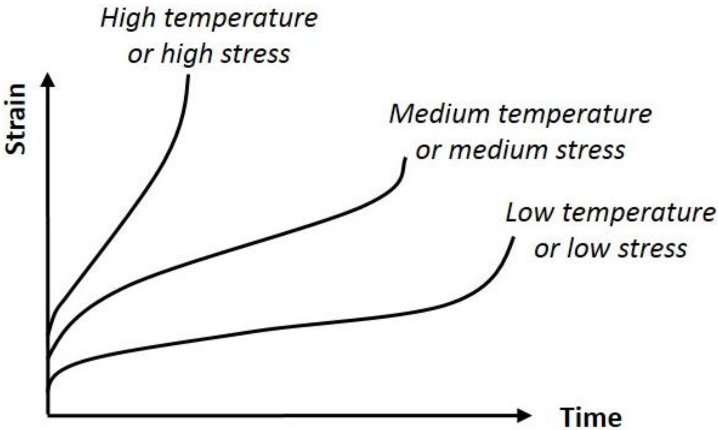


Figure 3.16: Effect of temperature or applied stress on creep curve

Since all mechanisms of steady-state creep are in some way dependent on diffusion, we expect that creep rate will have this exponential dependence on temperature:

$$\dot{\varepsilon} \propto exp(-Q/RT)$$

Where Q is the amount of heat due to absolute temperature T and R is universal gas constant.

Creep occurs faster at higher temperatures. However, what constitutes a high temperature is different for different metals.

The applied stress provides a driving force for dislocation movement and diffusion of atoms. As the stress is increased, the rate of deformation also increases.

In general, it is found that $\dot{\varepsilon} = \sigma^n$, where n is the stress exponent.

3.6 Fracture

A fracture is the (local) separation of an object or material into two, or more pieces under the action of stress. In crystalline materials, individual crystals fracture without the body actually separating into two or more pieces. Depending on the substance which is fractured, a fracture reduces strength (most substances) or inhibits transmission of light (optical crystals).

Fracture strength, also known as breaking strength, is the stress at which a specimen fails via fracture. This is usually determined for a given specimen by a tensile test. Ductile materials have a fracture strength lower than the ultimate tensile strength (UTS), whereas in brittle materials the fracture strength is equivalent to UTS.

Types of fracture

Brittle fracture

In brittle fracture, no apparent plastic deformation takes place before fracture. In brittle crystalline materials, fracture can occur by cleavage as the result of tensile stress acting normal to crystallographic planes with low bonding (cleavage planes). In amorphous solids, by contrast, the lack of crystalline structure results in a conchoidal fracture, with cracks preceding normal to the applied tension.

Ductile fracture

In ductile fracture, extensive plastic deformation (necking) takes place before fracture. The terms rupture or ductile rupture describe the ultimate failure of tough ductile materials loaded in tension. Rather than cracking, the material "pulls apart" generally leaving a rough surface. In this case, there is slow propagation and an absorption of a large amount of energy before fracture. The strain at which the fracture happens is controlled by the purity of the materials. At room temperature, pure iron can undergo deformation up to 100% strain before breaking, white cast iron or high carbon steels can barely sustain 3% of strain.

The basic steps of ductile fracture are void formation, void coalescence (also known as crack formation), crack propagation, and failure, often resulting in a cup and cone shaped failure surface.

REVIEW QUESTIONS

1. What is tensile test? Describe stress-strain curve highlighting different points on the curve.
2. What is true stress-strain diagram? Differentiate between ductile and brittle material with the help of engineering stress-strain diagram.
3. What is hardness test? Illustrate different types of hardness testing methods.
4. Explain Rockwell, Vickers, Knoop and micro-hardness test.
5. What is impact test? Explain two types of impact test.
6. What is fatigue test? Describe different types of fatigue loading.
7. How is fatigue test carried out? Explain S-N curve and endurance limit.
8. What is creep failure? How creep test is carried out.
9. Explain creep rupture curve and effects of stress and temperature on creep failure.

CHAPTER 4
DEFORMING PROCESS OF MATERIALS

Deformation processes transform solid materials from one shape into another. The initial shape is usually simple (e.g., a billet or sheet blank) and is plastically deformed between tools, or dies, to obtain the desired final geometry and tolerances with required properties. A sequence of such processes is generally used to form material progressively from a simple geometry into a complex shape. Deformation processes are frequently used in conjunction with other unit operations, such as casting, machining, grinding, and heat treating, to complete the transformation from raw material to finished and assembly-ready discrete parts. Deformation processes, along with machining, have been at the core of modem mass production, because they involve primarily metal flow and do not depend on long-term metallurgical rate processes.

4.1 Cold work

4.1.1 Cold work and its types

A metal is said to be cold worked, if it is mechanically processed below the recrystallization temperature of the metal and its grains are in a distorted condition after plastic deformation is completed. Cold working produces an improved surface finish and closer dimensional tolerance and because of this characteristic cold working process is generally used in making end-use products. Since, recrystallization does not take place in cold working, the grains are permanently distorted. During cold working residual stresses are set up. As their presence is undesirable a suitable heat treatment is generally necessary to neutralize these stresses and restore the metal to its original structure.

All the properties of a metal that are dependent on the lattice structure are affected by plastic deformation or cold working. The following properties are affected by cold work significantly:

- Tensile Strength
- Hardness
- Yield Strength
- Ductility
- Work Hardening

The various cold working operations are:

- **Drawing**: Drawing is an operation in which the cross-section of a bar, rod or a wire is reduced by pulling it through a die opening. Although the presence of tensile stresses is obvious in drawing, compressive stress also plays a significant role because the metal is squeezed down as it passes through the die opening.

- **Bending:** Bending is a sheet-metal operation which is defined as the straining of the metal around a straight axis. During bending, the metal on the inside of the neutral plane is compressed, while the metal on the outside of the neutral plane is stretched.

- **Shearing:** Shearing is a sheet-metal cutting operation along a straight line between two cutting edges. Shearing is typically used to cut large sheets into smaller sections for subsequent press-working operations.

- **Hobbing:** Hobbing is a deformation process in which a hardened steel form is pressed into a soft steel (or other soft metal) block. The process is often is used to make mould cavities for plastic moulding and die casting. The hardened steel form is called the *hob*.

- **Peening:** Peening is the process of working a metal's surface to improve its material properties, usually by mechanical means such as hammer blows, by blasting with shot (shot peening), or blasts of light beams with laser peening.

- **Cold Extrusion:** Extrusion is a compression forming process in which the work metal is forced to flow through a die opening to produce a desired cross-sectional shape. The process can be likened to squeezing toothpaste out of a toothpaste tube. Cold extrusion is done at room temperature or near room temperature.

4.1.2 Strain hardening and stress-strain curve

Work (or strain) hardening is a phenomenon which results in an increase in hardening and strength of a metal (specimen) subjected to plastic deformation at temperature lower than the recrystallization range (cold working). When a material is subjected to plastic deformation, a certain amount of work done on it is stored internally as strain energy. This additional energy in a crystal results in strengthening or work hardening of solids. Thus work hardening may be defined as increased hardness accompanying plastic deformation. This increase in hardening is accompanied by an increase in both tensile and yield strength.

4.1.3 Properties versus degree of cold work

Work hardening reduces ductility and plasticity. It is used in many manufacturing processes such as rolling of bars and drawing of tubes. Tensile strength, yield strength and hardness are increased, while ductility is decreased. Although both strength and hardness increase, the rate of change is not the same.

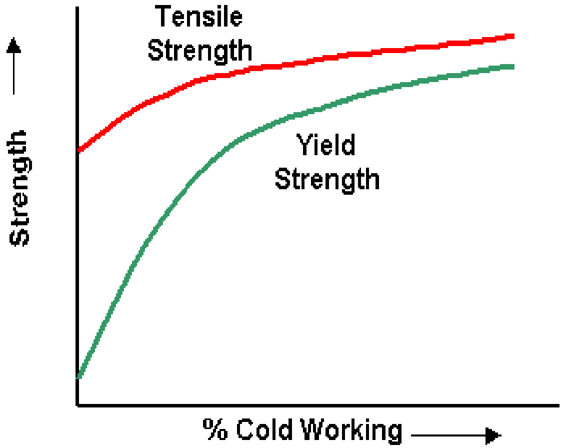


Figure 4.1: Effect of cold working in strength

Hardness generally increases most rapidly in the first 10 percent reduction (cold work), whereas the tensile strength increases more or less linearly. The yield strength increases more rapidly than the tensile strength, so that, as the amount of plastic deformation is increased, the gap between the yield and tensile strengths decreases. This is important in certain forming operations where appreciable deformation is required.

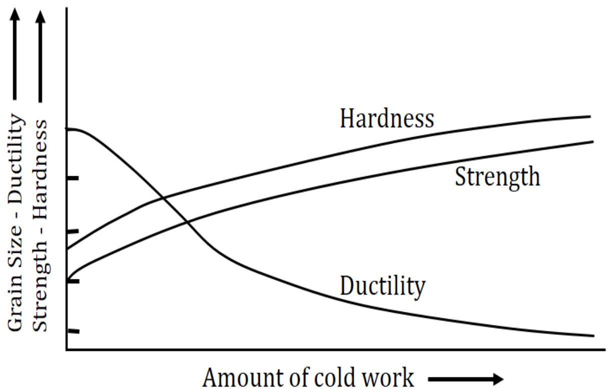


Figure 4.2: Effect of cold working in strength

Ductility follows a path opposite to that of hardness. A large decrease in the first 10 percent reduction and then a decrease at a slower rate is observed.

4.1.4 Microstructure and residual stress in cold worked metals

A small portion of the applied stress-perhaps about 10% is stored in the form of residual stresses within the structure. The residual stresses increase the total energy of the structure. The presence of dislocations increases the total internal energy of the structure. The higher the extent of cold working, the higher would be the level of total internal energy of the material. Residual stresses generated by cold working may not always be desirable and can be relieved by a heat treatment known as stress-relief annealing. In some instances residual compressive stresses at the surface of materials is created to enhance their mechanical properties.

4.2 Treatment after cold work

4.2.1 Annealing

As a result of cold working, the hardness, tensile strength, and electrical resistance increase, while ductility decreases. There is also a large increase in the number of dislocations, and certain planes in the crystal structure are severely distorted. Most of the energy used to cold work the metal is dissipated in heat, and a finite amount of energy is stored in the crystal structure as internal energy associated with the lattice defects created by the deformation.

Full annealing is the process by which the distorted cold worked lattice structure is changed back to one which is strain free through the application of heat. This process is carried out entirely in the solid state and is usually followed by slow cooling in the furnace from the desired temperature. The annealing process may be divided into three stages:

- Recovery
- Recrystallization
- Grain Growth

4.2.2 Three stages of annealing

Recovery

This is primarily a low temperature process, and the property changes produced do not cause appreciable change in microstructure or the properties, such as tensile strength, yield strength, hardness and ductility. The principal effect of recovery is the relief of internal stresses due to cold working.

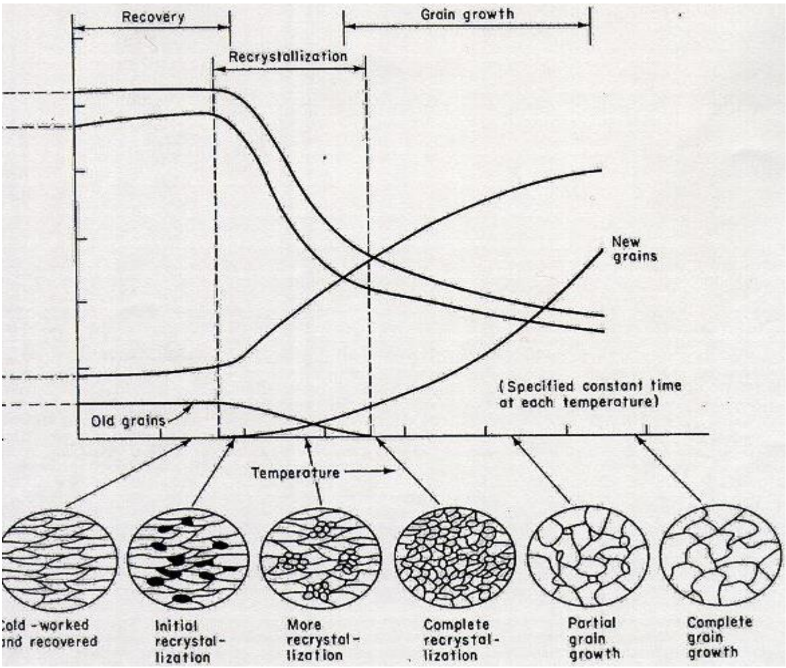


Figure 4.3: Effect of annealing on tensile strength, hardness, ductility and grain size

When the load which causes plastic deformation is released, all the elastic deformation does not disappear. This is due to the different orientation of crystals, which will not allow some of them to move back when the load is released. As the temperature is increased, there is some spring back of these elastically displaced atoms which relieve most of the internal stresses. Electrical conductivity is also increased appreciably during the recovery stage.

Since the mechanical properties of the metal are essentially unchanged, the main purpose of heating in the recovery range is stress relieving cold worked alloys to prevent stress corrosion cracking or to minimize the distortion produced by residual stresses. Commercially, this low temperature treatment in the recovery range is known as stress relief annealing or process annealing.

Recrystallization

As the temperature of the recovery range is reached, minute new crystals appear in the microstructure. These new crystals have the same composition and lattice structure as the original undeformed grains and are not elongated but are uniform in dimensions. The new crystals generally appear at the most drastically deformed portions of the grain, usually the grain boundaries and slip planes. The cluster of atoms from which the new grains are formed is called a nucleus. Recrystallization takes place by a combination of nucleation of strain free grains and the growth of these nuclei to absorb the entire cold worked material.

Pure metals have low recrystallization temperatures as compared with alloys. Zinc, tin and lead have recrystallization temperatures below room temperature. This means that these metals cannot be cold worked at room temperature since they crystallize spontaneously, reforming a strain free structure.

The greater the prior deformation, the lower the temperature for the start of recrystallization. Increasing the annealing time decreases the recrystallization temperature for the start of recrystallization. During recrystallization stage, there is a significant drop in tensile strength, hardness and a large increase in the ductility of the material.

Grain Growth

In this stage the tensile strength and hardness continue to decrease but at a much less rate than the recrystallization stage. The major change observed during this stage is the growth of the grain boundaries and reaching the original grain size.

4.3 Hot work

4.3.1 Hot work process and its types

Hot working refers to processes where metals are plastically deformed above their recrystallization temperature. Being above the recrystallization temperature allows the material to recrystallizes during deformation. This is important because recrystallization keeps the materials from strain hardening, which ultimately keeps the yield strength and hardness low and ductility high. This contrasts with cold working. Hot working process can be considered as simultaneous combination of cold working and annealing. Any work hardening effect caused by plastic deformation is neutralized immediately by the effect of high temperature.

The various hot working operations are:

- **Rolling:** Rolling is a deformation process in which the thickness of the work is reduced by compressive forces exerted by two opposing rolls. The rolls rotate in opposite directions to pull and simultaneously squeeze the work between them.

- **Forging:** Forging is a deformation process in which the work is compressed between two dies, using either impact or gradual pressure to form the part. A forging machine that applies impact an impact load is called a *forging hammer*, while one that applies gradual pressure is called a *forging press*.
- **Pipe Welding:** Pipe welding is the process of connecting metal pipes by heating them and joining them together.
- **Rotary piercing:** Rotary piercing is a hot working metal-working process for forming thick-walled seamless tubing. The process starts by feeding a blank in between the opposing rolls. Then, the rolls push the blank over the mandrel to form a uniform inner diameter of the tubing.

- **Hot Extruding:** Hot extrusion is a hot working process, which means it is done above the material's recrystallization temperature to keep the material from work hardening and to make it easier to push the material through the die. Most hot extrusions are done on horizontal hydraulic presses.

The advantages of hot-working operations are:

1. Decrease in yield strength, therefore it is easier to work and uses less energy or force.
2. Increase in ductility.
3. Elevated temperatures increase diffusion which can remove or reduce chemical in homogeneities.
4. Pores may reduce in size or close completely during deformation.
5. In steel, the weak, ductile, face-centered cubic austenite microstructure is deformed instead of the strong body-centered cubic ferrite microstructure found at lower temperatures.

4.3.2 Comparison between cold working and hot working

Cold Working	Hot working
1. Cold working is done at a temperature below the value required for recrystallization, so no appreciable recovery takes place during deformation.	1. Hot working is done at a temperature above recrystallization temperature, so it can be regarded as a simultaneous occurrence of deformation and recovery process.
2. Hardening is not eliminated as working is done at a temperature below recrystallization, so it is always accompanied by strain hardening.	2. Hardening due to plastic deformation is completely eliminated by recovery and recrystallization. This is true, however, only if the rate of crystallization is higher than rate of deformation.
3. Cold working decreases the value of elongation, reduction area and impact values.	3. Mechanical properties like elongation, reduction of area and impact values are improved.
4. Embrittlement does not occur due to less diffusion and no reaction of oxygen at lower temperature.	4. Reactive metals get severely embrittled by oxygen and hence must be protected from the action of oxygen by using inert atmosphere.
5. Surface decarburization in steels does not occur.	5. Surface decarburization is likely to occur at higher temperatures unless protected by a proper atmosphere.

REVIEW QUESTIONS

1. What is cold working? Give the examples of different cold working process.
2. What is strain hardening? Explain the effect of degree of cold work on properties of materials.
3. What is annealing? Explain the three stages of annealing.
4. What is hot working? Give the examples of different hot working process.
5. Compare and differentiate between hot working and cold working process.

CHAPTER 5

SOLIDIFICATION, PHASE RELATIONS AND STRENGTHENING MECHANISM

5.1 Solidification

Solidification is a phase change in which a liquid turns into a solid when its temp is lowered below its freezing point. Almost all metals and alloys, at one point in their processing are a liquid, when then solidifies as it cools below the freezing temperature the material can be used in the as-solidified condition or may be further processed by mechanical working or heat treatment. The structures produced during solidification affect the mechanical properties and influence the type of further processing needed to achieve the final properties. In pure metals, the grain size and shape may be controlled by solidification.

5.1.1 Nucleation and grain growth

Solidification requires two steps- nucleation and growth. Nucleation occurs when a small piece of solid forms from the liquid. Growth of solid occurs as toms from the liquid are attached to the tiny solid until no liquid remains. We expect a material to solidify when the liquid cools below the freezing temperature. As the temperature falls below the freezing temperature, the energy associated with the crystalline structure of solid is increasingly less than the energy of the liquid, making the solid more and more stable. However when the solid forms, an interface is created, separating the solid from the liquid; an increase in energy is associated with this interface.

The total change in free energy produced when the solid forms is the sum of the decrease in volume free energy and the increase in surface free energy.

Change in free energy in solid-liquid interface,

$$\Delta G = \frac{4}{3}\pi r^3 \Delta g + 4\pi r^2 \sigma$$

Spontaneous growth above radius, $r^* = \frac{2\sigma}{\Delta S_f \Delta T}$

Activation energy, $\Delta G = \frac{16\pi\sigma^3}{3\Delta S_f^{\,2}(\Delta T)^2}$

Where $\frac{4}{3}\pi r^3$ = volume of spherical solid,

$4\pi r^2$ = surface area of spherical solid,

σ = surface free energy,

Δg = free energy change per unit volume.

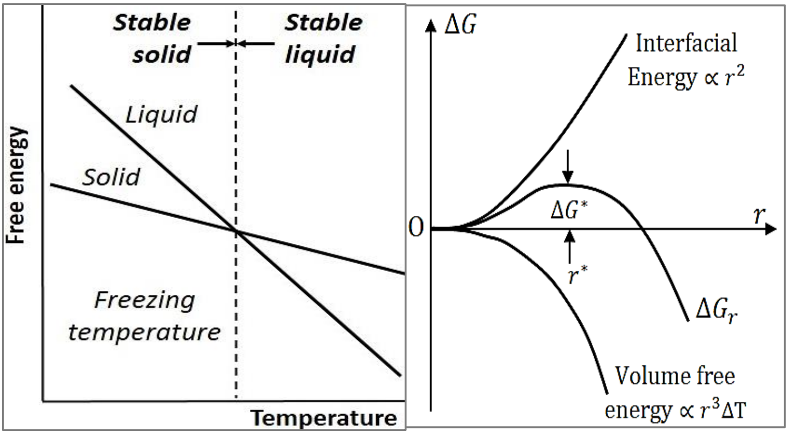


Figure 5.1: Free energy versus temperature plot for a pure metal

The total change in the free energy depends on the size of the solid. The growth of a very small solid particle, called an embryo, requires an increase in free energy. Instead of growing, the embryo re-melts; causing the free energy to decrease, and the metal remains liquid.

Since the liquid is present below the equilibrium freezing temperature, it is under-cooled. The under-cooling is the equilibrium freezing temperature minus the actual temp of the liquid. Nucleation has not occurred and growth cannot begin, even though the temperature is below the equilibrium freezing temperature.

When a large solid particle grows, the free energy decreases. The solid is now stable, nucleation has occurred, and growth of solid particle, which is now called a nucleus, begins. Nucleation only occurs when enough atoms spontaneously cluster together to produce a solid with a radius greater than the critical radius r^*, corresponding to a maximum on the total free energy curve.

Homogenous nucleation

As the liquid cools further below the equilibrium freezing temp, there is a greater probability that atoms will cluster together to form an embryo larger than the critical radius. In addition, there is a large volume free energy difference between the liquid and the solid, which reduces the critical size of the nucleus. Homogenous nucleation occurs when the under-cooling becomes large enough to permit the embryo to exceed the critical size. As the under-cooling increases, the critical radius required for the nucleation decreases.

As an approximation, homogenous nucleation occurs when $\partial T = 0.2T_m$

Where T_m = Equilibrium freezing temperature in K.

$\partial T = T_m - T$ is the under-cooling when the liquid temp is T.

Heterogeneous nucleation

Nucleation on preexisting surfaces is known as heterogeneous nucleation. This process is dependent on the contact angle (ɵ) for the nucleating phase and the surface on which nucleation occurs. The same type of phenomenon occurs in solid-state transformations.

A solid forming on an impurity can assume the critical radius with a smaller increase in the surface energy. Thus, heterogeneous nucleation can occur with relatively low undercooling.

5.1.2 Crystal growth/ dendrite formation

Once solid nuclei have formed, growth occurs as atoms are attached to the solid surface. In pure metals, growth depends on how heat is removed from the solid-liquid system. Two types of heat must be removed – the specific heat of the liquid and the latent heat of fusion. The specific heat must be removed first, usually by conduction into the surrounding mould, until the liquid cools to the freezing temp. The latent heat of fusion, which represents the energy that is evolved as the disordered liquid structure transforms to a more stable crystal structure, must be removed from the solid-liquid interface before solidification is completed. The manner in which we remove the latent heat of fusion determines the growth mechanism and final structure.

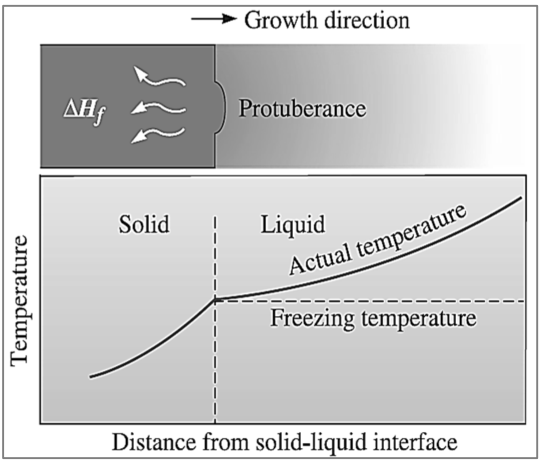


Figure 5.2: Formation of protuberance

When a well-inoculated liquid cools slowly under equilibrium conditions, the temp of the liquid metal is greater than the freezing temperature. The latent heat of fusion can only be removed by conduction from the solid-liquid interface through the solid to the surroundings.

Any small protuberance (refer fig. 5.2), that begins to grow on the interface is surrounded by liquid metal above the freezing temperature. This means that the formation of protuberance begins when the liquid temperature is above the freezing temperature. The mechanism, known as planar growth, occurs by the movement of a smooth solid-liquid interface into the liquid.

However, when nucleation is poor, the liquid may under-cool to a temp below the freezing temp. Under these conditions, a small solid protuberance called a dendrite, which forms at the interface, is encouraged to grow. As the solid dendrite grows, the latent heat of fusion is conducted into the under-cooled liquid, raising the temp of the liquid towards the freezing temp. Secondary and tertiary dendrite arms can also form on the primary stalks to speed the evolution of the latent heat. Dendritic growth continues until the under-cooled liquid warms to the freezing temp. Any remaining liquid then solidifies by planar growth.

In pure metals, dendritic growth normally represents only a small fraction of the total growth.

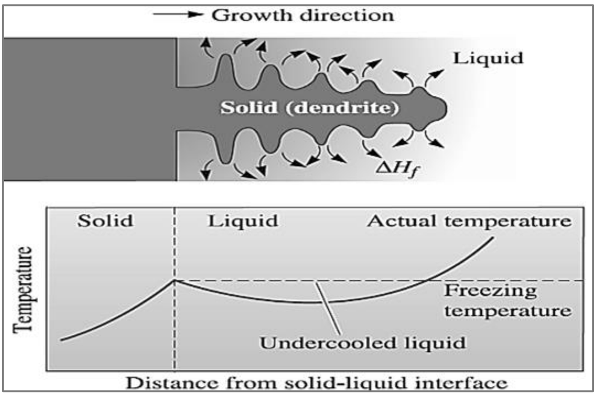


Figure 5.3: Dendrite formation

5.1.3 Cooling curve

A cooling curve is a line graph that represents the change of phase of matter, typically from a gas to a solid or a liquid to a solid. The independent variable (X-axis) is time and the dependent variable (Y-axis) is temperature. Below is an example of a cooling curve used in castings.

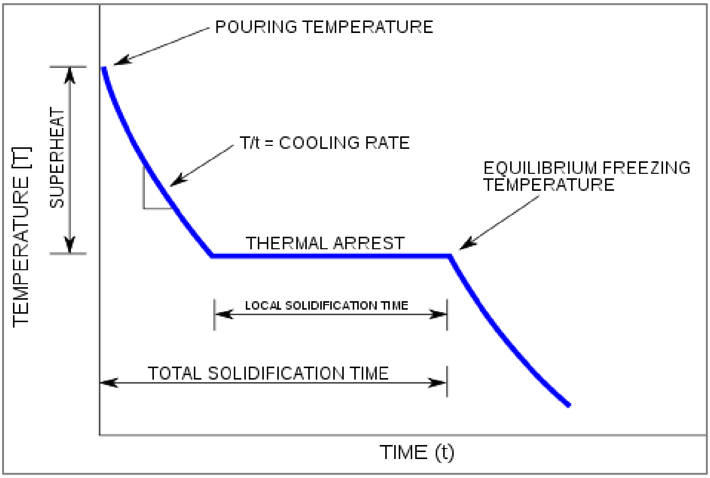


Figure 5.4: Cooling Curve

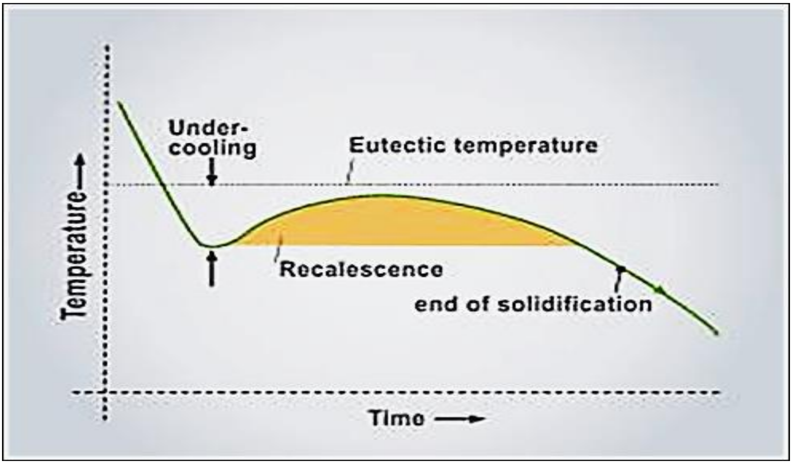


Figure 5.5: Cooling curve (undercooled solidification)

The initial point of the graph is the starting temperature of the matter, here noted as the pouring temperature. When the phase change occurs there is a thermal arrest, i.e. the temperature stays constant. This is because the matter has more internal energy as a liquid or gas than in the state that it is cooling to. The amount of energy required for a phase change is known as latent heat. The cooling rate is the slope of the cooling curve at any point.

In the part of the curve where the temperature decreases, the kinetic energy also decreases while the potential energy stays the same. However, at the phase transition, where the curve is flat, the kinetic energy stays the same while the potential energy decreases.

5.1.4 Solidification time

The rate at which the solid grows depends on the cooling rate, or rate of heat extraction. A fast cooling produces rapid solidification or short solidification times. The time required for a simple casting to solidify can be calculated using Chvorinov's rule.

$$T_S = B\ (V/A)^2$$

Where, T_S is solidification time, V is the volume of the casting, A is the surface area of the casting in contact with the mould, and B is the mould constant. The mould constant depends on the properties and initial temperature of both the metal and the mould. At most always, a shorter solidification time produces a finer grain size and a stronger casting.

5.1.5 Cast metal structure

The solidification stage has been shown to be crucial in the determination of microstructure in castings, in respect of the nature of the macro structural zones, primary grain size and structural features within the grains. A dendritic structure is characterized by the primary grain size itself and by the spacing of the dendrite arms, which governs the form and distribution of micro segregates and secondary constituents, whether precipitated during or after freezing. The spacing can also influence the grain size and other characteristics developed from the primary structure by solid state transformations and affects the time required to accomplish changes sought by heat treatment.

Melt conditions

The melt condition relates to superheating and the influence of temperature. This operates through its effect on the presence and survival of nuclei and, with certain exceptions, notably some magnesium base alloys, high superheat produces coarsening; in cast iron it increases the tendency to produce white or chill rather than gray structures.

Figure below summaries four of the dozens of casting processes. The processes are divided into several groups – sand moulds, ceramics moulds and metal moulds. The process using metal moulds tend to give the highest strength castings due to rapid solidification. Ceramics moulds, because they are good insulators, give the slowest cooling and lowest strength changes.

a) Green Sand moulding in which clay-bonded sand is packed around a pattern. Sand cores can produce internal cavities in the casting.

b) The permanent mould process in which metal is poured into an iron or steel mould.

c) Die casting in which metal is injected at high pressures into a steel die

d) Investment casting, in which a wax pattern is surrounded by a ceramic; after the wax is melted and drained, metal is poured into the mould.

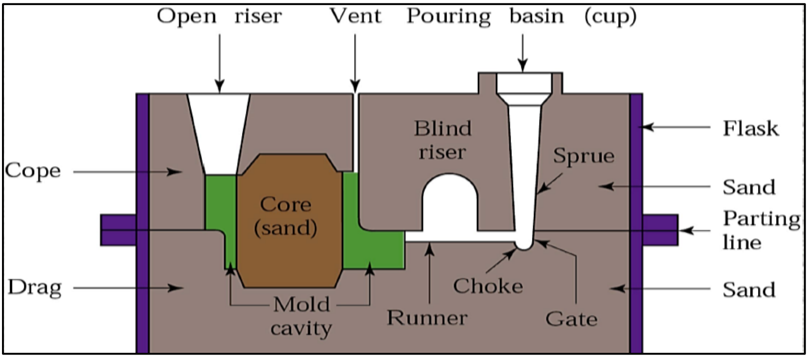


Figure 5.6: Sand casting

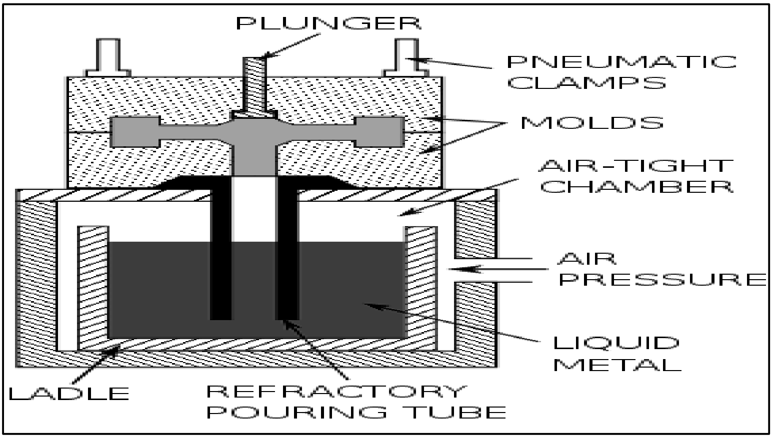


Figure 5.7: Permanent die casting

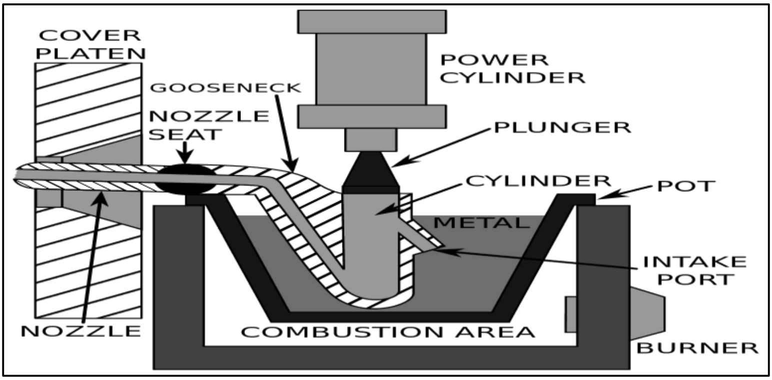


Figure 5.8: Die casting

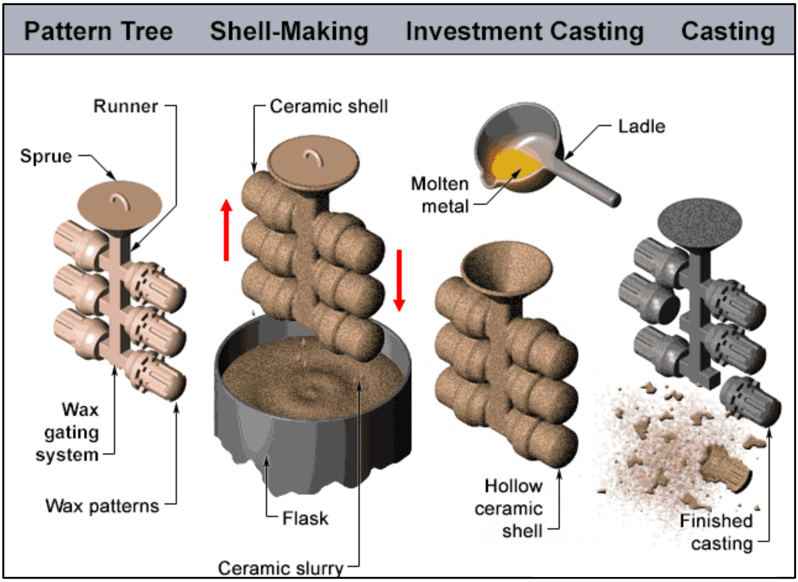


Figure 5.7: Investment casting

5.1.5 Solidification defects

A properly designed casting, a properly prepared mould and correctly melted metal should result in a defect free casting. However, if proper control is not exercised in the foundry-sometimes it is too expensive - a variety of defects may result in a casting.

These defects may be the result of:

- Improper pattern design.
- Improper mould and core pattern.
- Improper melting practice.
- Improper pouring practice.
- Moulding and core making materials.
- Improper gating system.
- Improper metal composition.
- Inadequate melting temperature.

It creates a deficiency or imperfection. Exceeding quality limits imposed by design and service casting defects are mainly three categories.

They are:

- Surface defects
- Internal defects
- Visible defects

Surface defects

Due to design and quality of sand moulds and general cause is poor ramming.

1. Blow

Blow is relatively large cavity produced by gases which displace molten metal form.

2. Scar

Due to improper permeability or venting. A scare is a shallow blow. It generally occurs on flat surface; whereas a blow occurs on a convex casting surface. A blister is a shallow blow like a scar with thin layer of metal covering it.

3. Scab

This defect occurs when a portion of the face of a mould lifts or breaks down and the recess thus made is filled by metal. When the metal is poured into the cavity, gas may be disengaged with such violence as to break up the sand which is then washed away and the resulting cavity filled with metal. The reasons can fine sand texture, low permeability of sand, high moisture content of sand and uneven mould ramming.

4. Drop

Drop or crush in a mould is an irregularly shaped projection on the cope surface of a casting. This defect is caused by the break-away of a part of mould sand as a result of weak packing of the mould, low strength of the moulding sand, malfunctioning of moulding equipment, strong jolts and strikes at the flask when assembling the mould. The loose sand that falls into the cavity will also cause a dirty casting surface, either on the top or bottom surface of the casting, depending upon the relative densities of the sand and the liquid.

5. Penetration

It is a strong crust of fused sand on the surface of a casting which results from insufficient refractoriness of moulding materials, a large content of impurities, inadequate mould packing and poor quality of mould washes. When the molten metal is poured into the mould cavity, at those places when the sand packing is inadequate, some metal will flow between the sand particles for a distance into the mould wall and get solidified. When the casting is removed, this lump of metal remains attached to the casting. Of course, it can be removed afterwards by chipping or grinding.

6. Buckle

A buckle is a long, fairly shallow, broad, vee depression that occurs in the surface of flat castings. It extends in a fairly straight line across the entire flat surface. It results due to the sand expansion caused by the heat of the metal, when the sand has insufficient hot deformation. It also results from poor casting design providing too large a flat surface in the mould cavity. Buckling is prevented by mixing cereal or wood flour to sand.

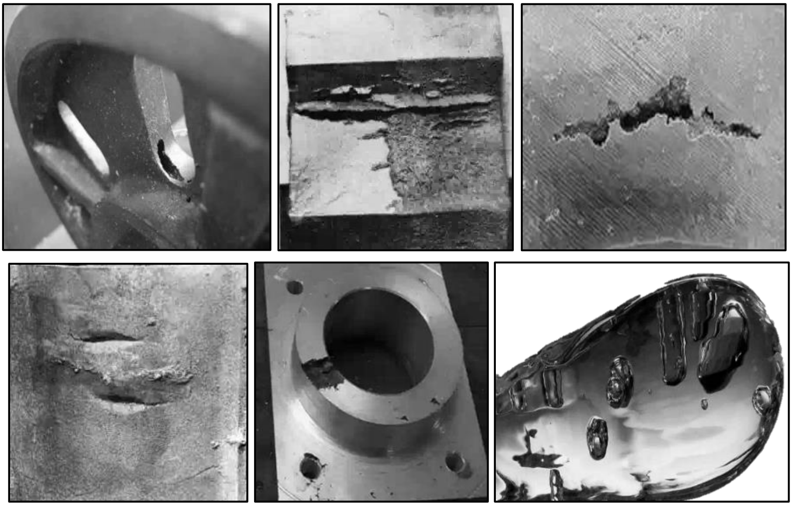


Figure 5.9: (clockwise from top left) (a) Blow (b) Penetration (c) Scar (d) Drop (e) Scab (f) Buckle

Internal defects

1. Blow holes

Blow holes, gas holes or gas cavities are well rounded cavities having a clean and smooth surface. They appear either on the casting surface or in the body of a casting. These defects occur when an excessive evolved gas is not able to flow through the mould. So, it collects into a bubble at the high points of a mould cavity ad prevents the liquid metal from filling that space. This will result in open blows.

2. Pin holes

Pin holes are small gas holes either at the surface or just below the surface. When these are present, they occur in large numbers and are fairly uniformly dispersed over the surface. This defect occurs due to gas dissolved in the alloy and the alloy not properly degassed.

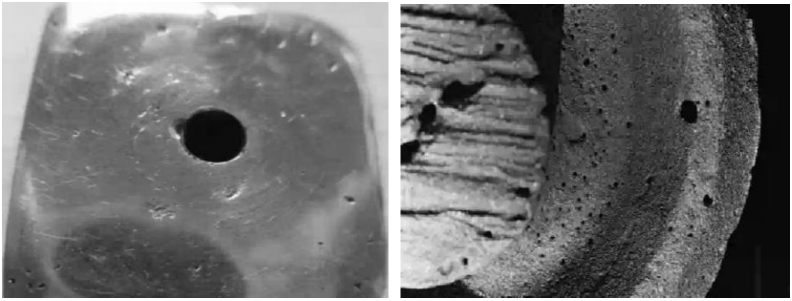


Figure 5.10: a) blow holes b) pin holes

Visible defects

1. Wash

A cut or wash is a low projection on the drag face of a casting that extends along the surface, decreasing in height as it extends from one side of the casting to the other end. It usually occurs with bottom gating castings in which the moulding sand has insufficient hot strength, and when too much metal is made to flow through one gate into the mould cavity.

2.**Rat tail**

A rat tail is a long, shallow, angular depression in the surface of a flat rating and resembles a buckle, except that, it is not shaped like a broad vee. The reasons for this defect are the same for buckle.

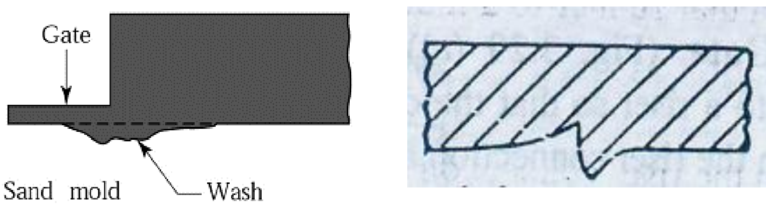


Figure 5.11: a) Wash b) Rat tail

3. Hot tear

Hot tears are hot cracks which appear in the form of irregular crevices with a dark oxidized fracture surface. They arise when the solidifying met does not have sufficient strength to resist tensile forces produced during solidification. This defect can be avoided by improving the design of the casting and by having a mould of low hot strength and large hot deformation.

4. Shrinkage

A shrinkage cavity is a depression or an internal void in a casting that results from the volume contraction that occurs during solidification.

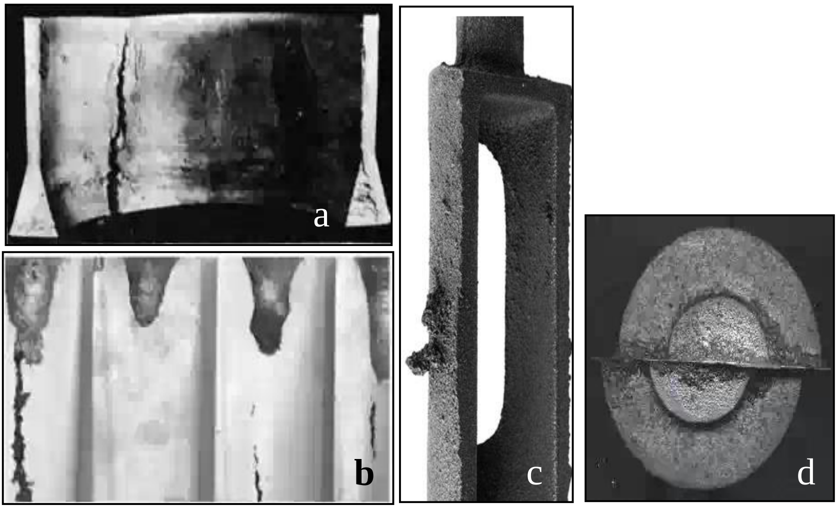


Figure 5.12: (a) Hot Tear (b) Shrinkage (c) Swell (d) Shift

5. Swell

A swell is a slight, smooth bulge usually found on vertical faces of castings, resulting from liquid metal pressure. It may be due to low strength of mould because of too high a water content or when the mould is not rammed sufficiently.

6. Shift

Mould shift refers to a defect caused by a sidewise displacement of the mould cope relative to the drag, the result of which is a step in the cast product at the parting line. Core shift is similar to mould shift, but it is the core that is displaced, and (the displacement is usually vertical. Core shift and mould shift are caused by buoyancy of the molten metal.

7. Misrun or cold sheet or short run

This defect is incomplete cavity filling. The reasons can be inadequate metal supply, too low mould or melt temperature, improperly designed gate or length to thickness ratio of the casting is too large. When molten metal is flowing from one side in a thin section, it may lose sufficient heat resulting in loss of its fluidity, such that the leading edge of the stream may freeze before it reaches the end of the cavity.

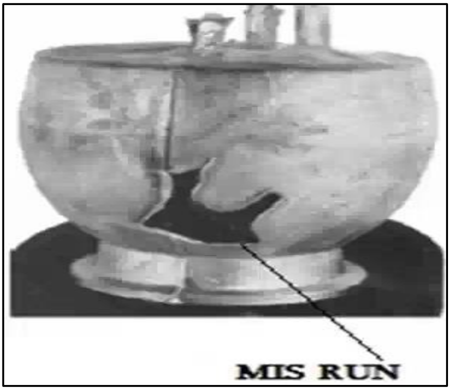


Figure 5.13: Misrun

5.1.6 Solid solution and solid solution strengthening

The mixture of two or more liquid material i.e. solute and solvent is called solution. The solution after solidification e.g. steel (alloy of iron and carbon) is called solid solutions. The solute atoms are introduced into the matrix (solvent atoms). Solid solution is compositionally homogenous; the solute (impurity) atoms are randomly distributed throughout the matrix. There are two types of solid solutions.

1. **Substitutional Solid Solution**: The solute and solvent atoms are similar in size, rendering the solute atoms to occupy lattice point of the solvent atoms.
2. **Interstitial Solid Solution**: The solute atoms are of similar size than the solvent atom, rendering the solute atoms to occupy the interstitial sites in the solvent lattice.

Solid solution strengthening is a technique to strengthen and harden metals by alloying with impurity atoms that go into either substitutional or interstitial solid solution. Solid solution strengthening distorts the lattice, offers resistance to dislocation movement which is greater with interstitial elements which cause asymmetric lattice distortion, e.g., carbon in steel.

Factors affecting solubility of solute atom

The solubility of the solute atoms in the host matrix (solvent) can be determined by several factors:

1. **Atomic size factor**: Solid solution is appreciable when the difference in atomic radii between the two atoms is $\leq$15%, otherwise it creates substantial lattice distortion.
2. **Crystal structure**: Similar crystal structure of metals of both atom types are preferred.

3. **Electro negativity**: The more electropositive one element and the more electronegative the other, the more tendency to form an inter-metallic compound than solid solution.
4. **Valency**: A metal will have more of a tendency to dissolve another metal of higher valency than one of a lower valency.

Effects of solid solution strengthening on properties

1. The yield strength, tensile strength and hardness of the alloy are greater than those of pure metals.
2. Almost always, the ductility of the alloy is less than the pure metal.
3. Electrical conductivity of the alloy is much lower than that of pure metal.
4. The resistance to creep or loss of strength at elevated temperature is improved by solid solution strengthening.

5.1.7 Hume-Rothery rules

The Hume-Rothery rules, known after William Hume Rothery, are a set of basic rules describing the conditions under which an element could dissolve in a metal, forming a solid solution. There are two sets of rules, one which refers to substitutional solid solutions, and another which refers to interstitial solid solutions.

Substitutional solid solutions rule

For substitutional solid solutions, the Hume-Rothery rules are:

1. The atomic radii of the solute and solvent atoms must differ by no more than 15%.

$$\% \text{ difference} = \left(\frac{r_{solute} - r_{solvent}}{r_{solvent}} \right) \times 100\% \leq 15\%$$

2. The crystal structure of solute and solvent must match.

3. Complete solubility occurs when the solvent and solute have the same valency. A metal will dissolve a metal of higher valency to greater extent than one of lower valency.
4. The solute and solvent should have the similar electronegativity. If the electronegativity difference is too great, the metals will tend to form intermetallic compounds instead of solid solutions.

Interstitial solid solution rules

For interstitial solid solutions, the Hume- Rothery rule are

1. Solute atoms must be smaller than the interstitial sites in the solvent lattice.
2. The solute and solvent should have similar electronegativity.

5.2 Phase relation and equilibrium

The mechanical properties of materials can be controlled by addition of point defects such as substitutional and interstitial atoms. Particularly in metals, the point defects disturb the atomic arrangement in the lattice and interfere with the movement of dislocations or slip. The point defects therefore cause the material to be solid solution strengthened. In addition, the introduction of point defects changes the composition of the material and influences solidification behavior. This effect can be examined by introducing the equilibrium phase diagram.

5.2.1 Phase

A phase has the following characteristics

a) A phase has the same structure or atomic arrangement throughout

b) A phase has roughly the same composition & properties throughout

c) There is a definite interface between the phase and any surrounding or adjoining phases. E.g. solid state, liquid state, gaseous state

5.2.2 Phase rule

The Gibbs phase rule can be stated as follows:

$$\mathbf{F = C - P + 2}$$

Where, F is the degree of freedom.

C is the number of components, and

P is the number of phases in the system.

Degree of freedom refer to the number of independent variables associated with the system. The total number of variables are the external variables (temperature and pressure) plus the composition variables. The degree of freedom F is equal to or less than the total number of variables.

Components refer to the independent chemical species that constitute an alloy. For a plain carbon steel, iron and carbon are the chemical species; so the components are Fe and C (graphite).

5.2.3 Phase diagram

The properties of a material depend on the type, number, amount, and form of the phases present, and can be changed by altering these quantities. In order to make these changes, it is essential to know the conditions under which these quantities exist and the conditions under which a change in phase will occur. A phase diagram shows the state or phase of the solution at various conditions of temp., pressure and composition. It shows the phases and their compositions at any combination of temperature and alloy composition.

The best method to record the data related to phase changes in many alloy systems is in the form of *phase diagrams*, also known as *equilibrium diagrams* or *constitutional diagrams*. In order to specify completely the state of a system in equilibrium, it is necessary to specify three independent variables. These variables, which are externally controllable, are *temperature, pressure* and *composition*. With pressure assumed to be constant at atmospheric value, the equilibrium diagram indicates the structural changes due to variation of temperature and composition. Phase diagrams show the phase relationships under equilibrium conditions, that is, under conditions in which there will be no change with time. Equilibrium conditions may be approached by extremely slow heating and cooling, so that if a phase change is to occur, sufficient time is allowed.

Phase diagrams are usually plotted with temperature as the ordinate, and the alloy composition as the abscissa as shown in figure alongside. The data for the construction of equilibrium diagrams are determined experimentally by a variety of methods, the most common methods are metallographic method, X-ray diffraction technique, thermal analysis, etc.

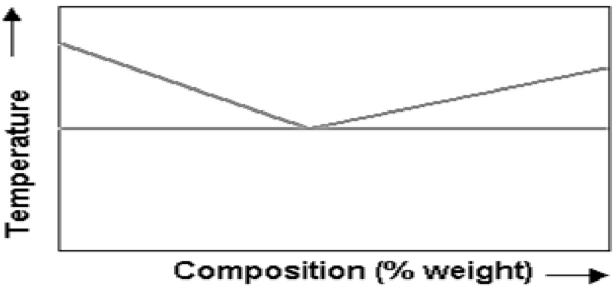


Figure 5.14: Phase diagram

Cooling curve of a pure metal

Under equilibrium conditions, all metals exhibit a definite melting or freezing point. If a cooling curve is plotted for a pure metal. It will show a horizontal line at the melting or freezing temperature.

Cooling Curve of a Solid Solution

A solid solution is a solution in the solid state and consists of two kinds of atoms combined in one type of space lattice. A solution is composed of two parts: a solute and a solvent. The solute is the minor part of the solution or the material which is dissolved, while the solvent constitutes the major portion of the solution. When solidification of the solution starts, the temperature may be higher or lower than the freezing point of the pure solvent. Most solid solutions solidify over a range in temperature. Figure below shows the cooling curve for the solidification of a solid solution.

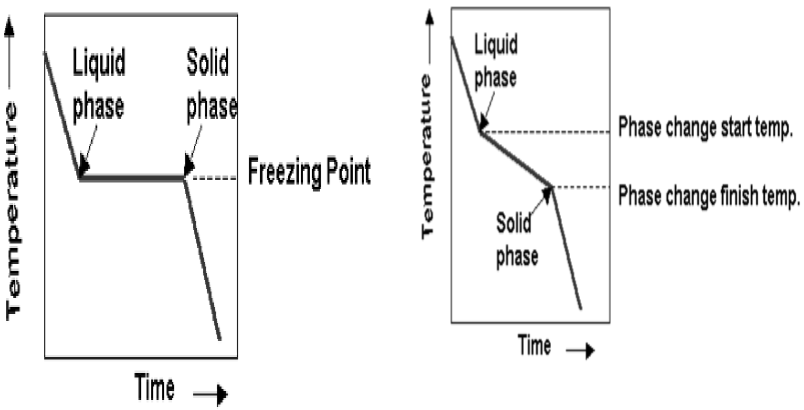


Figure 5.15: Cooling curve of a) pure metal b) solid solution

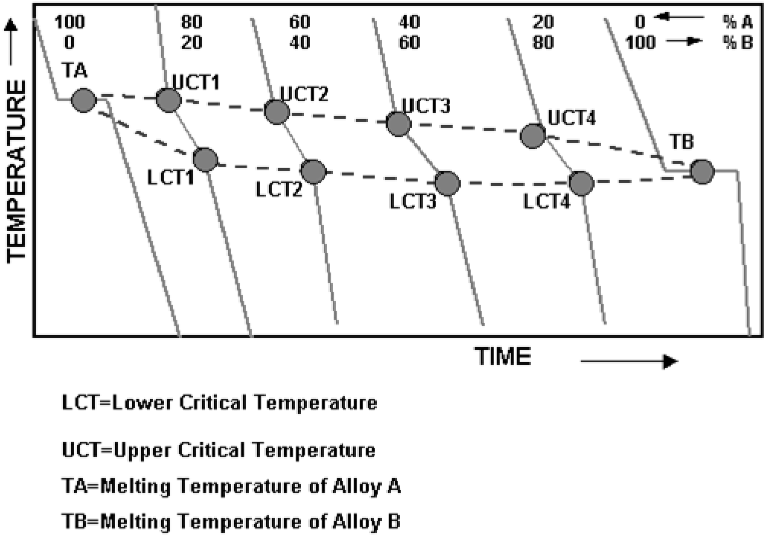


Figure 5.16 a): Phase diagram of metal A on alloying metal B

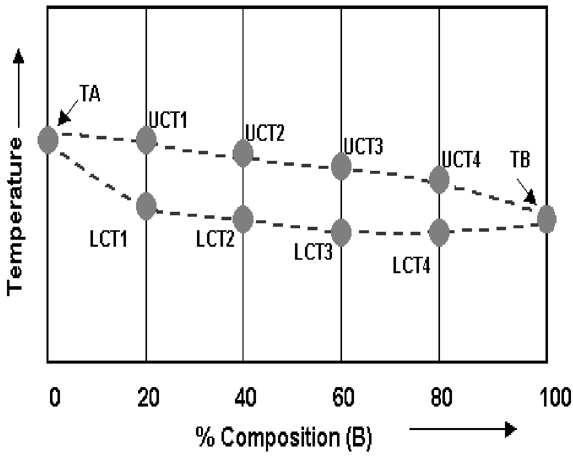


Figure 5.16 b): Phase diagram of metal A on alloying metal B

5.2.4 Inverse lever rule

Inverse lever rule is used to determine the percentage of each phase that is present at various temperatures. The length of line XY represents the sum of the two phases as 100 percent. The inverse lever rule states that liquid phase can be calculated by taking the length of the line XZ and dividing this by XY. The solid phase can be calculated by taking the length of the line ZY and dividing it by XY. To get the percentages, the values are multiplied by 100.

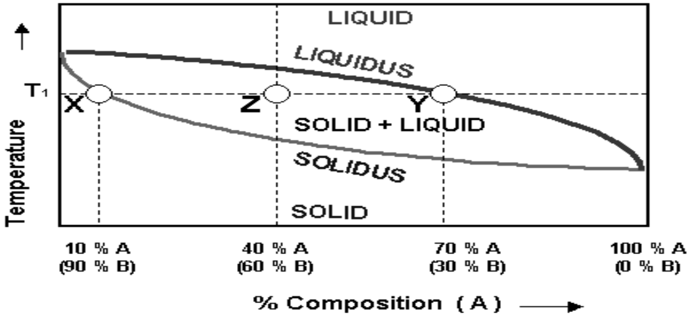


Figure 5.17: Phase diagram for inverse lever rule

Sample Calculation:

Alloy with 40 % A and 60 %B at room temperature is heated to a temperature T_1. Calculate the liquid and solid phase and indicate the composition of each phase.

Liquid Phase = (XZ / XY) x 100 = [(40-10) / (70-10)] x100%

= 50%

Solid Phase = (ZY / XY) x 100 = [(70-40) / (70-10)] x100 = 50%

At temperature T_1, the solid composition is found by the intersection (point X) of isothermal line T_1 and the solidus curve.

Solid composition: 10 % A and 90 % B

The liquid composition at temperature T_1 is found by the intersection (point Y) of isothermal line T_1 and the liquidus curve.

Liquid composition: 70% A and 30 % B

5.2.5 Phase reactions

Eutectic system

Eutectic system is a phase transformation that takes place when a single-phase liquid transforms directly to a two-phase solid.

$$L \rightarrow S1 + S2$$

$$Liquid \rightarrow Solid\ 1 + Solid\ 2$$

This is a reversible phase transformation.

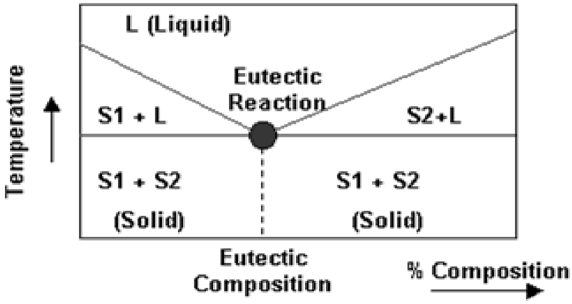


Figure 5.18: Eutectic system

Eutectoid system

The eutectoid phase transformation occurs when a single-phase solid transforms directly to two-phase solid.

$$S1 \rightarrow S2 + S3$$

$$Solid\ 1 \rightarrow Solid\ 2 + Solid\ 3$$

This is a reversible phase transformation.

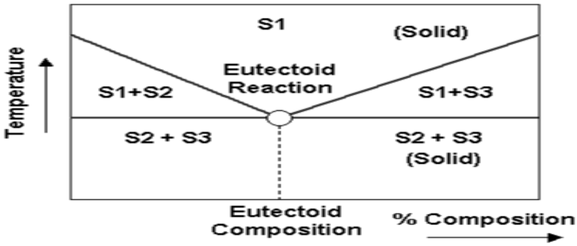


Figure 5.19: Eutectoid system

Peritectic system

The peritectic phase transformation occurs when a liquid and solid phase of fixed proportions reacts at fixed temperature to yield a single solid phase.

$$S1 + L = S2 \qquad i.e., solid\ 1 + liquid = solid\ 2$$

This transformation exists in the Iron-Iron Carbide System.

Peritectoid system

The peritectoid phase transformation occurs when two solids reacts at fixed temperature to yield a single solid phase.

$$S1 + S2 = S3 \qquad i.e., solid\ 1 + solid\ 2 = solid\ 3$$

Monotectic system

The monotectic phase transformation occur when a liquid transforms directly to a second liquid and a solid.

$$Liquid\ 1 = Liquid\ 2 + solid$$

This transformation is reversible phase transformation and occurs during cooling.

Syntectic system

The syntectic phase transformation occurs when two liquids reacts at fixed temperature to yield a single solid phase on cooling.

$$L1 + L2 = S1 \qquad i.e., liquid\ 1 + liquid\ 2 = solid\ 1$$

5.3 Strengthening mechanism

Since no two elements have the same atomic diameter, solute atoms will be either smaller or longer than the solvent atoms. Due to the difference in atomic size, lattice distortion is produced when one element is added to the other. Smaller atoms will produce a local tensile stress field and larger solute atoms will produce a local compressive field in the crystal. In both cases, the stress field of a moving dislocation interacts with the stress field of the solute atom. This increases the stress required to move the dislocation through the crystal, and hence strengthens the solution.

5.3.1 Alloys strengthening by exceeding solubility limit

When impurities (i.e. the solute atoms) are added into the solution of a metal or an alloy, the solute atoms go into interstitial or substitutional positions and increase the strength of the solution. The impurity atoms cause lattice strain which can anchor dislocations. This occurs when the strain caused by the alloying element compensates that of the dislocation, thus achieving a state of low potential energy. It costs strain energy for the dislocation to move away from this state (which is like a potential well). The scarcity of energy at low temperatures is why slip is hindered and solution is strengthened. Pure metals are almost always softer than their alloys.

5.3.2 Age hardening or precipitation hardening

Precipitation hardening, also called *age hardening*, is a heat treatment technique used to increase the yield strength of malleable materials, including most structural alloys of aluminum, magnesium, nickel, titanium, and some stainless steels. In super-alloys, it is known to cause yield strength anomaly providing excellent high temperature strength. Precipitation hardening relies on changes in solid solubility with temperature to produce fine particles of an impurity phase, which impede the movement of dislocations, or defects in a crystal's lattice. Since dislocations are often the dominant carriers of plasticity, this serves to harden the material. The impurities play the same role as the particle substances in particle-reinforced composite materials. Just as the formation of ice in air can produce clouds, snow, or hail, depending upon the thermal history of a given portion of the atmosphere, precipitation in solids can produce many different sizes of particles, which have radically different properties. Unlike ordinary tempering, alloys must be kept at elevated temperature for hours to allow precipitation to take place. This time delay is called *aging*.

5.3.3 Residual stress during quenching and heating

Residual stresses are stresses that remain in a solid material after the original cause of the stresses has been removed. Residual stress may be desirable or undesirable. For example, laser peening imparts deep beneficial compressive residual stresses into metal components such as turbine engine fan blades, and it is used in toughened glass to allow for large, thin, crack- and scratch-resistant glass displays on smartphones. However, unintended residual stress in a designed structure may cause it to fail prematurely.

When undesired residual stress is present from prior metalworking operations, the amount of residual stress may be reduced using several methods. These methods may be classified into thermal and mechanical (or non-thermal) methods. All the methods involve processing the part to be stress relieved as a whole.

Thermal method

The thermal method involves changing the temperature of the entire part uniformly, either through heating or cooling. When parts are heated for stress relief, the process may also be known as stress relief bake. Cooling parts for stress relief is known as cryogenic stress relief and is relatively uncommon.

i) Stress relief bake: Most metals, when heated, experience a reduction in yield strength. If the material's yield strength is sufficiently lowered by heating, locations within the material that experienced residual stresses greater than the yield strength (in the heated state) would yield or deform. This leaves the material with residual stresses that are at most as high as the yield strength of the material in its heated state.

Stress relief bake should not be confused with annealing or tempering, which are heat treatments to increase ductility of a metal. Although those processes also involve heating the material to high temperatures and reduce residual stresses, they also involve a change in metallurgical properties, which may be undesired.

For certain materials such as low alloy steel, care must be taken during stress relief bake so as not to exceed the temperature at which the material achieves maximum hardness (See Tempering in alloy steels).

ii) Cryogenic stress relief: Cryogenic stress relief involves placing the material (usually steel) into a cryogenic environment such as liquid nitrogen. In this process, the material to be stress relieved will be cooled to a cryogenic temperature for a long period, then slowly brought back to room temperature.

Non-thermal methods

Mechanical methods to relieve undesirable surface tensile stresses and replace them with beneficial compressive residual stresses include shot peening and laser peening. Each works the surface of the material with a media: shot peening typically uses a metal or glass material; laser peening uses high intensity beams of light to induce a shock wave that propagates deep into the material.

REVIEW QUESTIONS

1. What is solidification? Explain the mechanism solidification process.
2. Differentiate between homogenous and heterogeneous nucleation.
3. Explain solidification time and cooling curve with figure.
4. Explain different types of casting process.
5. What are the causes of solidification defects? Explain different types of solidification defects.
6. What is solid solution? What are the factors affecting solubility of solute atom?
7. Explain the term phase, phase rule and phase diagram.
8. What is inverse lever rule? How it is used to find out the composition of any element present in the alloy?
9. Explain different types of phase reactions.
10. What is age hardening?

CHAPTER 6

IRON-IRON CARBIDE DIAGRAM AND HEAT TREATMENT OF STEELS

6.1 Iron-iron carbide diagram

The alloys of iron are the most extensively used metals in industry. The principal of these are steel and cast iron, which are alloys of iron with carbon. Various alloying elements are added to obtain steel and cast iron with the required properties. Alloys with a carbon content up to 2 percent are known as steels whereas those having carbon above 2 percent are called cast irons. The iron-carbon diagram provides the most prominent example of heat-treatment and transformation. The primary constituent of iron-carbon diagram is the metal iron.

6.1.1 Iron-carbon equilibrium diagram

An equilibrium, phase or constitutional diagram is a graphic representation of the effects of temperature and composition upon the phase present in an alloy. Iron carbon equilibrium diagram indicates the phase changes that occur during heating and cooling and the nature and amount of the structural component that exist at any temperature. Besides, it establishes a correlation between the microstructure and properties of steel and cast iron and provides a basis for the understanding of principles of heat treatment.

Figure 6.1 shows the iron-iron carbide (Fe-Fe_3C) phase diagram. In this diagram, the carbon composition (weight percent) is plotted along the horizontal axis and temperature along the vertical axis. The diagram shows the phases present at various temperatures for very slowly cooled iron-carbon alloys with carbon content up to 6.7%. This diagram gives us information about the following points:

- Solid phases in the phase diagram
- Invariant reactions in the diagram
- Critical temperatures
- Eutectoid, hypo-eutectoid and hyper-eutectoid steels

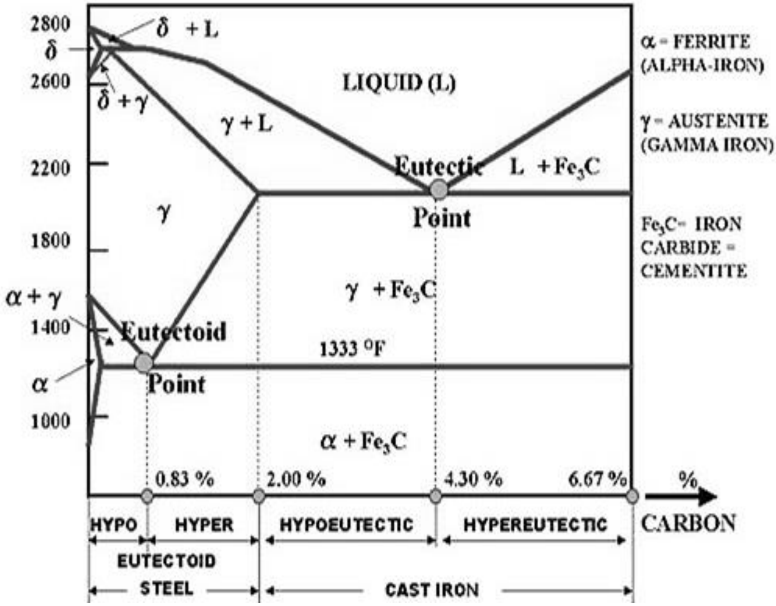


Figure 6.1: Iron- iron carbide diagram

6.1.2 Constituents of iron

The different microscopic constituent of iron and steel are as follows:

1. **Austenite**

Austenite is the solid solution of carbon and/or other alloying elements i.e., Mn, Ni etc., in gamma iron. Carbon is in interstitial solid solution whereas Mn, Ni, Cr etc. are in substitutional solid solution with iron. Austenite can dissolve maximum 2% carbon at 1130^{0}C.Austenite is normally not stable at room temperature. Under certain conditions, however, it is possible to obtain austenite at room temperature as in austenite stainless steel. Austenite is non-magnetic and soft. It is also known as (γ) gamma-iron, which is an interstitial solid solution of carbon dissolved in iron with a face-centered cubic crystal (F.C.C) structure. Average properties of austenite are:

Tensile strength: 150,000 psi.

Elongation: 10 % in 2 in gage length.

Hardness Rockwell: C 40

Toughness: High

2. Ferrite

Ferrite is B.C.C. iron phase with very limited solubility for carbon. The maximum solubility is 0.025% carbon at 723^{0}C. Ferrite dissolved only 0.08% carbon at room temperature. Ferrite is the softest structure that appears on the Fe-C equilibrium diagram. It is also known as (α) alpha -iron. It is the softest structure on the iron-iron carbide diagram. Average properties are:

Tensile Strength: 40,000 psi

Elongation: 40 % in 2 in gage length

Hardness: Less than Rockwell B 90.

Toughness: Low

3. Cementite

Cementite or iron carbide, chemical formula Fe_3C, contains 6.67% carbon by weight. It is a typical hard and brittle interstitial compound at low tensile strength but high compressive strength. Cementite is the hardest structure that appears on the iron-carbon equilibrium diagram. Its crystal structure is Orthorhombic.

4. Ledeburite

It is the eutectic mixture of austenite and cementite. It contains 4.3 % carbon and represents the eutectic form of cast iron. Ledeburite exists when the carbon content is greater than 2 %, which represents the dividing line on the equilibrium diagram between steel and cast iron. It is formed at about 1130^0C.

5. Pearlite

It is the eutectoid mixture containing 0.83 % carbon and is formed at 1333^oF on very slow cooling. It is very fine plate like or lamellar mixture of ferrite and cementite. The structure of pearlite includes a white matrix (ferritic background) which includes thin plates of cementite. Average properties are:

Tensile Strength: 120,000 psi

Elongation: 20 % in 2 in gage length

Hardness: Rockwell C 20 or BHN 250-300

6. Delta iron

Delta iron exists between 2552 and 2802^oF. It may exist in combination with the melt to about 0.50 % Carbon, in combination with austenite to about 0.18 % carbon and in a single phase state out to about 0.10 % carbon. Delta iron has the body centered cubic (B.C.C) crystal structure and is magnetic.

7. Bainite

Bainite is the constituent produced in a steel when austenite transforms at a temperature below that at which pearlite is produced and above that at which martensite is formed. Bainite is produced by austempering. Thus, bainite is a decomposition product of austenite, consisting of an aggregate of ferrite and carbide. Bainite is an isothermal transformation product and cannot be produced by continuous cooling.

8. Martensite

Martensite is a metastable phase of steel, formed by transformation of austenite below the martensite temperature in TTT (Time temperature transformation) diagram. Martensite is an interstitial supersaturated solid solution of carbon is α-iron and has a body-centered-tetragonal lattice. The decomposition of austenite below 320^0C starts the formation of martensite. Martensite, normally, is product of quenching.

9. Troostite

Troostite (nodular) is a mixture of radial lamellae of ferrite and cementite and therefore differs from pearlite only in the degree of fineness and carbon content. It is composed of the cementite phase in a ferrite matrix that cannot be resolved by a light microscope. It is less hard and brittle than martensite. It is also produced by cooling the metal slowly until transformation begins and then rapidly to prevent its completion. It has a dark appearance on etching.

10. Sorbite

Sorbite is the microstructure consisting of ferrite and finely divided cementite, produced on tempering martensite above approximately 4500C. It is produced when steel is heated at a fairly rapid rate from the temperature of the solid solution to normal room temperature. It has good strength and is practically pearlite. Its properties are intermediate between those of pearlite and troostite.

6.1.3 Phase transformation

Phase transformation (also called phase change) that occur in an alloy determine its micro-structural features. A phase change may involve a change in the state of aggregation e.g. liquid to solid, in the crystal structure e.g. B.C.C. to F.C.C. or composition e.g. 0.77 percent carbon in austenite to 0.02 percent carbon in ferrite and 6.67 percent carbon in cementite. There are two basic mechanism of phase transformations. In the nucleation and growth mechanism, a tiny particle indicated in the structure and composition to the product phase nucleates and then grows in size. In spinodal decomposition, compositional fluctuations occur in the parent phase. These fluctuations grow in intensity with time, until two discrete phases corresponding to the equilibrium compositions form. Many metallurgical transformation occur by nucleation and growth.

The reactions that can be observed in iron-iron carbide diagram are described below.

Peritectic reaction (at 1495ºC)

In equilibrium at the temperature 1495^0C, δ-iron having composition 0.1 percent carbon and the liquid phase of composition 0.51 percent carbon, austenite is formed.

At this temperature, the alloys undergo a peritectic reaction i.e.

Liquid (0.51%C) + δ - iron (0.1%C) = γ- iron (0.18%C)

Eutectic Reaction (at 1148 ºC)

In equilibrium at the temperature 1148ºC, the liquid phase of composition 0.51 percent carbon decomposes to form 2.0 percent carbon austenite and cementite of 6.67% carbon.

At this temperature, the alloys undergo a eutectic reaction i.e.

Liquid (4.30%C) →γ (2.00%C) + Fe_3C (6.67%C)

Eutectoid Reaction (at 723 ºC)

In equilibrium at the temperature 723ºC, γ- iron of composition 0.80 percent carbon decomposes to form 0.025 percent carbon α iron and cementite of 6.67% carbon.

At this temperature, the alloys undergo a eutectoid reaction i.e.

$$\gamma\ (0.80\%\ C) \rightarrow \alpha\ (0.025\%\ C) + Fe_3C\ (6.67\%\ C)$$

6.1.4 Classification of steel and cast iron

Steels are iron with carbon content percent less than 2.14 and cast irons are iron with carbon content percent more than 2.14. Steels are also further classified as follows.

- Hypo-eutectoid steel (< 0.8 %C)
- Hyper-eutectoid steel (0.8% < %C ≤ 2.0%)

Commercially steels are classified as listed below.

- Commercial pure irons (% C < 0.008)
- Low-carbon steels (0.008 - %C - 0.3)
- Medium carbon steels (0.3 - %C - 0.8)
- High-carbon steels (0.8 - %C - 2.14)

Cast irons are also further classified as follows.

- Hypo-eutectic cast iron (2.0% < %C < 4.3%)
- Hyper-eutectic cast iron (4.3% < %C < 6.67%)

6.1.5 Applications of iron-iron carbide diagram

With the core knowledge gained from the iron-carbon phase, one can investigate different types of structures derived from heat treatment of steel. The iron-carbon diagram provides a valuable foundation on which knowledge of both plain carbon and alloy steels in their immense variety is gained. The iron-iron carbide diagram is useful in finding out what will be the crystal structure and the physical and chemical properties of iron at known carbon percentage and temperature.

6.1.6 Limitations of iron – iron carbide diagram

1. Information about transformation of austenite at varying cooling rates is missing.
2. Design of heat treatment for desired properties is not possible.
3. It does not give the answer to the question "How much time is required to start the transformation or how long does the transformation take to complete?".

6.2 Simple heat treatments

Heat-treatment may be defined as an operation involving the heating of solid metals to definite temperatures, followed by cooling at suitable rates in order to obtain certain physical properties which are associated with change in the nature, form, size and distribution of the micro-constituents. Heat treatment is a very important process in the various fabrication and manufacturing operations.

All heat treatment processes, therefore, comprise the following three stages:

i. Heating the metal to a pre-decided temperature.

ii. Holding it at that temperature for sufficient time so that structure of the metal become uniform throughout.

iii. Cooling the metal at a predetermined rate in a suitable media so as to force the metal to acquire a desired internal structure and this, obtain the desired properties to the required extent.

After heating at particular temperature, the steel is hold at temperature for sufficient time to allow necessary changes, that holding time is known as soaking time. The time of holding at that temperature depends on the type of steel, size and shape of the parts to be heat-treated. After the component is held at the hardening temperature for a desired length of time it is taken out for cooling or quenching in order to obtain a hard martensite structure. The medium used for quenching depends upon the chemical composition of the steel, the hardness required, permissible degree of distortion and the complexity of the component. Water, brine, molten salt are some of the quenching media in use. The grade of steel, section thickness, distortion allowed and the properties to be imparted to the component govern the quenching methods to be adopted.

The purpose of heat treatment is to achieve one or more of the following objectives:

i) To relieve internal stresses set up during cold-working, casting, welding and hot-working operations.
ii) To improve machinability.
iii) To change grain size.
iv) To improve mechanical properties like hardness, toughness, strength, ductility etc.
v) To soften metals for further treatment as wire drawing and cold rolling.
vi) To modify the structure to increase wear, heat and corrosion resistance.
vii) To modify magnetic and electrical properties.

The different types of heat treatment done in steels are shown in the given figure.

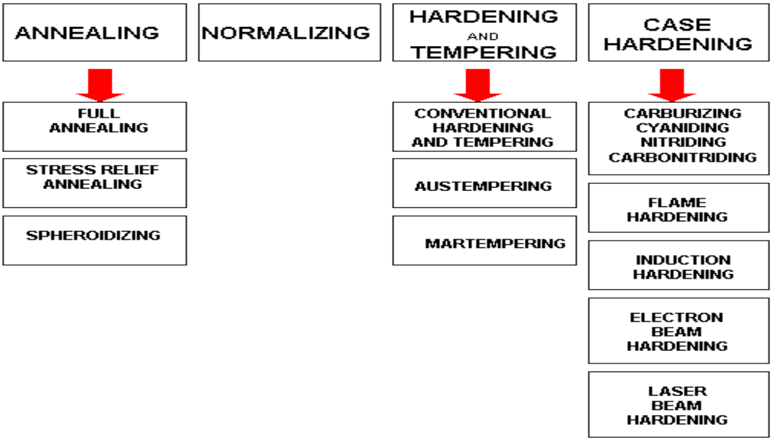


Figure 6.2: Different types of heat treatment

6.2.1 Annealing

Annealing is a heat process whereby a metal is heated to a specific temp/color and then allowed to cool slowly. This softens the metal which means it can be cut and shaped more easily. Mild steel, is heated to a red hot state and allowed to cool slowly. However, metals such as aluminum will melt if heated for too long. Annealed metals are relatively soft and can be cut and shaped more easily. They bend easily when pressure is applied. As a rule, they are heated and allowed to cool slowly.

The different types of annealing are:

- Full Annealing
- Process Annealing
- Stress-Relief Annealing
- Spheroidisation

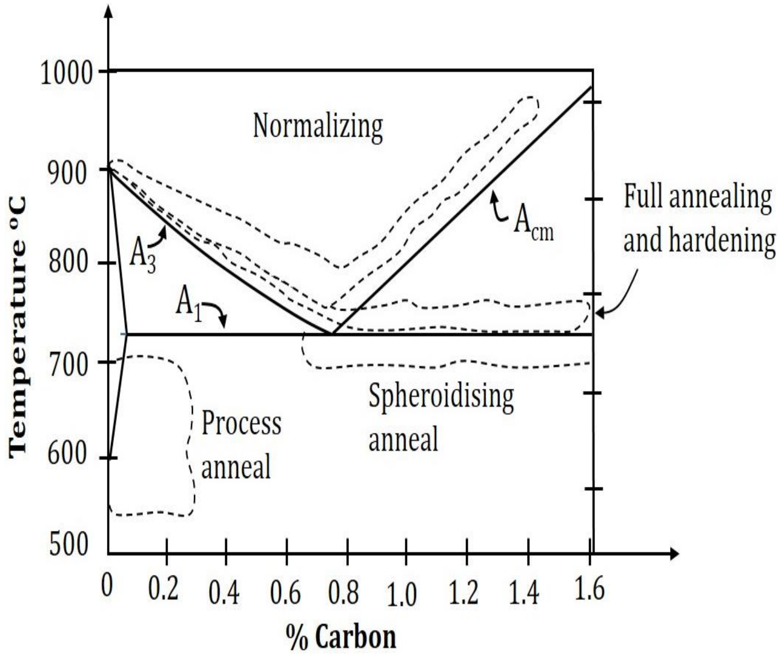


Figure 6.3: Annealing

Full Annealing

Full Annealing is the process of slowly raising the temp about 50^0C (90^0F) above the Austenite temp line A_3 or line A_{CM} in the case of hypo-eutectoid steels (steels with less than 0.77%).It is held at this temp for sufficient time for all the materials to transform into Austenite or Austenite-Cementite as the case may be. It is then slowly cooled at the rate of about 20^0C/hr. (36^0F/hr.) in a furnace to about 50^0C (90^0F) into the Ferrite-Cementite. At this point, it can be cooled in a room temp air with natural convection.

The grain structure has coarse Pearlite with Ferrite or Cementite (depending on whether hypo or hyper-eutectoid). The steel becomes soft and ductile.

Process Annealing

Process Annealing is used to treat work-hardened parts made out of low carbon steels (less than 0.25%). This allows the parts to be soft enough to undergo further cold working without fracturing. Process Annealing is done by raising the temp to just below the Ferrite-Austenite region, line A_1 on the diagram. This temp is about 727^0C (1341^0F) so heating it to about 700^0C (1292^0F) should suffice. This is held long enough to allow recrystallization of the ferrite phase, and then cooled in still air. Since the material stays in the same phase throughout the process, the only change that occurs in the size, shape and distribution of the grain structure. This process is cheaper than either full annealing or normalizing since the material is not heated to a very high temp or cooled in a furnace.

Stress-Relief Annealing

Stress-Relief Annealing is used to reduce residual stresses in large castings, welded parts and cold-formed parts. Such parts tend to have stresses due to thermal cycling or work hardening. Parts are heated to temp of up to $600\text{-}650^0$C and held for an extended time about 1 hour and more and then slowly cooled in still air.

Spheroidisation

Spheroidisation is an annealing process used for high carbon steels (carbon more than 0.6%) that will be machined or cold formed subsequently. This is done by one of the following ways:

1. Heat the part to a temperature just below the Ferrite- Austenite line, line A1 or below the Austenite - Cementite line, especially below the 727^0C (1340^0F) line. Hold the temperature for a prolonged time and follow by fairly slow cooling.

2. Cycle multiple times between temperature slightly above and slightly below the 727^0C (1340^0F) line, say for example between 700 and 750^0C and cool slowly.
3. For tool and alloy steels, heat to 750 to 800^0C (1382 – 1472^0F) and hold for several hours followed by slow cooling.

Figure 6.4: a) Spheroidite b) Austenite c) Cementite

All these methods result in a structure in which all the Cementite is in the form of small globules (spheroids) dispersed throughout the ferrite matrix. This structure allows for improved machining in continuous cutting operations such as lathes and screw machines. Spheroidisation also improves resistance to abrasion.

Applications of Annealing

Annealed metals are relatively soft and can be cut and shaped more easily. They bend easily when pressure is applied. As a rule they are heated and allowed to cool slowly. Thus applications include decorative items, Low-friction applications (locks, gears, and doorknobs), plumbing/electronics, musical instruments for acoustic properties & in bells and cymbals.

6.2.2 Normalizing

Normalizing aims to give the steel a uniform and fine-grained structure. The process is used to obtain a predictable microstructure and an assurance of the steel’s mechanical properties.

During normalizing, the material is heated to a temperature approximately equivalent to the hardening temperature (800-920°C). At this temperature new austenitic grains are formed. The austenitic grains are much smaller than the previous ferritic grains. After heating and a short soaking time the components are cooled freely in air (gas). During cooling, new ferritic grains are formed with a further refined grain size. In some cases, both heating and cooling take place under protective gas to avoid oxidation and decarburization.

Applications of Normalizing

By completing the normalization of metal (e.g. steel), the strength of the piece of steel can be improved, compared to steel that is not normalized or treated with another heating process. The ductile performance of the steel can also be improved through normalization; this is possible without reducing the hardness or strength as can happen with other heat treatments. By using the normalization process, modifications can be made to the grain structure of the steel to make it usable for specific purposes. Steels with low levels of carbon do not have their rate of ductility altered by using the normalization process. It is usually done for steel parts that require maximum amount of impact strength.

Normalization is mainly used on carbon and low alloyed steels to normalize the structure after forging, hot rolling or casting. The hardness obtained after normalizing depends on the steel dimension analysis and the cooling speed used (approximately 100-250 HB).

Comparison between Annealing and Normalizing

In general, the main purpose of annealing heat treatment is to soften the steel, regenerate overheated steel structures or just remove internal tensions. It basically consists of heating to austentising temperature (800ºC and 950ºC depending on the type of steel), followed by slow cooling. Normalization is an annealing process. The objective of normalization is to intend to leave the material in a normal state, in other words with the absence of internal tensions and even distribution of carbon. For the process the high temperatures are maintained until the complete transformation of austenite with air cooling. It is usually used as a post-treatment to forging, and pre-treatment to quenching and tempering. Induction is used in most applications of annealing and normalizing in compared to conventional ovens.

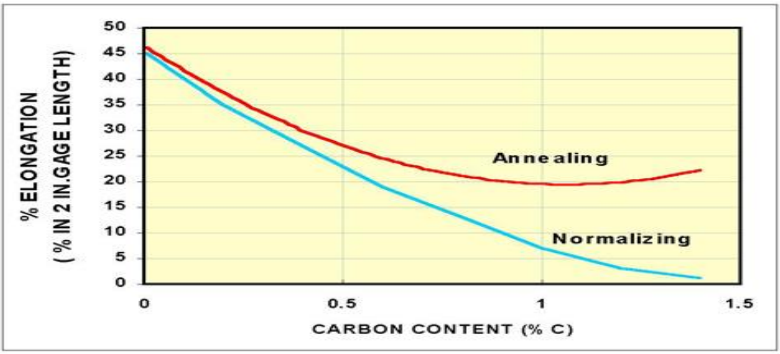


Figure 6.5: Effect of annealing and normalizing in ductility

As indicated in figure 6.5, annealing and normalizing do not present a significant difference on the ductility of low carbon steels. As the carbon content increases, annealing maintains the % elongation around 20%. On the other hand, the ductility of the normalized high carbon steels drop to 1 to 2 % level.

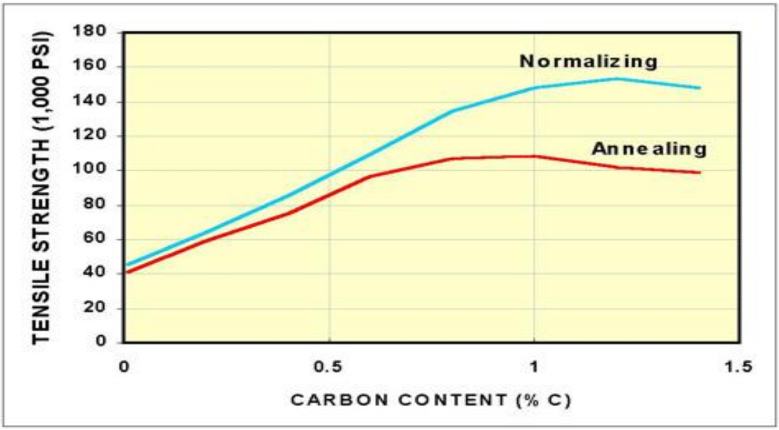


Figure 6.6: Effect of annealing and normalizing on tensile strength

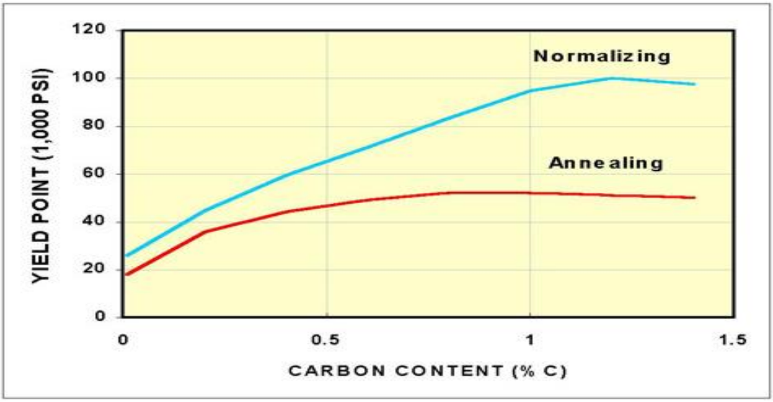


Figure 6.7: Effect of annealing and normalizing on yield point

Figures 6.6 and 6.7 show that the tensile strength and the yield point of the normalized steels are higher than the annealed steels. Normalizing and annealing do not show a significant difference on the tensile strength and yield point of the low carbon steels. However, normalized high carbon steels present much higher tensile strength and yield point than those that are annealed.

6.2.3 Quenching

Quenching of a steel is done from the austenitic range to produce martensite. The cooling rate at the centre of the cross-section should exceed the critical cooling rate for the steel to get full hardening. Hypoeutectoid steels are heated above the AC_3 temperature to ensure full austenitisation and to avoid the soft ferrite in the final micro-structure. For hypereutectoid steels, this is not necessary, as the undissolved cementite on heating to just above AC_1 is itself very hard. Moreover by heating above AC_m and also due to larger carbon content in solution in austenite. Tempering is almost always necessary to remove the residual stresses and to reduce the brittleness of martensite. For example a carbon tool for cutting is tempered only slightly to relieve the quenching stresses, so that a large part of the hardness achieved by quenching is retained.

Quenching Method

The component is held at the hardening temperature for a desired length of time, then it is taken out for cooling or quenching, in order to obtain a hard martensite structure. The rate of cooling must be controlled. So that the formation of soft pearlite or bainite is prevented. The various quenching methods are

1. **Direct Quenching:**

   In this method, components held at the hardening temperature for the required length of time and directly quenched in water or oil to get the hard martensite structure.

The disadvantages of this method are the high distortion and the presence of cracks due to very high rate of cooling in the martensite range. However this method is suitable for mild steel, low and medium carbon steels.

Quenching medium

The various mediums in which quenching can be done are:

- **Water:** Quenching can be done by plunging the hot steel in water. The water adjacent to the hot steel vaporizes, and there is no direct contact of the water with the steel. This slows down cooling until the bubbles break and allow water contact with the hot steel. As the water contacts and boils, a great amount of heat is removed from the steel. With good agitation, bubbles can be prevented from sticking to the steel, and thereby prevent soft spots. Water is a good rapid quenching medium, provided good agitation is done. However, water is corrosive with steel, and the rapid cooling can sometimes cause distortion or cracking.

- **Salt Water:** Salt water is a more rapid quench medium than plain water because the bubbles are broken easily and allow for rapid cooling of the part. However, salt water is even more corrosive than plain water, and hence must be rinsed off immediately.
- **Oil:** Oil is used when a slower cooling rate is desired. Since oil has a very high boiling point, the transition from start of martensite formation to the finish is slow and this reduces the likelihood of cracking. Oil quenching results in fumes, spills, and sometimes a fire hazard.

- **Polymer quench:** Polymer quenches that will produce a cooling rate in between water and oil. The cooling rate can be altered by varying the components in the mixture-as these are composed of water and some glycol polymers. Polymer quenches are capable of producing repeatable results with less corrosion than water and less of a fire hazard than oil. But, these repeatable results are possible only with constant monitoring of the chemistry.

- **Cryogenic quench:** Cryogenics or deep freezing is done to make sure there is no retained Austenite during quenching. The amount of martensite formed at quenching is a function of the lowest temperature encountered. At any given temperature of quenching there is a certain amount of martensite and the balance is untransformed Austenite. This untransformed Austenite is very brittle and can cause loss of strength or hardness, dimensional instability, or cracking. Quenches are usually done to room temperature. Most medium carbon steels and low alloy steels undergo transformation to 100% martensite at room temperature. However, high carbon and high alloy steels have retained Austenite at room temperature. To eliminate retained Austenite, the quench temperature has to be lowered. This is the reason to use cryogenic quenching.

Hardness and Hardenability

Hardness is a function of the carbon content of the steel. Hardening of a steel requires a change in structure from the body-centered cubic structure found at room temperature to the face-centered cubic structure found in the austenitic region.

Hardenability, which is a measure of the depth of full hardness achieved, is related to the type and amount of alloying elements. Different alloys, which have the same amount of Carbon content, will achieve the same amount of maximum hardness; however, the depth of full hardness will vary with the different alloys. The reason to alloy steels is not to increase their strength, but increase their hardenability – the ease with which full hardness can be achieved throughout the material.

The hardenability of an iron metal or alloy to form martensite when cooled at various rates is measured by its hardenability. A material with higher hardenability will form martensite in large sections on quenching. It is not related to actual hardness of the martensite formed, which is almost solely dependent upon the carbon content of the iron metal or alloy. As a matter of fact hardenability is in effect the reciprocal of the critical cooling rate.

Jominy End Quench Test

One standard procedure that is widely utilized to determine hardenability is the Jominy end quench test.

In Jominy test, the standard specimen consists of a cylindrical rod 4" long 1" in diameter. The test specimen is first heated to a suitable austentising temperature and held there long enough to obtain a uniform austenite structure. It is then placed in a jig and stream of water is allowed to strike one end of the specimen. The cooling conditions within the Jominy bar during quenching changes very little across the diameter but vary rapidly at that end and progressively less rapidly at points towards the opposite end.

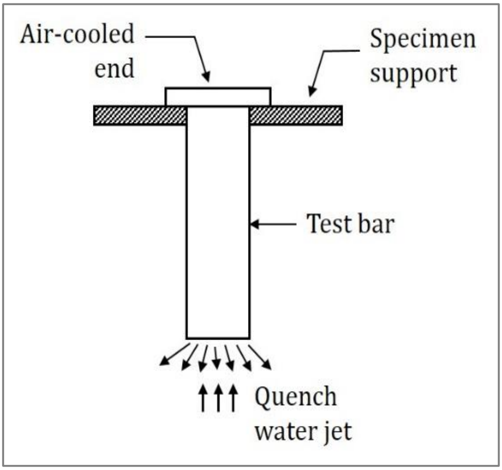


Figure 6.8: Jominy test

A plot is made of the hardness reading against the distance from the quenched end of the bar. Such a plot is shown if figure 6.8.

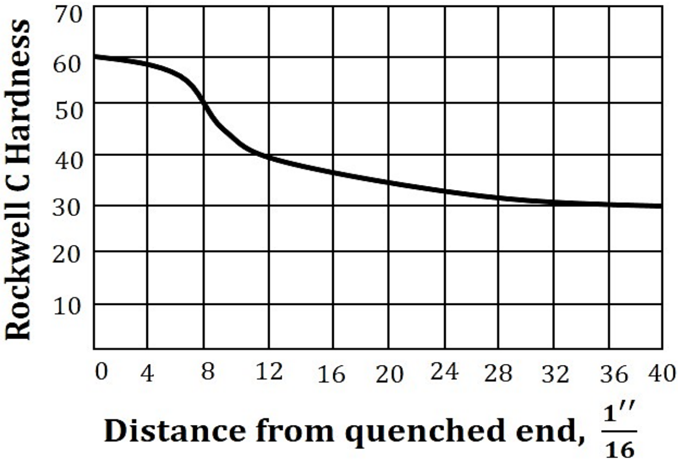


Figure 6.9: Plot of Rockwell C Hardness versus distance from the quenched end of the bar

The addition of alloys or coarsening of the austenitic grain structure increase the hardenability of steel. Any steel that has low critical cooling rate will harden deeper than one that has a high cooling rate of quenching.

Time-Temperature-Transformation (TTT) diagram

T (Time) T (Temperature) T (Transformation) diagram is a plot of temperature versus the logarithm of time for a steel alloy of definite composition. It is used to determine when transformations begin and end for an isothermal (constant temperature) heat treatment of a previously austenitised alloy. When austenite is cooled slowly to a temperature below LCT (Lower Critical Temperature), the structure that is formed is Pearlite. As the cooling rate increases, the pearlite transformation temperature gets lower. The microstructure of the material is significantly altered as the cooling rate increases.

By heating and cooling a series of samples, the history of the austenite transformation may be recorded. TTT diagram indicates when a specific transformation starts and ends and it also shows what percentage of transformation of austenite at a particular temperature is achieved.

Cooling rates in the order of increasing severity are achieved by quenching from elevated temperatures as follows: furnace cooling, air cooling, oil quenching, liquid salts, water quenching, and brine. If these cooling curves are superimposed on the TTT diagram, the end product structure and the time required to complete the transformation may be found.

In figure 6.10 the area on the left of the transformation curve represents the austenite region. Austenite is stable at temperatures above LCT but unstable below LCT. Left curve indicates the start of a transformation and right curve represents the finish of a transformation. The area between the two curves indicates the transformation of austenite to different types of crystal structures. (Austenite to pearlite, austenite to martensite, austenite to bainite transformation.)

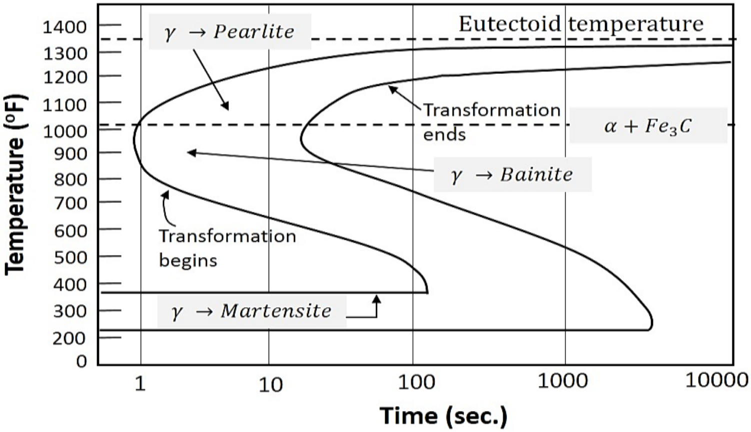


Figure 6.10: Time-Temperature Transformation Curve

Figure 6.11 represents the upper half of the TTT diagram. As indicated in figure, when austenite is cooled to temperatures below LCT, it transforms to other crystal structures due to its unstable nature. A specific cooling rate may be chosen so that the transformation of austenite can be 50 %, 100 % etc.

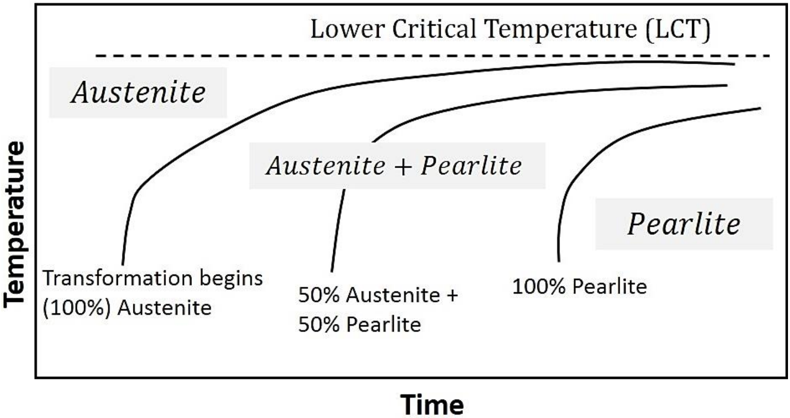


Figure 6.12: Upper half of TTT Curve

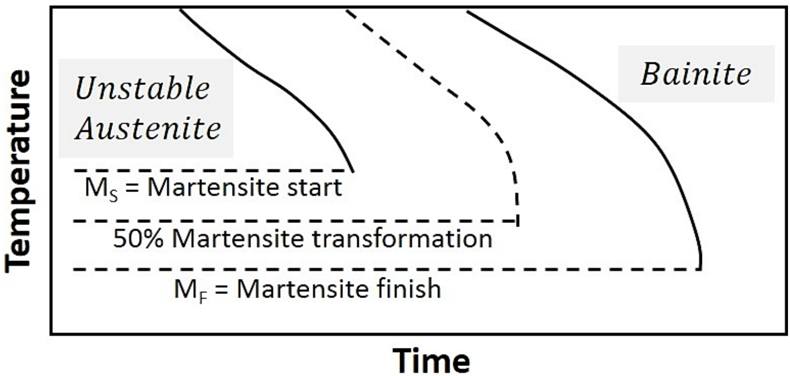


Figure 6.11: Lower half of TTT Curve

If the cooling rate is very slow such as annealing process, the cooling curve passes through the entire transformation area and the end product of this the cooling process becomes 100% Pearlite. In other words, when slow cooling is applied, all the Austenite will transform to Pearlite. If the cooling curve passes through the middle of the transformation area, the end product is 50 % Austenite and 50 % Pearlite, which means that at certain cooling rates we can retain part of the Austenite, without transforming it into Pearlite.

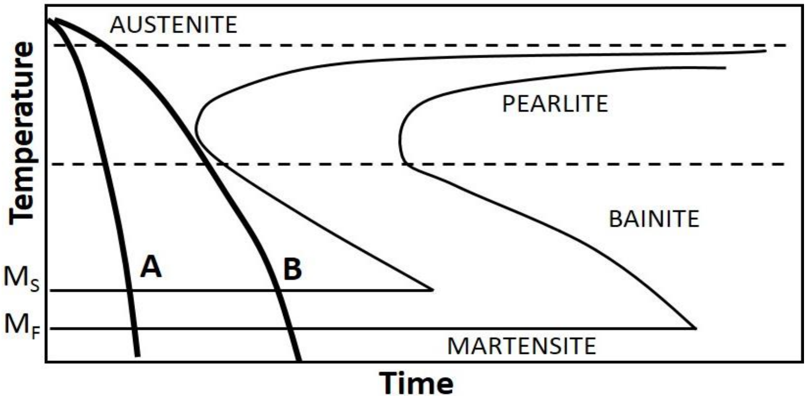


Figure 6.13: Transformation with different cooling rates

In the figure 6.14, the cooling rates A and B indicate two rapid cooling processes. In this case, curve A will cause a higher distortion and a higher internal stresses than the cooling rate B. The end product of both cooling rates will be martensite. Cooling rate B is also known as the Critical Cooling Rate, which is represented by a cooling curve that is tangent to the nose of the TTT diagram. Critical cooling rate is defined as the lowest cooling rate which produces 100% martensite while minimizing the internal stresses and distortions.

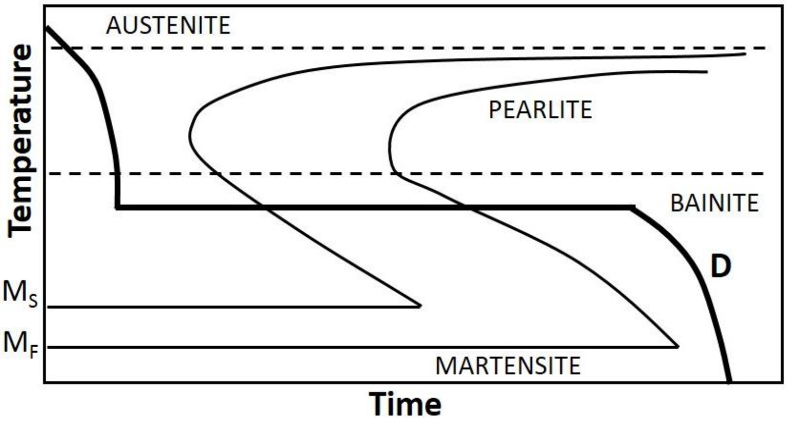


Figure 6.15: Transformation with an interruption

In figure 6.15, a rapid quenching process is interrupted (horizontal line represents the interruption) by immersing the material in a molten salt bath and soaking at a constant temperature followed by another cooling process that passes through bainite region of TTT diagram. The end product is bainite, which is not as hard as martensite. As a result of cooling rate D; more dimensional stability, less distortion and less internal stresses are created.

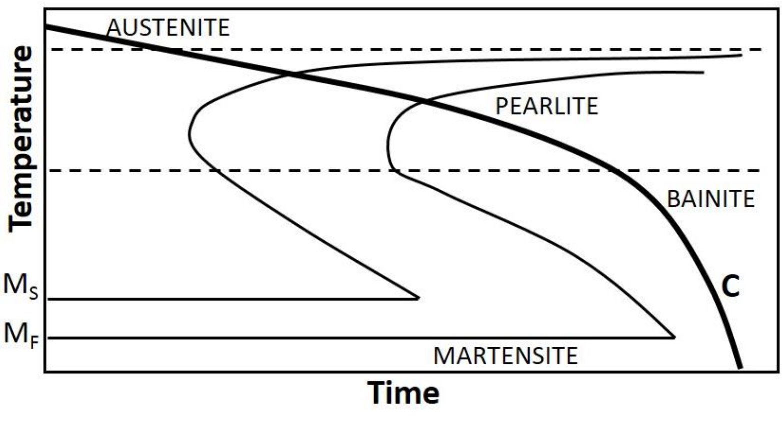


Figure 6.16: Transformation with a slow cooling process

In figure 6.16, cooling curve C represents a slow cooling process, such as furnace cooling. An example for this type of cooling is annealing process where all the austenite is allowed to transform to Pearlite as a result of slow cooling.

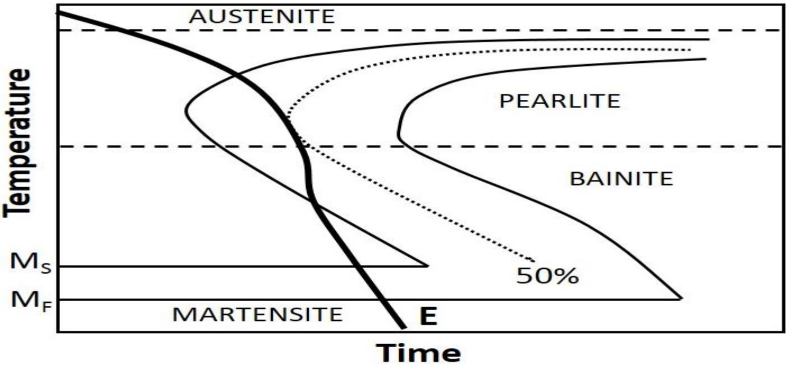


Figure 6.17: Transformation with an insufficient cooling rate

Sometimes the cooling curve may pass through the middle of the Austenite-Pearlite transformation zone. In figure 6.17, cooling curve E indicates a cooling rate which is not high enough to produce 100% martensite. This can be observed easily by looking at the TTT diagram. Since the cooling curve E is not tangent to the nose of the transformation diagram, austenite is transformed to 50% Pearlite (curve E is tangent to 50% curve). Since curve E leaves the transformation diagram at the martensite zone, the remaining 50 % of the austenite will be transformed to martensite.

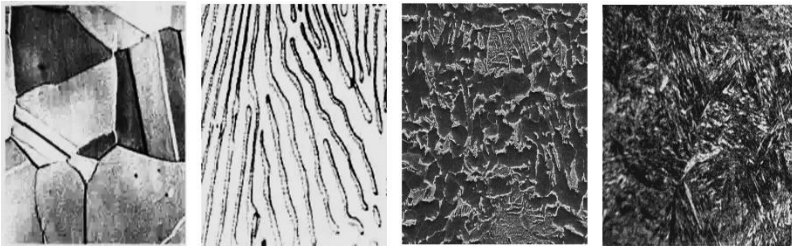


Figure 6.18: (from left to right) Microstructures of Austenite, Pearlite, Bainite and Martensite

Continuous cooling transformation (CCT) diagrams

CCT diagrams are generally more appropriate for engineering applications as components are cooled (air cooled, furnace cooled, quenched etc.) from a processing temperature as this is more economic than transferring to a separate furnace for an isothermal treatment.

CCT measure the extent of transformation as a function of time for a continuously decreasing temperature. In other words a sample is austenitised and then cooled at a predetermined rate and the degree of transformation is measured, for example by dilatometry. Obviously a large number of experiments is required to build up a complete CCT diagram.

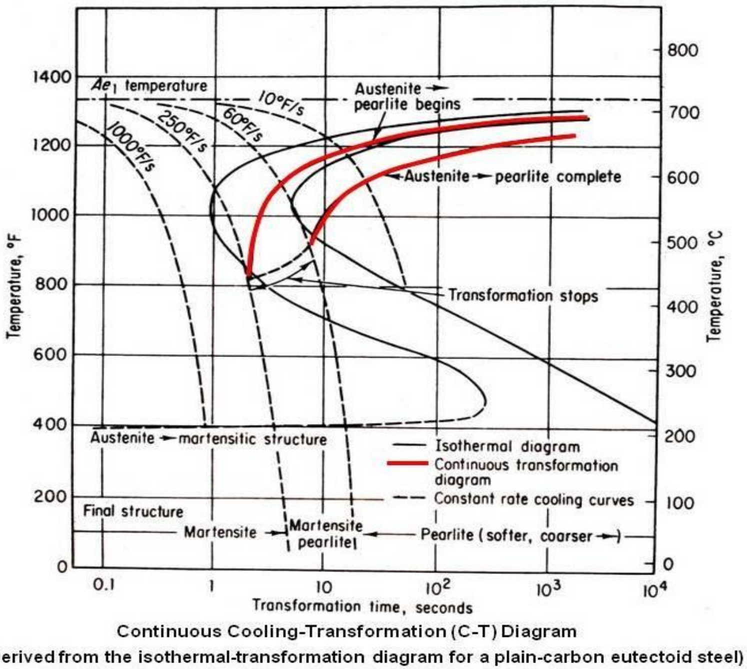


Figure 6.19: Continuous cooling transformation diagram

6.2.4 Tempering

Tempering is a low temperature heat treatment process normally performed after neutral hardening, double hardening, atmospheric carburizing, carbonitriding or induction hardening in order to reach a desired hardness/toughness ratio.

The tempering temperature may vary, depending on the requirements and the steel grade, from 160°C to 500°C or higher. Tempering is normally performed in furnaces which can be equipped with a protective gas option. Protective gas will prevent the surface from oxidation during the process and is mainly used for higher temperatures. For some types of steels the holding time at the tempering temperature is of great importance; an extended holding time will correspond to a higher temperature. Depending on the steel grade a phenomenon known as temper brittleness can occur in certain temperature intervals. Tempering inside this temperature interval should normally be avoided. These areas are shown in the steel suppliers steel catalogues, as well as the most suitable temperature depending on hardness requirements.

Tempering is a heat treatment that reduces the brittleness of a steel without significantly lowering its hardness and strength. All hardened steels must be tempered before use.

The various hardening methods are:

(1) Conventional heat, quench and temper process

(2) Stepped quenching or martempering

(3) Isothermal quenching or austempering

Conventional heat, quench and temper process

In this process, austenite is transformed to martensite as a result of rapid quench from furnace to room temperature. Then, martensite is heated to a temperature which gives the desired hardness. One serious drawback is the possibility of distorting and cracking the metal as a result of severe quenching required to form martensite without transforming any of the austenite to pearlite. During quenching process, the outer area is cooled quicker than the center. Thinner parts are cooled faster than parts with greater cross-sectional areas. What this means is that transformations of the austenite are proceeding at different rates. As the metal cools, it also contracts and its microstructure occupies less volume. Extreme variations in size of metal parts complicate the work of the heat treater and should be avoided in the designing of metal parts. This means there is a limit to the overall size of parts that can be subjected to such thermal processing.

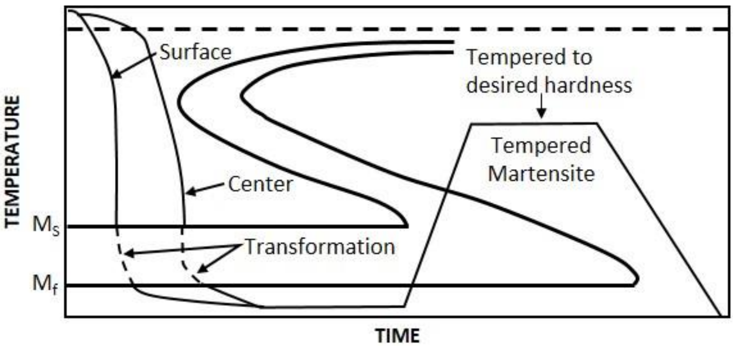


Figure 6.19: Conventional quench and tempering method

Stepped quenching or martempering (marquenching)

To overcome the restrictions of conventional quenching and tempering, Martempering process can be used. Martempering or Marquenching permits the transformation of austenite to martensite to take place at the same time throughout the structure of the metal part. This is shown in Figure. By using interrupted quench, the cooling is stopped at a point above the martensite transformation region to allow sufficient time for the center to cool to the same temperature as the surface.

Then cooling is continued through the martensite region, followed by the usual tempering.

Martempering has the following advantages over conventional quenching:

1. Less volume changes occur due to the presence of a large amount of retained austenite and possibility of self-tempering of martensite.
2. Less warping since the transformations occur simultaneously in all parts of the article.
3. Less chances of quenching cracks appearing in the article.

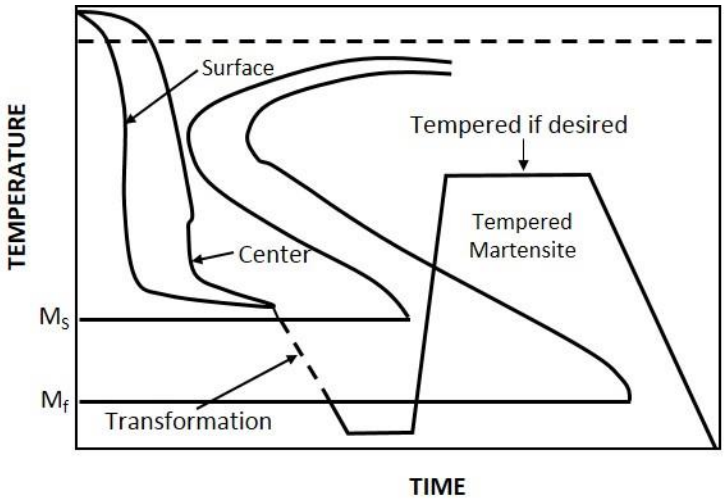


Figure 6.20: Martempering

Isothermal quenching or austempering

It is performed in the same manner principally as martempering but with a longer holding time at hot bath temperature (above the martensite point) to ensure a sufficiently complete austenite decomposition (usually with a circular troostite or bainite).

This method is used to overcome the restrictions of conventional quench and tempering. The quench is interrupted at a higher temperature than for Martempering to allow the metal at the center of the part to reach the same temperature as the surface. By maintaining that temperature, both the center and the surface are allowed to transform to bainite and are then cooled to room temperature.

Molten salts are usually used as a medium in martempering and austempering. The lower the temperature of the salt bath, the higher the cooling rate it provides.

Austempering process is commercially used for thin steel sections to obtain products free from cracks and with good impact resistance.

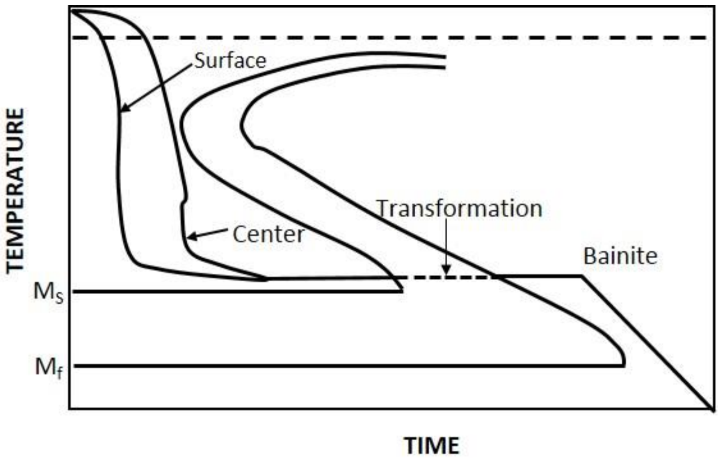


Figure 6.21: Austempering

Advantages of Austempering

1. Less distortion and cracking than Martempering.
2. No need for final tempering (less time consuming and more energy efficient).
3. Improvement of toughness (impact resistance is higher than the conventional quench and tempering).
4. Improved ductility.

6.2.5 Surface hardening processes

It is desirable that a steel to be used should have a hardened surface to resist wear and tear. At the same time, it should have soft and tough interior or core so that it is able to absorb any shocks etc. This is achieved by hardening the surface layers of the article, while the rest of it is left as much. This type of treatment is applied to gears, ball bearings, railway wheels etc.

6.2.5.1 Carburizing

Carburizing is a heat treatment process in which iron or steel absorbs carbon liberated when the metal is heated in the presence of a carbon bearing material, such as charcoal or carbon monoxide, with the intent of making the metal harder. Depending on the amount of time and temperature, the affected area can vary in carbon content. Longer carburizing times and higher temperatures lead to greater carbon diffusion into the part as well as increased depth of carbon diffusion.

When iron or steel is cooled rapidly by quenching, the higher carbon content on the outer surface becomes hard via the transformation from austenite to martensite, while the core remains soft and tough as a ferritic and/or pearlite microstructure.

There are three different types of carburizing processes:

1. Pack Carburizing
2. Gas Carburizing
3. Liquid Carburizing

1. Pack Carburizing

In this process, the part that is to be carburized is packed in a steel container so that it is completely surrounded by granules of charcoal. The charcoal is treated with an activating chemical such as Barium Carbonate ($BaBO_3$) that promotes the formation of Carbon Dioxide (CO_2). This gas in turn reacts with the excess carbon in the charcoal to produce carbon monoxide.

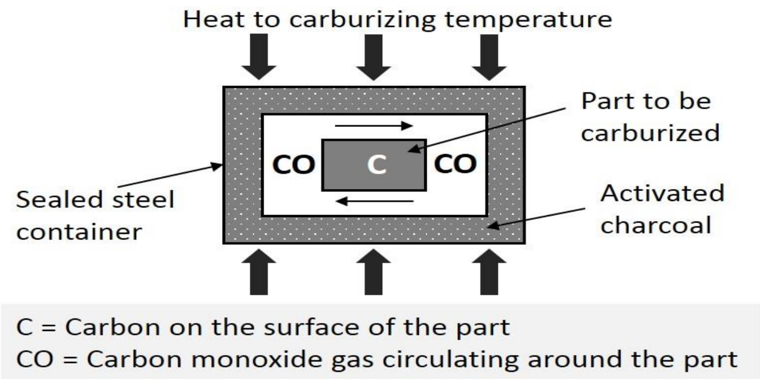


Figure 6.22: Pack carburizing

Carbon Monoxide reacts with the low-carbon steel surface to form atomic carbon which diffuses into the steel. Carbon Monoxide supplies the carbon gradient that is necessary for diffusion. The carburizing process does not harden the steel. It only increases the carbon content to some predetermined depth below the surface to a sufficient level to allow subsequent quench hardening.

Carbon Monoxide reaction: $CO_2 + C \rightarrow 2\ CO$

Reaction of Cementite to Carbon Monoxide:

$$2\ CO + 3\ Fe \rightarrow Fe_3C + CO_2$$

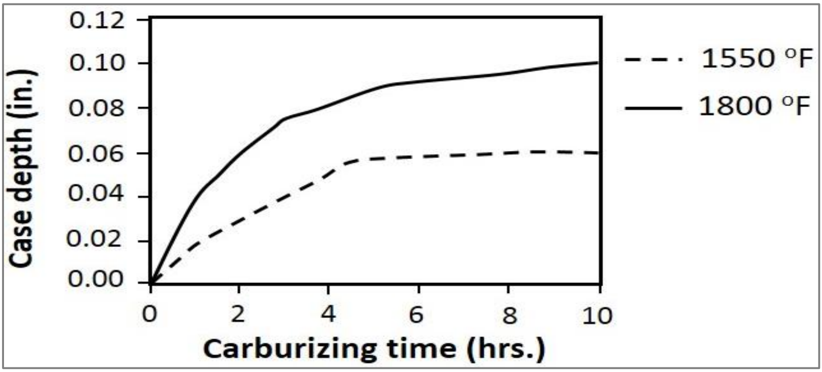


Figure 6.23: Carburizing time at different depths

2. Gas Carburizing

Gas carburizing can be done with any carbonaceous gas, such as methane, ethane, propane, or natural gas. Most carburizing gases are flammable and controls are needed to keep carburizing gas at 1700^0F from contacting air (oxygen). The advantage of this process over pack carburizing is an improved ability to quench from the carburizing temperature. Conveyor hearth furnaces make quenching in a controlled atmosphere possible.

3. Liquid Carburizing

Liquid carburizing can be performed in internally or externally heated molten salt pots. Carburizing salt contains cyanide compounds such as sodium cyanide (NaCN). Cycle times for liquid cyaniding is much shorter (1 to 4 hours) than gas and pack carburizing processes. Disadvantage is that the salt has to be disposed, which proves to be very costly and it is hazardous to environment as well.

6.2.5.2Nitriding

In this process, nitrogen is diffused into the surface of the steel being treated. The reaction of nitrogen with the steel causes the formation of very hard iron and alloy nitrogen compounds. The resulting nitride case is harder than tool steels or carburized steels. The advantage of this process is that hardness is achieved without the oil, water or air quench. As an added advantage, hardening is accomplished in a nitrogen atmosphere that prevents scaling and discoloration. Nitriding temperature is below the lower critical temperature of the steel and it is set between 925°F and 1050°F. The nitrogen source is usually Ammonia (NH_3).

At nitriding temperature the ammonia dissociates into Nitrogen and Hydrogen according to the chemical reaction: $2NH_3 \rightarrow 2N + 3H_2$

The nitrogen diffuses into the steel and hydrogen is exhausted. A typical nitriding setup is illustrated in figure 6.24.

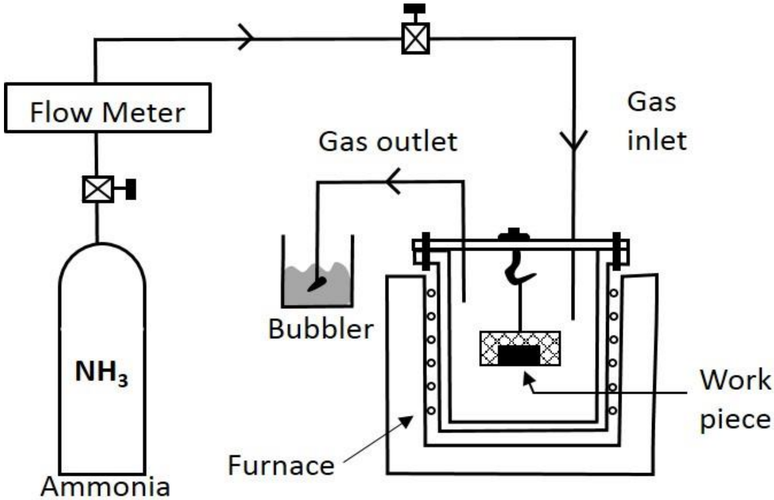


Figure 6.24: Nitriding

The white layer shown in figure alongside has a detrimental effect on the fatigue life of nitrided parts, and it is normally removed from parts subjected to severe service. Two stage gas-nitriding processes can be used to prevent the formation of white layer. White layer thickness may vary between 0.0003 and 0.002 in. which depends on nitriding time. The most commonly nitrided steels are chromium-molybdenum alloy steels and nitralloys. Nitrided steels are very hard and grinding operations should not be performed after nitriding. White layer is removed by lapping.

Figure 6.25: Nitrided parts

6.2.5.3 Carbonitriding

This process involves the diffusion of both carbon and nitrogen into the steel surface. The process is performed in a gas atmosphere furnace using a carburizing gas such as propane or methane mixed with several percent (by volume) of ammonia.

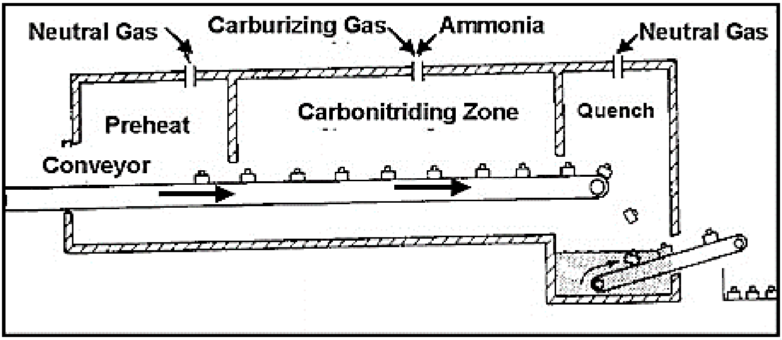


Figure 6.26: Carbonitriding

Methane or propane serve as the source of carbon, the ammonia serves as the source of nitrogen. Quenching is done in a gas which is not as severe as water quench. As a result of les severe quench, there is less distortion on the material to be treated. A typical Carbonitriding system is shown in the following slide. Case hardness of HRC 60 to 65 are achieved at the surface but not as high as nitrided surfaces. Case depths of 0.003 to 0.030 in can be accomplished by carbonitriding. One of the advantages of this process is that it can be applied to plain carbon steels which give significant case depths. Carbonitriding gives less distortion than carburizing. Carbonitriding is performed at temperatures above the transformation temperature of the steels (1400^{o}F to 1600^{o}F).

6.2.5.4 Cyaniding

It is similar to carbonitriding, and involves the diffusion of both carbon and nitrogen into the surface of the steel. The source of the diffusing element in this method is a molten cyanide salt such as sodium cyanide. It is a supercritical treatment involving temperatures in the range of 1400^{o}F to 1600^{o}F. Case depths are between 0.010 in. and 0.030 in.

During cyaniding, the cyanide salts are oxidized with the liberation of atomic carbon and nitrogen, which diffuse into the steel. In medium-temperature cyaniding, the cyanide layer formed, containing 0.6–0.7 percent C and 0.8–1.2 percent N, has a thickness of 0.15 to 0.6 mm, while in high-temperature cyaniding (a method often used instead of carburizing), the cyanide layer, containing 0.8–1.2 percent C and 0.2–0.3 percent N, has a thickness of 0.5 to 2 mm.

Diffusion times are less than one hour. Water or oil quench is required to remove residual cyanide. This type of cases present a significant distortion. Advantage of this method is the short time it requires to complete the diffusion, otherwise it should be avoided because of high distortion. The major drawback of cyaniding is that cyanide salts are poisonous and the process itself is costly.

6.2.6 Major defects due to faulty heat treatment

Following are some of the major defects found in metal or alloy due to faulty heat treatment:

1. Overheating: Prolonged heating of a metal or alloy above the A_3 line (refer figure 6.1) leads to the formation of very large actual grains. It is called overheating. On cooling a metal, this yields a structure containing coarse crystalline martensite. Such structure has reduced ductility and toughness. It is possible to retrieve an overheated metal or alloy by usual annealing. For considerable overheating, double annealing may be used.

2. Burning: Heating a metal or an alloy to still higher temperatures near melting point for a longer time leads to burning. This results in the formation of iron oxide inclusions along the grain boundaries. Burnt metal or alloy has a stony fracture and such metal or alloy is irretrievable and is rejected.

3. Oxidation: Sometimes a metal or alloy is oxidized due to oxidizing atmosphere in the furnace. It is characterized by a thick layer of scale on the surface of a metal or alloy. It can be prevented by using controlled atmosphere in the furnace or using molten salt baths.

4. Decarburization: It is the loss of carbon in the surface layers of the metal or alloy. Decarburization results in lower hardness and lower fatigue life. It is caused by the oxidizing furnace atmosphere. In order to prevent decarburization, the metal or alloy should be heated in a neutral or reducing atmosphere or in boxes with cast iron chips or in molten salt baths.

5. Cracks: The cracks occur in quenching when the internal tensile stresses exceed the resistance of the metal or alloy to separation. The tendency of a metal or alloy to crack formation increases with carbon content, hardening temperature and cooling rate in the temperature interval of martensite transformation. It also increases with the hardenability of metal or alloy. Another reason of crack formation is the concentration of local stresses.

To prevent the formation of cracks, it is advisable to (i) avoid stress concentrations such as sharp corners on projections, acute angles, and abrupt changes from thicker to thinner cross-sections etc., (ii) conduct quenching from lowest possible temperature, (iii) cool the metal or alloy slowly in the martensite interval of temperatures by quenching in two media and stepped quenching (i.e. martempering), (iv) apply isothermal quenching.

6. Distortion and warping: Distortion or deformation consisting in changes in the size and shape of heat-treated work, is due to thermal and structural stresses. Asymmetrical distortion of work is often called warping in heat-treating practice. It is usually observed in the case of non-uniform heating or over-heating for hardening, when the work is quenched in the wrong position and when the cooling rate is too high in the temperature interval of martensite transformation. An elimination of these causes should substantially reduce warping.

REVIEW QUESTIONS

1. Explain the phases and transformation reactions in Iron-iron carbide diagram with figure.

2. Classify steels and cast irons. What are the applications and limitations of Iron-iron carbide diagram?
3. What is heat treatment? What are the objectives of heat treatment? List out different types of heat treatment performed in steels.
4. What is annealing? Why it is done? Explain different types of annealing.
5. What is normalizing? What are the applications of normalizing? Compare between normalizing and annealing.
6. What is quenching? Briefly introduce the various mediums in which quenching can be done?
7. Differentiate between hardness and hardenability. Explain Jominy end quench test.
8. Explain TTT diagram with different cooling curves.
9. Briefly introduce CCT diagrams.
10. What is tempering? Explain different types of tempering methods.
11. Why is surface hardening performed? List the different types of surface hardening process.
12. What is carburizing? Explain three types of carburizing process.
13. Explain and differentiate between nitriding, carbonitriding and cyaniding process.
14. What are the major defects due to faulty heat treatment?

CHAPTER 7

TYPES OF STEELS AND CAST IRON

7.1 Alloy Steel

Steel in itself is an alloy of iron with mainly carbon along with elements like chromium, manganese, nickel, silicon, vanadium, molybdenum, etc. Alloy steel refers to the alloying of steel with these different elements in a significant amount. Alloying of steel is done in order to acquire desired characteristics.

For example: Alloying of steel with sulphur increases the machinability of steel however it reduces weldability of steel. Likewise alloying of other elements with steel can improve hardenability, provide solid solution strengthening of ferrite, improve corrosion resistance etc.

Types of Alloy Steels

- HSLA Steel
- Stainless Steel
- Tool Steel

7.1.1 High Strength Low Alloy Steels (HSLA)

High Strength Low-Alloy steels or micro-alloyed steels are designed to provide better mechanical properties, better resistance to environmental corrosion than conventional carbon steels. They are designed to meet the required mechanical properties than acquire a certain chemical composition. Hence, the composition of the alloy depends on the mechanical property to be acquired. HSLA in plate or sheet form have low carbon content (0.05-0.25%) to increase the formability and weldability of steel with manganese content up to 2%. Chromium, nickel, copper, nitrogen, vanadium, molybdenum, niobium, titanium are also added in small amounts for various combinations.

HSLA steels are classified as rolled mild carbon steels enhanced with mechanical properties obtained by the addition of small alloys. HSLA steels are mainly used in pipelines for oils and gases, heavy-duty highway and off road vehicles, storage tanks, power transmission towers, light poles, etc. The use of HSLA steels is determined on the basis of its purpose but the most important thing to consider is its favourable strength to weight ratio.

7.1.2 Stainless Steels

All true stainless steels contain a minimum of 11% Chromium which makes them excellent to corrosion. The secret lies in the protective surface layer of chromium oxide that forms when the steel is exposed to oxygen. Chromium causes the austenitic region to shrink and hence the ferrite region increases in size making it a ferrite stabilising agent.

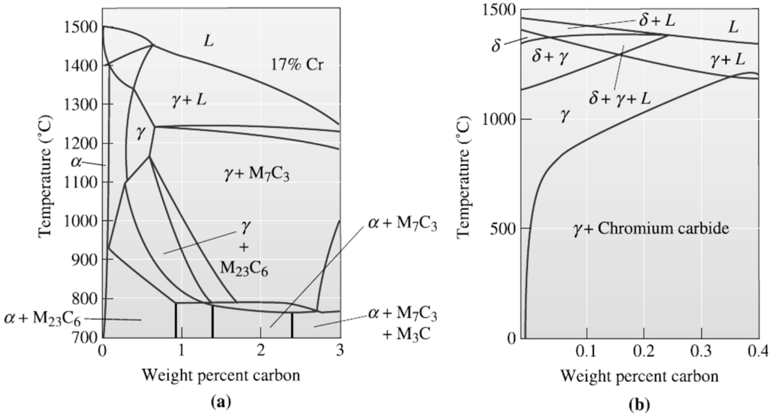


Figure 7.1: (a) The effect of 17% chromium on the iron-carbon phase diagram. At low carbon contents, ferrite is stable at all temperatures. Note that "M" stands for metal such as Cr and Fe or other alloying additions. (b) A section of the iron-chromium-nickel-carbon phase diagram at a constant 18% Cr - 8% Ni. At low carbon contents, austenite is stable at room temperature.

Types of Stainless Steels

1. Ferritic Stainless Steels

These contain 12% to 25% Cr and less than 0.1% carbon. Because of B.C.C structure, the ferritic stainless steels have good strengths and moderate ductility derived from solid solution strengthening and strain hardening. Ferritic stainless steels are magnetic. They are not heat treatable. They have excellent corrosion resistance, moderate formability and are relatively less expensive.

2. Martensitic Stainless Steels

These contain 12% to 25% Cr and 0.1% to 1.0%C. A 17%Cr-0.5%C alloy heated to 1200°C forms 100% austenite, which transforms to martensite on quenching in oil. The martensite is then tempered to produce high strengths and hardness. The combination of hardness, strength and corrosion resistance makes the alloy attractive for applications such as high quality knives, ball bearings and valves.

3. Austenitic Stainless Steels

Nickel, which is an austenite stabilizing element, increases the size of austenite field, while nearly eliminating ferrite from the iron-chromium-carbon alloys. If the carbon content is below, about 0.3%, the carbides do not form and the steel is virtually all austenite at room temperature. The F.C.C austenite stainless steels have excellent ductility, formability and corrosion resistance. These are non-magnetic, which is an advantage for many applications. Unfortunately, the high nickel and chromium contents make the alloy expensive. The 304 alloy containing 18% Cr and 8% Ni is most widely used grade of stainless steel.

Steel	% C	% Cr	% Ni	Others	Tensile Strength (psi)	Yield Strength (psi)	% Elongation	Condition
Austenitic								
201	0.15	17	5	6.5% Mn	95,000	45,000	40	Annealed
304	0.08	19	10		75,000	30,000	30	Annealed
					185,000	140,000	9	Cold-worked
304L	0.03	19	10		75,000	30,000	30	Annealed
316	0.08	17	12	2.5% Mo	75,000	30,000	30	Annealed
321	0.08	18	10	0.4% Ti	85,000	35,000	55	Annealed
347	0.08	18	11	0.8% Nb	90,000	35,000	50	Annealed
Ferritic								
430	0.12	17			65,000	30,000	22	Annealed
442	0.12	20			75,000	40,000	20	Annealed
Martensitic								
416	0.15	13		0.6% Mo	180,000	140,000	18	Quenched and tempered
431	0.20	16	2		200,000	150,000	16	Quenched and tempered
440C	1.10	17		0.7% Mo	285,000	275,000	2	Quenched and tempered
Precipitation hardening								
17-4	0.07	17	4	0.4% Nb	190,000	170,000	10	Age-hardened
17-7	0.09	17	7	1.0% Al	240,000	230,000	6	Age-hardened

Figure 7.2: Typical compositions and properties of stainless steels

7.1.3 Tool Steels

Tools steels are high carbon steels that obtain high hardness by quench and temper heat treatment. Hardenability and stability at high temperatures are the results of alloying elements. Water hardenable steels must be quenched rapidly to produce martensite and also soften rapidly even at low temperatures. Oil hardenable steels form martensite more easily, temper slowly but still soften at high temperatures. The alloy carbides are particularly stable, resist growth for spheroidisation and are important in establishing the high temperature resistance of steels. Tool steels are mainly used in making cutting tools for machining, die casting as high strength, hardness, toughness and temperature resistance is needed off them. With a carbon content between 0.7% and 1.5%, tool steels are manufactured under carefully controlled conditions to produce the required quality. The manganese content is often kept low to minimize the possibility of cracking.

7.1.4 Weldability of steels

Weldability of steels refers to the ability of steels to be joined by welding. As steels are used as important structural components along with the strength of steels, joints must also be strong for the integrity of structures. Many low carbon steels weld easily. Welding of medium and high carbon steel is comparatively more difficult since martensite can form in the heat affected zone rather easily, thereby causing a weld with poor toughness. Several strategies such as preheating the material or minimizing incorporation of hydrogen have been developed to counter these problems. In low carbon steels the strength of the welded zones is higher than the base material itself. This occurs as a result of formation of pearlite microstructure during cooling. Retained austenite along the ferrite grain boundaries prevents recrystallization and thus helps retain a finer grain size which contributes to the strength of the welded joints. During welding, metal near the weld starts to heat up above the A_1temperature (refer fig. 6.1) and austenite forms.

During cooling, austenite transforms to a new structure depending on the cooling rate.

7.1.5 Embrittlement of steels

Embrittlement is the complete loss or partial loss of ductility in material where an embrittled product characteristically fails by fracture without appreciable deformation. The most common embrittlement is usually encountered in galvanized steel related to aging phenomena, cold working, and absorption of hydrogen. Embrittlement or loss of ductility in steel is often associated with strain-aging. Strain-aging refers to the delayed increase in hardness and strength, and loss of ductility and impact resistance which occur in susceptible steels as a result of the strains induced by cold working. The aging changes proceed slowly at room temperature, but proceed at an accelerated rate as the aging temperature is raised and may occur rapidly at the galvanizing temperature of approximately 455°C (850°F).

Hydrogen Embrittlement may also occur due to the possibility of atomic hydrogen being absorbed by the steel. The susceptibility to hydrogen Embrittlement is influenced by the type of steel, its previous heat treatment, and degree of previous cold work. In the case of galvanized steel, the acid pickling reaction prior to galvanizing presents a potential source of hydrogen. However, the heat of the galvanizing bath partially expels hydrogen that may have been absorbed. In practice hydrogen Embrittlement of galvanized steel is usually of concern only if the steel exceeds approximately 1100 MPa (150 ksi) in ultimate tensile strength, or if it has been severely cold worked prior to pickling. Loss of ductility of cold-worked steels is dependent on many factors including the type of steel (strength level, aging characteristics), thickness of steel, and degree of cold work, and is accentuated by areas of stress concentration such as caused by notches, holes, fillets of small radii, sharp bends, etc.

7.2 Cast iron

Cast iron is essentially an iron-carbon alloy modified by the presence of silicon, phosphorus, sulphur, and manganese in varying amounts. The cast iron contains a greater proportion of carbon than is found in steel. Cast iron may have 2.5 percent to 5.5 percent carbon content. The carbon present in its natural state as graphite is normally finely dispersed throughout the metal in the form of small flakes. Whilst the cast iron can be cast, it can be forged. Cast iron is brittle and may easily be broken if a heavy hammering is done upon it. Small thin casting should not be dropped because of the risk of being cracked or fractured.

Cast iron is used in industries for casting different machine parts because of its low cost, good casting characteristics, high compressive strength, wear resistance and good damping qualities. A further good property of cast iron is that the free graphite in its structure seems to act as a lubricant. There are many applications of cast iron in engineering and machine tools, bodies of electrical machine, cylinders and beds of engine etc.

The cast iron may be classified as:

1. Gray cast iron
2. Ductile cast iron
3. White cast iron
4. Malleable cast iron
5. Alloy cast iron

7.2.1 Gray cast iron

When the carbon is not chemically combined with iron, it is known as free carbon or graphite carbon and the resulting cast iron is known as gray cast iron. Two conditions –a slow rate of cooling and the presence of silicon will cause the carbon to change from the combined form of the graphitic form to produce gray cast iron.

The gray cast iron is readily cast into a desired shape in a sand mould. It contains 2.3-3.8%C, 1.2-2.8% Si, 0.4-1.0% Mn, 0.15% and 0.1% S. It is marked by in presence of flakes of graphite in a matrix of ferrite, pearlite or austenite as shown in figure 7.3.

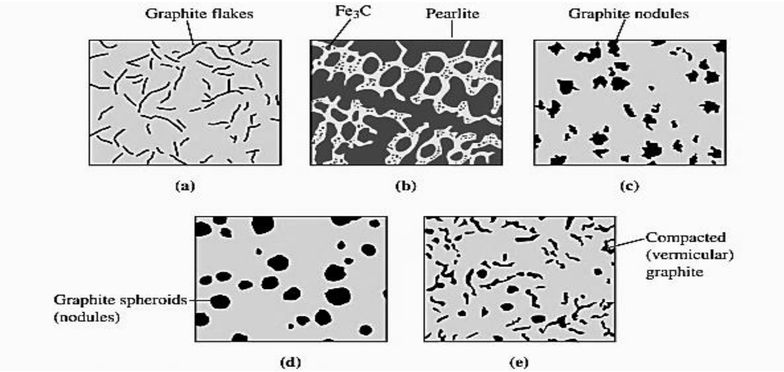


Figure 7.3: Cast iron: (a) gray iron, (b) white iron, (c) malleable iron, (d) ductile iron and (e) compacted graphite iron

The graphite flakes occupy about 10 percent of the metal volume. Length of flakes may vary from 0.05 to 0.1 mm. Gray cast iron, when fractured, gives gray appearance. It possesses lowest melting point of the ferrous alloys. It has high resistance to wear, high vibration damping capacity and high compressive strength.

Uses:

i. Automatic parts such as cylinder block and cylinder heads
ii. Machine cool structure e.g. bed, frame etc.
iii. Manhole covers
iv. Frames of electric motor
v. Sanitary wear
vi. Gas or water pipes
vii. Household appliances etc.

7.2.2 Ductile (nodular) cast iron

In nodular cast iron, graphite appears as rounded particles or nodules or spheroids form. The spheroidising elements when added to melt eliminate sulphur and oxygen (from the melt), which changes solidification characteristics and possibly account for the nodulisation. Ductile cast iron possesses very good machinability. The properties of nodular cast iron depend upon the metal composition and the cooling rate. Nodular or ductile cast iron contains 3.2-4.2% C, 1.1- 3.5% Si, 0.3-0.8% Mn, 0.08 % P and 0.2% S. it possesses excellent castability, wear resistance and damping capacity, high strength, high toughness, higher machinability than gray cast iron.

Uses:

i. Power transmission equipments.
ii. Farm implements and tractors.
iii. Earth moving machinery.
iv. Paper industries machinery.
v. Steel mill rolls and mill equipments.
vi. Pump and compressors.
vii. Valves and fittings.

7.2.3 White cast iron

When iron and carbon are chemically combined in the form of Cementite, the iron is known as the white cast iron. The presence of Cementite or iron carbon Fe_3C either free or lamellar pearlite makes the metal hard and brittle and the absence of graphite gives the fracture, of white color as shown in figure 7.3.

Quick cooling of iron from its moulded state does not allow so much graphite to form and a large proportion of the carbon is in combined form. The fracture of quickly cooled iron is white showing that there is less free carbon and when all or nearly all, carbon is combined, the metal is called white cast iron.

The white iron contains 1.8-3.6% C, 0.5- 2.0% Si, 0.2-0.8% Mn, 0.18 % P and 0.10% S. White cast iron under normal circumstances is brittle and not machinable but also has better high temperature strength, grain growth resistance and corrosion resistance.

Uses:

i. For producing malleable cast iron.
ii. For manufacturing those component parts which required a hard and abrasion resistance material.

7.2.4 Malleable cast iron

Malleable cast iron is an alloy in which all combined carbon is a special white cast iron has been changed to free or temper carbon by suitable heat treatment. Malleable cast iron is one which can be hammered and rolled to obtained different shapes. Fig 7.3 showing malleable cast iron having ferrite matrix and temper carbon (dark particle). The malleable cast iron is obtained from hard and brittle white iron through a controlled heat conversion process. An alloy malleable cast iron contains chromium and nickel and possesses high strength and corrosion resistance. It possesses good wear resistance, vibration damping capacity and high yield strength. Malleable cast iron contain 2-3%C, 0.6-1.3% Si, 0.2-0.6% Mn, 0.15 % P and 0.10% S.

Uses:

i. Rail road
ii. Agriculture implements
iii. Conveyor chain links
iv. Gear case
v. Rear axle housing
vi. Truck axle assembly parts
vii. Automatic crank shaft

7.2.5 Alloy cast iron

The main alloying elements used for producing alloy cast iron are nickel, chromium and molybdenum. Because the cast iron is supposed to be very hard, brittle lacking in tensile strength, it is alloyed with other metal to improve its properties. Two recent example of alloy case iron are acicular and spheroidal cast iron. Acicular cast iron has nickel and molybdenum in its composition and is being used for cast crankshaft. In spheroidal cast iron, the graphite content is converted from a flaky to a spheroidal form by the alloying of a small amount of magnesium and cerium. This change in the graphite from increases in the tensile strength producing the metal tough, which can be twisted and bent.

7.2.6 Applications of cast iron

- Cookware, such as skillets, pots and pans, is one of the common uses of cast iron. Once the cookware is cured, it seals in the seasoning of the foods cooked in them and even adds iron to the diet.
- Cooking utensils for stove-top and outdoor cooking are also constructed of cast iron. Many of these cast-iron cooking utensils include spoons, grill presses and tongs.
- Designers use cast iron to build furniture, particularly outdoor patio pieces. Cast iron furniture includes chairs and glass-top tables.
- Welded barbecue grills use heavy cast-iron materials and stainless steel. Larger models of these grills are often so heavy that two people have to carry them.

- Industrial and residential construction used cast iron on buildings, especially for beams and posts that hold up architectural structures. Cast iron is also used in pipe plumbing.

REVIEW QUESTIONS

1. What is alloy steels? List the different types of alloy steels. Briefly introduce High Strength Low Alloy (HSLA) steels with its properties and applications.
2. What is stainless steels? Differentiate different types of stainless steels on the basis of composition, properties and applications.
3. Write down the composition, properties and applications of tool steels.
4. What is weldability and embrittlement of steels?
5. What is cast iron? List out the different types of cast iron and applications of cast iron.
6. Differentiate different types of cast iron on the basis of micro-structure, composition, properties and applications.

CHAPTER 8

ENVIRONMENTAL EFFECTS

8.1 Corrosion

It is the gradual destruction of material, usually metals, by chemical reaction with its environment. Corrosion can also occur in materials other than metals, such as ceramics or polymers, although in this context, the term degradation is more common. Corrosion degrades the useful properties of materials and structures including strength, appearance and permeability to liquids and gases.

Many structural alloys corrode merely from exposure to moisture in air, but the process can be strongly affected by exposure to certain substances. Because corrosion is a diffusion-controlled process, it occurs on exposed surfaces. As a result, methods to reduce the activity of the exposed surface, such as passivation and chromate conversion, can increase a material's corrosion resistance.

Types of corrosion

1. Galvanic corrosion
2. Stress corrosion

8.1.1 Galvanic corrosion

Galvanic corrosion occurs when two different metals have physical or electrical contact with each other and are immersed in a common electrolyte, or when the same metal is exposed to electrolyte with different concentrations. In a galvanic couple, the more active metal (the anode) corrodes at an accelerated rate and the more noble metal (the cathode) corrodes at a retarded rate. When immersed separately, each metal corrodes at its own rate. The type of metal(s) to be used is readily determined by the galvanic series.

For example, zinc is often used as a sacrificial anode for steel structures. Galvanic corrosion is of major interest to the marine industry and also anywhere water (containing salts) contacts pipes or metal structures.

Factors such as relative size of anode, types of metal, and operating conditions (temperature, humidity, salinity, etc.) affect galvanic corrosion. The surface area ratio of the anode and cathode directly affects the corrosion rates of the materials. Galvanic corrosion is often utilized in sacrificial anodes.

8.1.2 Stress corrosion

Stress-corrosion cracking refers to cracking caused by the simultaneous presence of tensile stress and a specific corrosive medium. Many investigators have classified all cracking failures occurring in corrosive mediums as stress-corrosion cracking, including failures due to hydrogen embrittlement.

During stress-corrosion cracking, the metal or alloy is virtually unattacked over most of its surface, while fine cracks progress through it. This cracking phenomenon has serious consequences since it can occur at stresses within the range of typical design stress. Exposure to boiling $MgCl_2$ at 310°F (154°C) is shown to reduce the strength capability to approximately that available at 1200°F.

The two classic cases of stress-corrosion cracking are "season cracking" of brass, and the "caustic embrittlement" of steel. Both of these obsolete terms describe the environmental conditions present which led to stress-corrosion cracking. Season cracking refers to the stress-corrosion cracking failure of brass cartridge cases. During periods of heavy rainfall, especially in the tropics, cracks were observed in the brass cartridge cases at the point where the case was crimped to the bullet. It was later found that the important environmental component in season cracking was ammonia resulting from the decomposition of organic matter.

Many explosions of riveted boilers occurred in early steam-driven locomotives. Examination of these failures showed cracks or brittle failures at the rivet holes. These areas were cold-worked during riveting operations, and analysis of the whitish deposits found in these areas showed caustic, or sodium hydroxide, to be the major component. Hence, brittle fracture in the presence of caustic resulted in the term caustic embrittlement. While stress alone will react in ways well known in mechanical metallurgy (i.e., creep, fatigue, tensile failure) and corrosion alone will react to produce characteristic dissolution reactions; the simultaneous action of both sometimes produces the disastrous results.

8.2 Corrosion protection

Applied coatings

Plating, painting, and the application of enamel are the most common anti-corrosion treatments. They work by providing a barrier of corrosion-resistant material between the damaging environment and the structural material. Platings usually fail only in small sections, and if the plating is more noble than the substrate (for example, chromium on steel), a galvanic couple will cause any exposed area to corrode much more rapidly than an unplated surface would. For this reason, it is often wise to plate with active metal such as zinc or cadmium. Painting either by roller or brush is more desirable for tight spaces; spray would be better for larger coating areas such as steel decks and waterfront applications.

Reactive coatings

If the environment is controlled (especially in re-circulating systems), corrosion inhibitors can often be added to it. These form an electrically insulating or chemically impermeable coating on exposed metal surfaces, to suppress electrochemical reactions. Such methods obviously make the system less sensitive to scratches or defects in the coating, since extra inhibitors can be made available wherever metal becomes exposed.

Bio-film coatings

A new form of protection has been developed by applying certain species of bacterial films to the surface of metals in highly corrosive environments. This process increases the corrosion resistance substantially. Alternatively, antimicrobial-producing biofilms can be used to inhibit mild steel corrosion from sulfate-reducing bacteria.

Controlled permeability formwork

Controlled permeability formwork (CPF) is a method of preventing the corrosion of reinforcement by naturally enhancing the durability of the cover during concrete placement. CPF has been used in environments to combat the effects of carbonation, chlorides, frost and abrasion.

Cathodic protection

Cathodic protection (CP) is a technique to control the corrosion of a metal surface by making that surface the cathode of an electrochemical cell. Cathodic protection systems are most commonly used to protect steel, water, and fuel pipelines and tanks; steel pier piles, ships, and offshore oil platforms.

Sacrificial anode protection

Sacrificial anode in the hull of a ship. For effective CP, the potential of the steel surface is polarized (pushed) more negative until the metal surface has a uniform potential. With a uniform potential, the driving force for the corrosion reaction is halted. For galvanic CP systems, the anode material corrodes under the influence of the steel, and eventually it must be replaced. The polarization is caused by the current flow from the anode to the cathode, driven by the difference in electrochemical potential between the anode and the cathode.

Impressed current cathodic protection

For larger structures, galvanic anodes cannot economically deliver enough current to provide complete protection. Impressed current cathodic protection (ICCP) systems use anodes connected to a DC power source (such as a cathodic protection rectifier). Anodes for ICCP systems are tubular and solid rod shapes of various specialized materials. These include high silicon cast iron, graphite, mixed metal oxide or platinum coated titanium or niobium coated rod and wires.

Anodic protection

Anodic protection impresses anodic current on the structure to be protected (opposite to the cathodic protection). It is appropriate for metals that exhibit passivity (e.g., stainless steel) and suitably small passive current over a wide range of potentials. It is used in aggressive environments, e.g., solutions of sulfuric acid.

REVIEW QUESTIONS

1. What is corrosion? Explain galvanic and stress corrosion.
2. List the different methods of corrosion protection

CHAPTER 9
NON-FERROUS ALLOYS

9.1 Aluminium alloys

Aluminium alloys are alloys in which aluminium (Al) is the predominant metal. The typical alloying elements are copper, magnesium, manganese, silicon and zinc. There are two principal classifications, namely casting alloys and wrought alloys, both of which are further subdivided into the categories heat-treatable and non-heat-treatable.

Aluminium alloys can be classified into wrought and casting alloys on the basis of method of fabrication. Wrought alloys are formed by plastic deformation while casting alloys are formed by using casting process. Wrought alloys can be classified as heat treatable and non-heat treatable alloys. Non-heat treatable alloys include aluminium alloys with magnesium, manganese and silicon. While heat treatable wrought alloys are aluminium alloys with Copper, Lithium, Silicon and Zinc

Properties

- Aluminium and its alloys are easily formable and can be worked upon by rolling, extruding, drawing, machining and other mechanical processes.
- They have high thermal and electrical conductivity and do not show ductile to brittle transition at low temperatures.
- Cast aluminium alloys have low melting point, although they generally have lower tensile strengths than wrought alloys.
- Aluminium and its alloys lend themselves to a huge variety of surface treatments, which enhances its intrinsic qualities. For example, an anodisation of a few micrometres is enough to preserve

the optical or decorative properties of the materials, while improving resistance, especially to corrosion and stress.

- The tensile strength of pure aluminum is around 90 MPa but this can be increased to over 690 MPa for some heat-treatable aluminum alloys.

Uses

- Non-heat treatable aluminum alloys are used for making electrical components, foil, food packaging, beverage can bodies, architectural uses, filler metal welding, beverage can tops and marine components while heat treatable aluminum alloys are used for truck wheels, aircraft skins, pistons, canoes, railroad cars and aircraft frames.
- Casting alloys contain high amount of silicon to cause the eutectic reaction giving the alloys low melting points, good fluidity and good castability
- Aluminium alloys are widely used in engineering structures and components where light weight or corrosion resistance is required.
- Advanced aluminum alloys have enabled the design of high-speed ships, by lightening hulls by 40% to 50% over steel.

9.2 Magnesium alloys

Magnesium alloys are mixtures of magnesium with other metals, often aluminium, zinc, manganese, silicon, copper, rare earths metals and zirconium. Magnesium alloys have a hexagonal lattice structure, which affects the fundamental properties of these alloys. Plastic deformation of the hexagonal lattice is more complicated than in cubic latticed metals like aluminium, copper and steel. Therefore magnesium alloys are typically used as cast alloys.

Properties

- Magnesium is the lightest structural metal and its alloys containing Lithium have extremely light weight.

- Magnesium has low modulus of elasticity and poor resistance to fatigue, creep and wear.
- Magnesium alloys can be strengthened either by dispersion strengthening or age hardening.
- Some age hardened alloys such as those containing Zirconium, Thorium, Silver, and Cerium have good resistance to over aging temperatures.
- Magnesium alloys with less impurities and high Cerium content form a protective layer of MgO making it corrosion resistant.
- The machinability of magnesium alloys is the best of any commercial metal, and in many applications, the savings in machining costs more than compensate for the increased cost of the material.
- Magnesium alloys are easily machined, cast, forged, and welded.

Uses

- Magnesium alloys are used for aerospace applications, high speed machinery and materials handling equipment.
- Cast magnesium alloys are used for many components of modern cars, and magnesium block engines have been used in some high-performance vehicles.
- *Magnox* (alloy), whose name is an abbreviation for 'magnesium non-oxidising', is 99% magnesium and 1% aluminium, and used in the cladding of fuel rods in some nuclear power stations.
- They are now also used in jet-engine parts, rockets and missiles, luggage frames, portable power tools, and cameras and optical instruments.
- They are also used in pyrotechnics, especially in incendiary bombs, signals, and flares, and as a fuse for thermite.

9.3 Copper alloys

Copper alloys have higher densities than that of steels. Copper based alloys benefit by strengthening mechanisms as they improve the mechanical properties of copper alloys. Copper containing 0.1% impurities is used for electrical and microelectronics applications. Small amounts of cadmium, silver and Aluminium oxide improve their hardness without impairing conductivity. Cold working is used for strengthening of single phase copper based alloys.

Properties

- Even though their yield strength is higher than that of aluminium and magnesium alloys, they have low specific strengths.
- They are better resistant to creep, fatigue, and wear.
- Most of high ductility, corrosion resistance, electrical and thermal conductivity and can be joined and fabricated into complex shapes easily.
- The face centred cubic structure of copper provides excellent ductility and high strain hardening coefficient.
- Copper alloy corrode easily by reacting with sulphur and forms a green layer on the surface.

Uses

- Copper based alloys are used in pumps, valves and plumbing parts where their properties are used as advantages.
- Copper based alloys are used as decorative items because of the changing colours as it reacts with different elements.
- Copper and its alloys are widely used in deep draw and flat stamped products because they have excellent electrical and thermal performance, good resistance to corrosion, high ductility and relatively low cost.
- Integrated circuits and printed circuit boards use copper and its alloys in place of aluminum because of its superior electrical conductivity.

Types of copper alloys

1. Brass

Brass is an alloy made of copper and zinc; the proportions of zinc and copper can be varied to create a range of brasses with varying properties. Brass is a substitutional copper alloy. The relatively low melting point of brass (900 to 940 °C, depending on composition) and its flow characteristics make it a relatively easy material to cast. Different grades of brass, with different amounts of zinc, have different characteristics in surface smoothness, corrosion-resistance, ease and quality of machining, response to plating, ductility levels, chemical and temperature limits.

Uses: It is used for decoration for its bright gold-like appearance; for applications where low friction is required such as locks, gears, bearings, doorknobs, ammunition casings and valves; for plumbing and electrical applications; and extensively in brass musical instruments such as horns and bells for its acoustic properties. It is also used in zippers. Brass is often used in situations where it is important that sparks not be struck, as in fittings and tools around explosive gases.

2. Bronze

Bronze is principally an alloy of copper and tin. Bronze does not necessarily contain tin, and a variety of alloys of copper, including alloys with arsenic phosphorus, aluminum, manganese, and silicon, are commonly termed "bronze". Bronze is generally harder than wrought iron, with Vickers hardness of 60–258.Bronze is softer, weaker and less stiff than steel. Bronze resists corrosion (especially seawater corrosion) and metal fatigue more than steel and is a better conductor of heat and electricity than most steels.

Uses: Bronze was especially suitable for use in boat and ship fittings prior to the wide employment of stainless steel owing to its combination of toughness and resistance to salt water corrosion. Bronze is still commonly used in ship propellers and submerged bearings. It is also widely used for cast bronze sculpture. Many common bronze alloys have the unusual and very desirable property of expanding slightly just before they set, thus filling in the finest details of a mould. Bronze parts are tough and typically used for bearings, clips, electrical connectors and springs. Bronze is also used to make bronze wool for woodworking applications where steel wool would discolor oak.

9.4 Nickel and cobalt alloys

Nickel and cobalt have high strengths and high melting points and thus are used for corrosion protection and high temperature resistance. Solidified powders of nickel and cobalt based super alloys can be formed by using spray atomization followed by hot isostatic pressing which are then used for making rings that retain the turbine blades. Since Iron, Nickel and Cobalt are magnetic, they form very good magnetic materials.

Solidified powders of nickel and cobalt based super alloys can be formed by using spray atomization followed by hot isostatic pressing which are then used for making rings that retain the turbine blades. Since Iron, Nickel and Cobalt are magnetic, they form very good magnetic materials. Because of their biocompatibility, cobalt alloys are used for prosthetic devices.

A Ni-Fe alloy *Invar* displays approximately no expansion during heating and are hence used in producing bimetallic composite materials. Cobalt is used for absorbing shocks and vibrations in cutting tools. Copper added nickel alloy called *Monel* is used for its strength and corrosion resistance in salt water and at elevated temperatures. Solid solution strengthening, dispersion strengthening and precipitation hardening are used to make super alloys which basically consist of Iron, Nickel and Cobalt. These strengthening measures produce high strength at elevated temperatures, resistance to corrosion and resistance to creep up to 1000°C. Thus made super alloys are used in making vanes and blades for turbines and jet engines, heat exchangers, chemical reaction vessel components and heat treating equipment.

Typical alloys and their applications are listed in the table below:

Material	Tensile Strength (psi)	Yield Strength (psi)	% Elongation	Strengthening Mechanism	Applications
Pure Ni (99.9% Ni)	50,000	16,000	45	Annealed	Corrosion resistance
	95,000	90,000	4	Cold worked	Corrosion resistance
Ni-Cu alloys:					
Monel 400 (Ni-31.5% Cu)	78,000	39,000	37	Annealed	Valves, pumps, heat exchangers
Monel K-500 (Ni-29.5% Cu-2.7% Al-0.6% Ti)	150,000	110,000	30	Aged	Shafts, springs, impellers
Ni superalloys:					
Inconel 600 (Ni-15.5% Cr-8% Fe)	90,000	29,000	49	Carbides	Heat-treatment equipment
Hastelloy B-2 (Ni-28% Mo)	130,000	60,000	61	Carbides	Corrosion resistance
DS-Ni (Ni-2% ThO_2)	71,000	48,000	14	Dispersion	Gas turbines
Fe-Ni superalloys:					
Incoloy 800 (Ni-46% Fe-21% Cr)	89,000	41,000	37	Carbides	Heat exchangers
Co superalloys:					
Stellite 6B (60% Co-30% Cr-4.5% W)	177,000	103,000	4	Carbides	Abrasive wear resistance

Figure 9.1: Compositions, properties and applications for selected Nickel and Cobalt alloys

9.5 Titanium alloys

Titanium alloys are metals which contain a mixture of titanium and other chemical elements. Such alloys have very high tensile strength and toughness (even at extreme temperatures). They are light in weight, have extraordinary corrosion resistance and the ability to withstand extreme temperatures. However, the high cost of both raw materials and processing limit their use to military applications, aircraft, spacecraft, medical devices, connecting rods on expensive sports cars and some premium sports equipment and consumer electronics.

Auto manufacturers *Porsché* and *Ferrari* also use titanium alloys in engine components due to its durable properties in these high stress engine environments.

Although "commercially pure" titanium has acceptable mechanical properties and has been used for orthopaedic and dental implants, for most applications titanium is alloyed with small amounts of aluminium and vanadium, typically 6% and 4% respectively, by weight. This mixture has a solid solubility which varies dramatically with temperature, allowing it to undergo precipitation strengthening. This heat treatment process is carried out after the alloy has been worked into its final shape but before it is put to use, allowing much easier fabrication of a high-strength product.

Titanium has excellent mechanical properties. An adherent TiO_2 layer on the surface provides excellent resistance to corrosion and contamination below 535°C but above this temperature the oxide film breaks down and small atoms such as carbon, oxygen, hydrogen and nitrogen make titanium brittle.

Excellent corrosion resistance makes titanium suitable for chemical processing equipment, marine components and biomedical implants. Titanium alloys are also used in making airframes and jet engine components. When combined with niobium, it forms a superconductive intermetallic compound (mixture of elements producing its own crystal structure and mechanical properties). Titanium alloys are used for sports equipment such as the head of the golf club.

9.6 Refractory metals

Refractory metals are metals with really high melting points normally above 2000°C. This category consists of tungsten, molybdenum, tantalum and niobium. These metals are therefore used for their high temperature service such as filaments in light bulbs, rocket nozzles, nuclear power generators, tantalum and niobium based electronic capacitors and chemical processing equipment. However, their high densities limit their specific strengths.

Refractory metals rapidly oxidize between 200 and 425°C and are therefore contaminated or embrittled. Hence, special precautions are required during casting, welding, hot working or powder metallurgy. For tungsten filaments, preserving in vacuum protects it from oxidization.

Some metals can be coated with a silicide or aluminium coating. These coating materials are selected on the basis of their high melting temperature, compatibility with refractive metals, ability to form a diffusion barrier so that contaminants can't reach the underlying metals, and their coefficient of thermal expansion which is required to be similar to that of refractory metals. As refractory metals have B.C.C crystal structure, they have low ductile to brittle transition temperature. For niobium and tantalum this temperature is below room temperature whereas for molybdenum and tungsten, it's above room temperature. Therefore, niobium and tantalum can be easily formed while molybdenum and tungsten are hot worked to produce a fibrous micro structure improving their forming characteristics. Further alloying of refractory metals produces improved mechanical properties along with improving high temperature properties.

REVIEW QUESTIONS

1. Write short notes on:
   a) Aluminium alloys
   b) Magnesium alloys
   c) Nickel and cobalt alloys
   d) Titanium alloys
   e) Refractory metals
2. What are copper alloys? Write the composition, properties and applications about different types of copper alloys.

CHAPTER 10

ORGANIC AND COMPOSITE MATERIALS

10.1 Polymers

Polymers are materials consisting of giant macromolecules which are built of many units joined by chemical bonding. The process by which polymers are formed is called polymerization. Polymers are used in many applications including clothing, toys, home appliances, structural and decorative items, paints, adhesives, automobile tyres, car bumpers, biomedical devices, interiors, foams and packaging. Polymers are also used to make electrical components as they possess insulating ability and low dielectric constant. Polymers can also be used as substitutes for glass when they are transparent. Because of their lightweight, corrosion resistant property and their inexpensiveness they are used in several applications. Engineering polymers are therefore designed to improve strength at elevated temperatures but are expensive to manufacture.

10.1.1 Types of polymers

Polymers can be classified into three groups and their comparison can be done as follows:

1) Thermoplastic
2) Thermosetting
3) Elastomers

Thermoplastic polymers have flexible linear chain whether straight or branched while thermosetting polymers have rigid three dimensional network. Elastomers on the other hand are a mixture of thermosetting and thermoplastic polymers with spring like molecules. Polyethylene, polyurethane and rubber are examples of thermoplastic, thermosetting and elastomer respectively.

Thermoplastics

Thermoplastics behave in a ductile manner and are composed of long chains produced by joining together of monomers. The individual chains may be branched or single. The Vander Waals forces between atoms of each chain is very weak. Therefore, these chains tangle up in order to form a material and can be untangled by the application of tensile stress. They soften and melt upon heating and can therefore be easily recycled.

Thermosetting Polymers

Thermosetting polymers are composed of long chains of molecules, single or branched that are cross linked to each other to form a three dimensional network. Thermosetting polymers are therefore stronger but brittle when compared to thermoplastics. Unlike thermoplastics they do not melt on heating, they decompose and the cross links break making it harder for them to be recycled.

Elastomers

Elastomers are basically rubbers which can be plastically deformed up to 200%. The polymer chains consist of coil like molecules which can reversibly stretch by applying a force. Thermoplastic polymers have the ease of processing of thermoplastics and elastic behaviours of elastomers.

10.1.2 Addition and condensation polymerisation

The polymer is the only product in addition polymerisation. An addition polymer is a polymer which is formed by an addition reaction, where many monomers bond together via rearrangement of bonds without the loss of any atom or molecule. It involves the opening out of a double bond. The conditions of the reaction can alter the properties of the polymer. Reaction proceeds by a free radical mechanism. Oxygen is often used as the initiator in many reactions. An example of addition reaction is given below: In this reaction vinyl chloride monomer is changed to polyvinyl chloride by addition reaction.

Condensation polymerisation uses monomers that have two functional groups per molecule. These are said to be di-functional. Polymerisation occurs when these monomers react 'head-to-tail' to form a new bond that will eventually join the monomers together. A small molecule (often water) is eliminated during the reaction.

An example of condensation polymerization is given below:

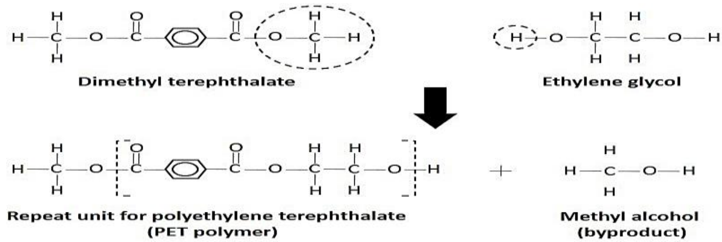


Figure 10.1: Condensation reaction for polyethylene terephthalate (PET)

10.1.3 Degree of polymerization

The degree of polymerization, or DP, is usually defined as the number of monomeric units in a macromolecule or polymer or oligomer molecule.

For a homo-polymer, there is only one type of monomeric unit and the number-average degree of polymerization is given by,

$$DP_n = \frac{average\ molecular\ weight\ of\ polymer}{molecular\ weight\ of\ repeating\ unit}$$

For most industrial purposes, degrees of polymerization in the thousands or tens of thousands are desired.

International Union of Pure and Applied Chemistry (IUPAC) defines degree of polymerization as "the number of monomeric units in a macromolecule an oligomer molecule, a block, or a chain."

10.1.3 Forming of polymers

Thermoplastics are melted from a solid and reshaped before cooling. Common forming methods include injection moulding, blow moulding and extrusion. Thermosets are not fully polymerised before forming. They are formed into the desired shape and then made into a solid through a chemical reaction to create cross-links in the polymer chains. Common forming methods include injection moulding, compression moulding and transfer moulding.

Injection moulding

Injection moulding is by far the most important method of forming plastic materials. Plastic pellets are fed into a rotating auger. The auger moves the pellets down a path towards the mould. As the pellets move they are pushed against the heated outer walls and are melted into a viscous liquid. At the end of the moulding machine the polymer liquid is injected into a fashioned mould and cooled until solid.

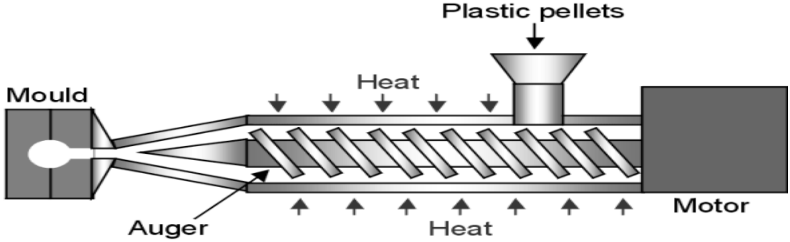


Figure 10.2: Injection moulding

Compression moulding Compression moulding is principally used for thermosetting plastics. Preheated resin is placed into a hot mould cavity. The upper section of the mould is subsequently forced down onto the resin to create the desired product shape. The applied pressure and heat forces the liquefied polymer to fill the cavity. Following the compression, a period of heating is required to force cross-linking of the thermosetting polymer.

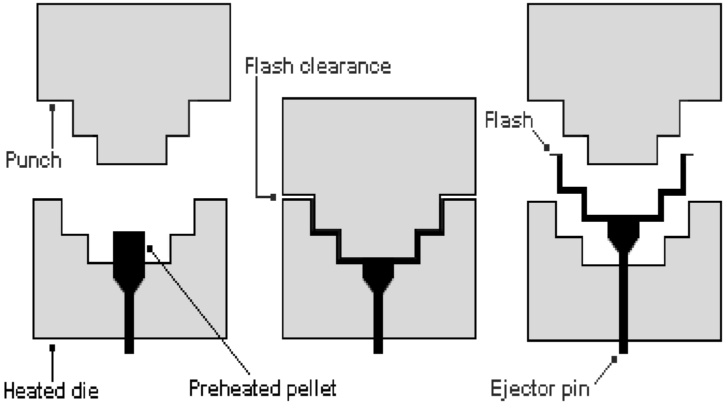


Figure 10.3: Compression moulding

Transfer moulding

Transfer Moulding is similar to compression moulding in that a carefully calculated, pre-measured amount of uncured moulding compound is used for the moulding process. The difference is, instead of loading the polymer into an open mould, the plastic material is pre-heated and loaded into a holding chamber called the pot. The material is then forced/transferred into the pre-heated mould cavity by a hydraulic plunger through a channel called sprue. The mould remains closed until the material inside is cured.

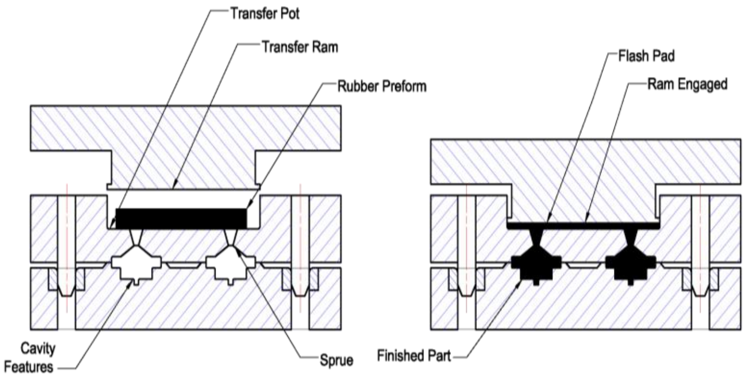


Figure 10.4: Transfer moulding

10.1.4 Additives to polymers

Additives to polymers are materials that are added to polymers in order to improve or change the visual, process, or environmental resistance and degradation properties.

Additives to polymers can be categorized based on their primary purpose, which will be either to add bulk or volume while controlling properties and costs, or modify the chemical or physical properties of the polymer or reinforce the mechanical properties of the polymer, making it stronger or higher in impacts strength.

10.1.5 Adhesives to polymers

Adhesives are materials, typically liquid or semi-liquid, that adhere or bond items together. Adhesives come from either natural or synthetic sources. Adhesives cure (harden) by either evaporating a solvent or by chemical reactions that occur between two or more constituents. Adhesives are also very useful for joining thin or dissimilar materials, minimizing weight, and providing a vibration-damping joint. A disadvantage of most adhesives is that most do not form an instantaneous joint, unlike many other joining processes, because the adhesive needs time to cure.

Types of Adhesives

1. **Non-reactive adhesives**: white glue, rubber cement, natural rubber, polychloroprene
2. **Reactive adhesives**: polyester resin, polyurethane resin, polyols, acrylic polymers
3. **Natural adhesives**: animal glue, Casein,
4. **Artificial adhesives**: epoxy, polyurethane, cyanoacrylate

10.2 Ceramics

Ceramic is an inorganic, non-metallic solid prepared by the action of heat and subsequent cooling. Ceramic materials may contain ionic, covalent or both bonds. The building criteria for the crystal structure are:

- maintain neutrality
- charge balance dictates chemical formula
- achieve closest packing (refer figure below)
- the condition for minimum energy implies maximum attraction and minimum repulsion. This leads to contact, configurations where anions have the highest number of cation neighbours and vice versa.

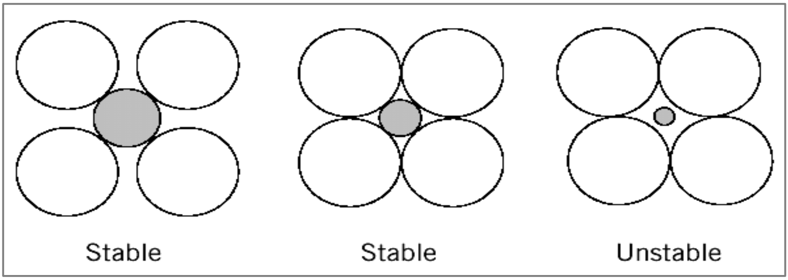


Figure 10.5: Ceramics

10.2.1 Imperfections in ceramics

Imperfections include point defects and impurities. The columbic forces are very large and any charge imbalance has a strong tendency to balance itself. To maintain charge neutrality several point defects can be created.

Non-stoichiometry (composition deviates from the one predicted by chemical formula) may occur when one ion type can exist in two valence states. To minimize energy, the effect of non-stoichiometry is a redistribution of the atomic charges. E.g. Fe^{2+}, Fe^{3+}

For example, in FeO, usual Fe valence state is 2+. If two Fe ions are in 3+ state, then a Fe vacancy is required to maintain charge neutrality which means there'll be fewer Fe ions and hence leads to non-stoichiometry.

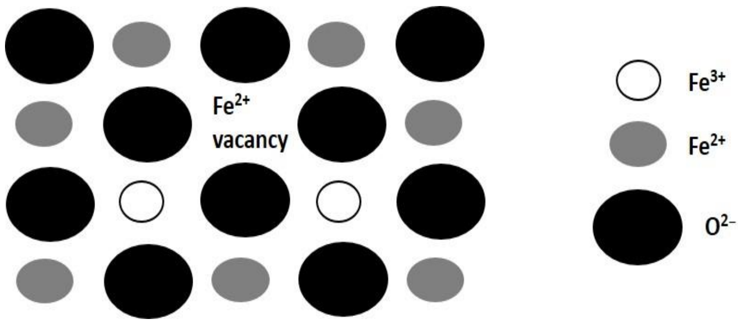


Figure 10.6: Imperfections

Charge neutral defects include the Frenkel and Schottky defects.

Frenkel defect is a pair of cation (positive ion) vacancy and a cation interstitial. It may also be an anion (negative ion) vacancy and anion interstitial. However, anions are larger than cations and placing large anions in an interstitial position requires a lot of energy in lattice distortion. A *Schottky-defect* is a pair of nearby cation and anion vacancies.

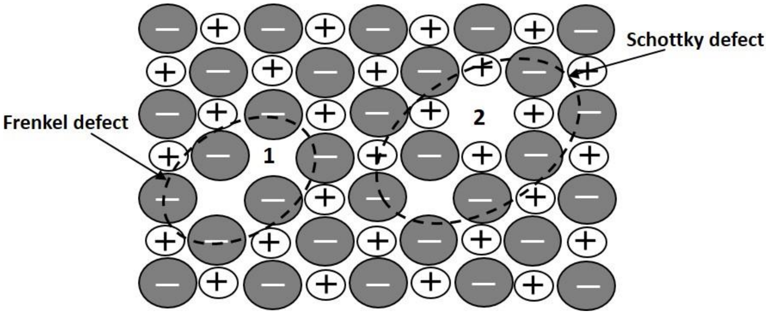


Figure 10.7: Frenkel and Schottky defects

Impurity atoms can exist as either substitutional or interstitial solid solutions.

- Substitutional ions substitute for ions of like type.
- Interstitial ions are small compared to host structure - anion interstitials are unlikely.
- Solubility is higher if ion radii and charges match closely.
- Incorporation of ion with different charge state requires compensation by point defects.

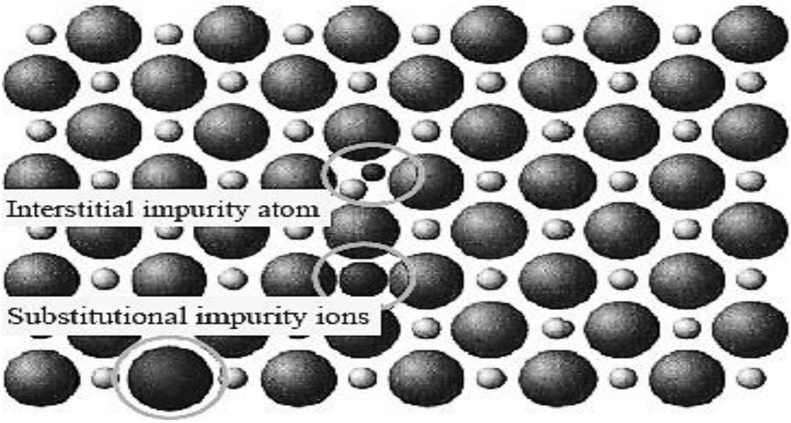


Figure 10.8: Interstitial and substitutional defects

10.2.2 Silicate ceramics

Silicate is mainly composed of silicon and oxygen, the two most abundant elements in earth's crust (rocks, soil, clay, sand). The basic building block is SiO_4 4- tetrahedron. Si-O bonding is largely covalent, but overall SiO4 block has charge of–4.

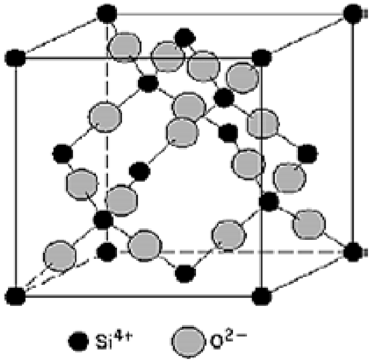


Figure 10.9: 3D network of SiO_4 tetrahedra in cristobalite

10.2.3 Glass

Glass is a non-crystalline (amorphous) ceramic. Fused silica is SiO_2 to which no impurities have been added. Other common glasses contain impurity ions such as Na^+, Ca^{2+}, Al^{3+}, and B^{3+}.

Basic unit of glass is shown in figure 10.10.

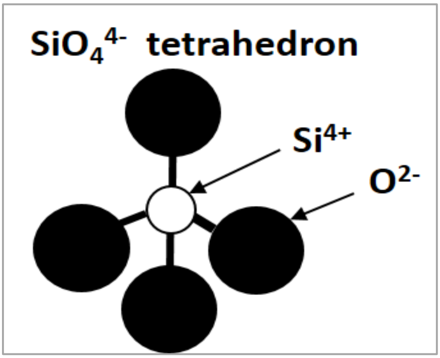


Figure 10.10: Basic unit of glass

A special characteristic of glasses is that solidification is gradual, through a viscous stage, without a clear melting temperature. The specific volume does not have an abrupt transition at a temperature but rather shows a change in slope at the *glass-transition temperature*. The melting point, working point, softening point and annealing point are defined in terms of viscosity, rather than temperature, and depend on glass composition.

Sheet glass forming

Sheet forming is a continuous casting process where sheets of glass (float glass) are formed by floating the molten glass on a pool of molten tin or lead or any other alloy with low melting point. This method gives the sheet uniform thickness and very flat surfaces. Modern windows are made of float glass, which are mostly soda-lime glass.

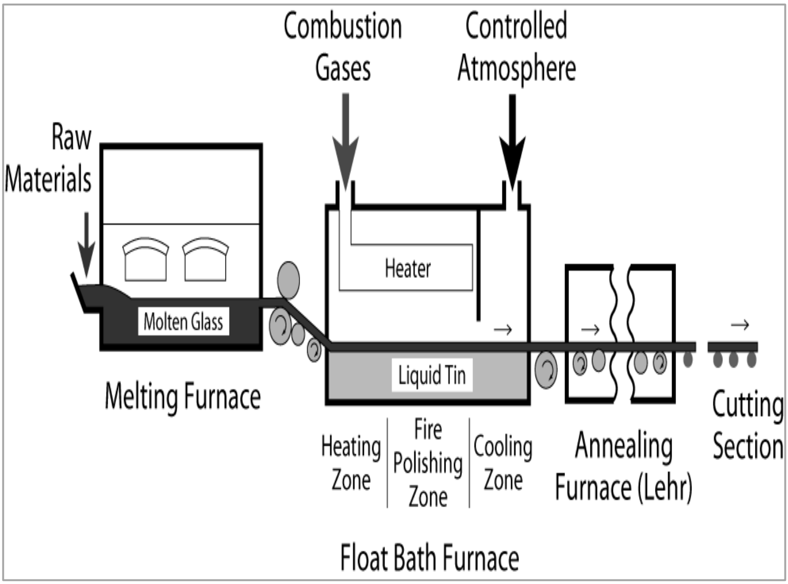


Figure 10.11: Sheet Glass Forming

10.2.4 Deformation and failure

The brittle fracture of ceramics limits applications. It occurs due to the unavoidable presence of microscopic flaws (micro-cracks, internal pores, and atmospheric contaminants) that result during cooling from the melt. The flaws need to crack formation, and crack propagation (perpendicular to the applied stress) is usually transgranular, along cleavage planes. The flaws cannot be closely controlled in manufacturing; this leads to a large variability (scatter) in the fracture strength of ceramic materials.

The compressive strength is typically ten times the tensile strength. This makes ceramics good structural materials under compression (e.g., bricks in houses, stone blocks in the pyramids), but not in conditions of tensile stress, such as under flexure.

Plastic deformation in crystalline ceramics is by slip, which is difficult due to the structure and the strong local (electrostatic) potentials. There is very little plastic deformation before fracture.

Non-crystalline ceramics, like common glass deform by viscous flow (like very high-density liquids). Viscosity decreases strongly with increases temperature.

10.2.5 Processing of ceramics

Ceramic processing is used to produce commercial products that are very diverse in size, shape, detail, complexity, and material composition, structure, and cost. The purpose of ceramics processing to an applied science is the natural result of an increasing ability to refine, develop, and characterize ceramic materials.

Ceramics are typically produced by the application of heat upon processed clays and other natural raw materials to form a rigid product. Ceramic products that use naturally occurring rocks and minerals as a starting material must undergo special processing in order to control purity, particle size, particle size distribution, and heterogeneity. These attributes play a big role in the final properties of the finished ceramic. Chemically prepared powders also are used as starting materials for some ceramic products. These synthetic materials can be controlled to produce powders with precise chemical compositions and particle size.

The next step is to form the ceramic particles into a desired shape. This is accomplished by the addition of water and/or additives such as binders, followed by a shape forming process. Some of the most common forming methods for ceramics include extrusion, slip casting, pressing, tape casting and injection moulding. After the particles are formed, these "green" ceramics undergo a heat-treatment (called firing or sintering) to produce a rigid, finished product. Some ceramic products such as electrical insulators, dinnerware and tile may then undergo a glazing process. Some ceramics for advanced applications may undergo a machining and/or polishing step in order meet specific engineering design criteria.

10.2.6 Applications of ceramics

Most ceramics materials do not exhibit a non-linear plastic region before failure. Instead, ceramics are known to be brittle and fail catastrophically. Their application in engineering applications has certainly been limited by their lack of toughness.

Ceramics are used in an array of applications:

- Compressive strength makes ceramics good structural materials (e.g., bricks in houses, stone blocks in the pyramids).
- High voltage insulators and spark plugs are made from ceramics due to its electrical conductivity properties. E.g. Due to good

thermal insulation, ceramic tiles are used in ovens and as exterior tiles on the Shuttle orbiter. Some ceramics are transparent to radar and other electromagnetic waves and are used in transmitters.

- Hardness, abrasion resistance, imperviousness to high temperatures and extremely caustic conditions allow ceramics to be used in special applications where no other material can be used.
- Chemical inertness makes ceramics ideal for biomedical applications like orthopaedic prostheses and dental implants.
- Glass-ceramics, due to their high temperature capabilities, leads to uses in optical equipment and fibre insulation.
- Ceramic materials that contain alumina and silicon carbide are extremely rigid and are used to sand various surfaces, cut metals, polish, and grind. Ceramics that consists of silica, zirconium oxide, magnesium oxide, silicon carbide & alumina are used in making refractories.

10.3 Composite materials

Sometimes two or more materials are combined together to produce a new material, which possesses much superior properties than any one of the constituent materials. Such a material is known as composite material. The common example of a natural composite is wood, which consists of long cellulose fibre held together by amorphous lignin. Some of the artificial (or synthetic) composite materials are cement concrete, glass, reinforced plastic, plywood, etc.

A type of composite that has been discussed is pearlitic steel, which combines hard, brittle cementite with soft, ductile ferrite to get a superior material.

Usually composites have two phases: matrix phase (continuous) or dispersed phase (particulates, fibres).Properties of composites depend on properties of phases, geometry of dispersed phase (particle size, distribution, orientation) and amount of phase.

10.3.1 Dispersion-strengthened composites

It is the process of use of very hard, small particles to strengthen metals and metal alloys. The effect is like precipitation hardening but not so strong. In dispersion strengthened composites, small particles on the order of 10^{-5} mm to 2.5 x 10^{-4} mm in diameter are added to the matrix material. These particles act to help the matrix resist deformation. This makes the material harder and stronger. Consider a metal matrix composite with a fine distribution of very hard and small secondary particles. The matrix material is carrying most of the load and deformation is accomplished by slip and dislocation movement. The secondary particles impede slip and dislocation and, thereby, strengthen the material. The mechanism is that same as precipitation hardening but effect is not quite as strong. However, particles like oxides do not react with the matrix or go into solution at high temperatures so the strengthening action is retained at elevated temperatures.

Classification of composites

The composite materials are of the following three types:

10.3.2 True particulate composites (particle-reinforced materials or agglomerated materials)

The materials in which the particles are condensed together to form an integral mass, are known as *agglomerated materials*. A common example of agglomerated material is cement concrete, which is formed by mixing coarse aggregate, fine aggregate, cement and water in different proportions. Other useful agglomerated material is grinding wheel (abrasive) which is formed by mixing asphalt, stone and resin of different sizes. Agglomerated materials are usually dispersion strengthened to increase their strength than the normal composites. The particles in these composite are larger than in dispersion strengthened composites. The particle diameter is typically on the order of a few microns. In this case, the particles carry a major portion of the load. The particles are used to increase the modulus and decrease the ductility of the matrix. Another example of particle reinforced composite is an automobile tire which has carbon black particles in a matrix of polyisobutylene elastomeric polymer. Particle reinforced composites are much easier and less costly than making fibre reinforced composites. With polymeric matrices, the particles are simply added to the polymer melt in an extruder or injection moulder during polymer processing. Similarly, reinforcing particles are added to a molten metal before it is cast.

10.3.3 Fibre-reinforced materials

The materials which are produced by combining some suitable material to provide additional strength which does not exist in a single material, are known as *reinforced materials.* The common example of reinforced material is reinforced cement concrete (R.C.C), glass fibre, reinforced plastics, etc. Fibre-reinforced composites use short or long fibre as their prime component, which are usually interwoven using appropriate binder to give a stiff product.

Fibreglass is likely the best known fibre reinforced composite but carbon-epoxy and other advanced composites all fall into this category. The fibres can be in the form of long continuous fibres, or they can be discontinuous fibres, particles, whiskers and even weaved sheets. Fibres are usually combined with ductile matrix materials, such as metals and polymers, to make them stiffer, while fibres are added to brittle matrix materials like ceramics to increase toughness. The length-to diameter ratio of the fibre, the strength of the bond between the fibre and the matrix, and the amount of fibre are variables that affect the mechanical properties. It is important to have a high length-to-diameter aspect ratio so that the applied load is effectively transferred form the matrix to the fibre.

10.3.4 Laminar composite materials (structural materials)

The materials which are produced by bonding two or more layers of different materials completely to each other, are known as *laminated materials* or *laminates*. The materials, constituting a laminated material, may be metallic or non-metallic depending upon the type of application. Sheets (panels) with different orientation of high strength directions are stacked and glued together, producing a material with more isotropic strength in the plane. Plywood, Tufnol are some of the examples of laminates. Strong, stiff end sheets are bonded to lightweight core structure, for instance honeycomb which provides strength to shear. It is used in roofs, walls, and aircraft structures.

10.3.5 Wood

Wood is a natural composite material made of complex array of cellulose called lignin. It has high tangential strength along longitudinal axis of tree trunk. Wood is used to make other composites like plywood and paper. Wood is machined according to different applications.

Properties

- Wood is a complex structure with no uniformity.

- Diagram of the cross-section of a tree trunk shows the following:
- Outer bark : Composed of dry, dead tissues
- Inner bark : Moist & soft layer
- Sapwood : Light colored outer wood; contains some living cells
- Heartwood : Dark, older wood that forms inner part of stem; contains non-living cells
- Pith : Soft tissue at center of the wood
- Classified as softwood and hardwood.
- Annual growth rings determine the age of trees.
- Moisture content is a parameter of wood. i.e.
- Wood moisture (wt. %) = (wt. of water in sample/wt. of dry wood sample) x 100%
- At green condition, moisture content ranges form 60-150% while at dry condition, it comes down to around 20-30%.
- Young's Modulus = 9 to 11 X 10^9 N/m^2

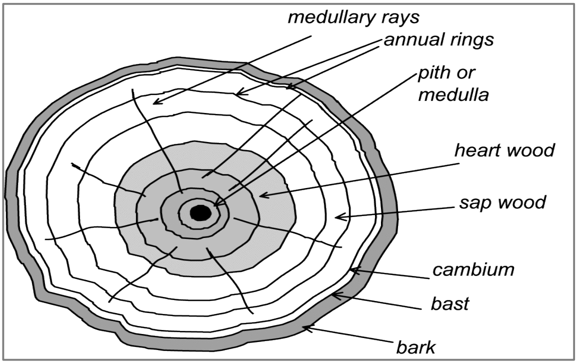


Figure 10.12: Transverse section of a tree trunk

Applications

- Wood is extensively used for making doors and window frames.
- Extensively used in in-door furniture.

- Used as floors and walls
- Used in aircrafts and water boats as major components plus decorative items.
- Used during construction for scaffolding purpose.
- Used for making musical instruments plus sports equipment.

10.3.6 Concrete

Concrete is a mixture of cement (or lime), sand, brick or stone ballast and water, which when placed in forms and allowed to cure, becomes hard like stone. The hardening is caused by the chemical reaction between cement and water. The cement and water form a paste which, upon hardening, binds the aggregates to a permanent mass. Cement is called the *binding material*. The stone or brick ballast is called the *coarse aggregate* as distinguished from the *fine aggregate* which is sand. The mortar used in concrete is called *matrix*. Cement concrete when used by itself is known as *mass concrete*.

Cement

- Raw materials: CaO, Sio_2, Al_2O_3, Fe_2O_3 are mixed in a rotatory kiln at temperature around 1400°C.
- Gypsum is added to cement controls its setting time.
- Different proportions of aforesaid ingredients differ grade of cement (Classified as Type I to Type V)

Aggregate

- Source include rocky mountains.
- Big rocks are cut using manpower, machines or explosives.
- Aggregates of standard sizes (coarse and fine) are sieved using appropriate sieves.

- With the components of concrete manufactured, they are mixed in different proportions.
- Ratio of 1:2:4 for cement: water: aggregate is the usual ratio for a concrete mixture.
- Sand is also mixed in most cases to be economical.

Properties

- Aggregate size, cement types and water content plus use of sand in concrete differs its properties.
- Quality in terms of reactance with sulfate, strength and setting time.
- Concrete has high compressive strength relative to its tensile strength.
- Steel reinforcing bars are used within the concrete structure to increase its tensile strength. Such concrete is called RCC (Reinforced Concrete Cement).
- Steel bars are first applied with tensile force.
- Concrete paste is poured on the steel bars.
- Tensile force is removed from the steel bar.
- Steel tries to be compressed which introduces compressive force which concrete can.

Applications

- Making beams, pillars and roofs for buildings.
- Construction of septic tanks, water pipes and water reserve tanks.

- Concrete blocks are used widely for making pavements and making walls.
- Used in making bank vaults because of immense strength.
- Used in water proofing of floors.

10.3.7 Asphalt

Asphalt is a black colored composite in liquid or semi-liquid phase. It is highly viscous and sticky. It is a petroleum product. It is also called bitumen or pitch. It is heated and mixed with fine aggregate to form asphalt aggregate. Asphalt aggregate is used extensively in road construction and waterproofing purpose.

Properties

- A petroleum product.
- A hydrocarbon with some oxygen, sulfur and other impurities.
- Highly viscous liquid/ semi-solid substance.
- Viscosity decreases with increase in temperature.
- Mechanical properties resemble with that of thermoplastic.
- Aggregate mix behaves like a strong bonded mass when set.
- Glossy but provides friction on road surfaces.

Applications

- Mixed with fine aggregate in 5% - 95% ratio to form asphalt concrete which is rolled onto road to form hard surface.
- Used to make runways for airplanes.
- Mastic asphalt is used to waterproof underground tanks and floors.
- Asphalt is also used to make lacquers and emulsions.
- Used to seal batteries during manufacture.

10.3.8 Nano-composites

A nano-composite is a multiphase solid material where one of the phases has one, two or three dimensions of less than 100 nanometres (nm), or structures having nano-scale repeat distances between the different phases that make up the material. In the broadest sense this definition can include porous media, colloids, gels and copolymers, but is more usually taken to mean the solid combination of a bulk matrix and nano-dimensional phase(s) differing in properties due to dissimilarities in structure and chemistry.These materials have high surface to volume ratio.The mechanical, electrical, thermal, optical, electrochemical, catalytic properties of the nano-composite will differ markedly from that of the component materials. Nano-composites have enhanced electrical, thermal, magnetic, optical plus mechanical properties with respect to macro-composites and can be made of ceramic, polymer or metal.

Size limits for these effects have been proposed, < 5 nm for catalytic activity, < 20 nm for making a hard magnetic material soft, <50 nm for refractive index changes, and < 100 nm for achieving super paramagnetism, mechanical strengthening or restricting matrix dislocation movement. Nano-composites are found in nature, for example in the structure of the abalone shell and bone. The use of nanoparticle-rich materials long predates the understanding of the physical and chemical nature of these materials.

Carbon-reinforced composite is one of its most useful exampleof nano composite.

REVIEW QUESTIONS

1. What are polymers? Briefly introduce different types of polymers.
2. What is degree of polymerisation? Differentiate between addition and condensation polymerisation?
3. Explain about the forming of polymers? What are adhesives and additives to polymers?
4. Write about the imperfections in ceramics? Briefly introduce silicate ceramics and glass.
5. Write about processing and application of ceramics?
6. What are composite materials? Write about true particulate, fibre-reinforced and laminar composite materials.
7. Why are wood, concrete and asphalt considered as composite materials? Explain them in brief.
8. What are nano-composites? Write down about their properties and applications.

GLOSSARY

Abrasive: A substance used for grinding, honing, lapping, super finishing, polishing, pressure blasting, or barrel finishing. It includes natural materials such as garnet, emery, corundum, and diamond and electric- furnace products like aluminium oxide, silicon carbide, and boron carbide.

Addition polymerization: Process by which polymer chains are built up by adding monomers together without creating a by-product.

Age hardening: Hardening by aging, usually after rapid cooling or cold working. See aging.

Aging: In a metal or alloy, a change in properties that generally occurs slowly at room temperature and more rapidly at higher temperatures.

Air-hardening: Term describing a steel containing sufficient carbon and other alloying elements to harden fully during cooling in air or other gaseous mediums from a temperature above its transformation range. The term should be restricted to steels that are capable of being hardened by cooling in air in fairly large sections, about 2 in. or more in diameter.

Allotropy: The reversible phenomenon by which certain metals may exist in more than one crystal structure. If not reversible, the phenomenon is termed “polymorphism.”

Alloy: A substance having metallic properties and being composed of two or more chemical elements of which at least one is an elemental metal.

Alloying element: An element which is added to a metal to effect changes in properties and which remains within the metal.

Alloy steel: Steel containing significant quantities of alloying elements (other than carbon and the commonly accepted amounts of manganese, silicon, sulphur, and phosphorus) added to effect changes in the mechanical or physical properties.

Alpha iron: The body-centered cubic from of pure iron, stable below 1670^0F.

Amorphous: Not having a crystal structure; non-crystalline.

Annealing: Heating to and holding at a suitable temperature and then cooling at a suitable rate, for such purposes as reducing hardness, improving machinability, facilitating cold working, producing a desired microstructure, or obtaining desired mechanical, physical, or other properties.

When applied to ferrous alloys, the term annealing, without qualification, implies full annealing.

When applied to nonferrous alloys, the term annealing implies a heat treatment designed to soften a cold-worked structure by recrystallization or subsequent grain growth or to soften an age-hardened alloy by causing a nearly complete precipitation of the second phase in relatively coarse form.

Any process of annealing will usually reduce stresses, but if the treatment is applied for the sole purpose of such relief, it should be designated stress relieving.

Apparent porosity: The percentage of a ceramic body that is composed of interconnected porosity.

Aspect ratio: The length of a fiber divided by its diameter.

Atomization: The dispersion of a molten metal into particles by a rapidly moving stream of gas or liquid.

Austempering: Quenching a ferrous alloy from a temperature above the transformation range, in a medium having a rate of heat abstraction high enough to prevent the formation of high-temperature transformation products, and then holding the alloy, until transformation is complete, at a temperature below that of pearlite formation and above that of martensite formation.

Austenite: A solid solution of one or more elements in face-centered cubic iron. The solute is generally assumed to be carbon.

Austenitic steel: An alloy steel whose structure is normally austenitic at room temperature.

Base metal: (1) The metal present in the largest proportion in an alloy; brass, for example, is a copper-base alloy. (2) The metal to be brazed, cut, or welded. (3) After welding, that part of the metal which was not melted.

Brass: An alloy consisting mainly of copper (over 50 percent) and zinc, to which smaller amounts of other elements may be added.

Bimetallic: A laminar composite material produced by joining two strips of metal with different thermal expansion coefficients, making the material sensitive to temperature changes.

Brazing: Joining metals by flowing a thin layer, capillary thickness, of nonferrous filler metal into the space between them. Bonding results from the intimate contact produced by the dissolution of a small amount of base metal in the molten filler metal, without fusion of the base metal. Sometimes the filler metal is put in place as a thin solid sheet or as a clad layer, and the composite is heated, as in furnace brazing. The term brazing is used where the temperature exceeds some arbitrary value, such as 800^0F; the term soldering is used for temperatures lower than the arbitrary value.

Brinell hardness test: A test for determining the hardness of a material by forcing a hard steel or carbide ball of specified diameter into it under a specified load. The result is expressed as the Brinell hardness number, which is the value obtained by dividing the applied load in kilograms by the surface area of the resulting impression in square millimetres.

Brittle fracture: Fracture with little or no plastic deformation.

Brittleness: The quality of a material that leads to crack propagation without appreciable plastic deformation.

Bronze: A copper-rich copper-tin alloy with or without small proportions of other elements such as zinc and phosphorus. Also, certain other essentially binary copper-base alloys containing no tin, such as aluminum bronze (copper-aluminum), silicon bronze (copper silicon), and beryllium bronze (copper-beryllium).

Burning: (1) Permanently damaging a metal or alloy by heating to cause either incipient melting or inter-granular oxidation. (2) In grinding, getting the work hot enough to cause discoloration or to change the microstructure by tempering or hardening.

Carbide: A compound of carbon with one or more metallic elements.

Carbonitriding: Introducing carbon and nitrogen into a solid ferrous alloy by holding above Ac_1 in an atmosphere that contains suitable gases such as hydrocarbons, carbon monoxide, and ammonia. The carbonitrided alloy is usually quench-hardened.

Carbonizing: Driving off the non-carbon atoms from a polymer fiber, leaving behind a carbon fiber of high strength. Also known as pyrolizing.

Carbon steel: Steel containing carbon up to about 2 percent and only residual quantities of other elements, except those added for deoxidation, with silicon usually limited to 0.60 percent and manages to about 1.65 percent. Also termed plain-carbon steel, ordinary steel, and straight-carbon steel.

Carburizing: Introducing carbon into a solid ferrous alloy by holding above Ac_1 in contact with a suitable carbonaceous material, which may be a solid, liquid, or gas. The carburized alloy is usually quench-hardened.

Case: In a ferrous alloy, the outer portion that has been made harder than the inner portion, or core, by case hardening.

Case hardening: Hardening a ferrous alloy so that the outer portion, or case, is made substantially harder than the inner portion, or core. Typical processes used for case hardening are carburizing, cyaniding, carbonitriding, nitriding, induction hardening, and flame hardening.

Casting: (1) An object at or near finished shape obtained by solidification of a substance in a mould. (2) Pouring molten metal into a mould to produce an object of desired shape.

Cast iron: An iron containing carbon in excess of the solubility in the austenite that exists in the alloy at the eutectic temperature. For the various forms gray cast iron, white cast iron, malleable cast iron and nodular cast iron, the word "cast" is often left out, resulting in "gray iron", "white iron," "malleable iron," and "nodular iron," respectively.

Cast structure: The internal physical structure of a casting evidenced by shape and orientation of crystals and segregation of impurities.

Cathodic protection: Partial or complete protection of a metal from corrosion by making it a cathode, using either a galvanic or impressed current.

Cementation: Bonding ceramic raw materials into a useful product, using binders that form a glass or gel without firing at high temperatures.

Cementite: A compound of iron and carbon, known chemically as iron carbide, and having the approximate chemical formula Fe_3C. It is characterized by an orthorhombic crystal structure.

Ceramic: An inorganic material with high melting temperature. Usually hard and brittle.

Ceramic bond: Bonding ceramic materials by permitting a glassy product to form at high firing temperatures.

Charpy test: A pendulum-type single-bow impact test in which the specimen, usually notched, is supported at both ends as a simple beam and broken by a falling pendulum. The energy absorbed, as determined by the subsequent rise of the pendulum, is a measure of impact strength or notch toughness.

Cladding: A laminar composite produced when a corrosion-resistant or high-hardness layer of a laminar composite is formed onto a less expensive or higher-strength backing.

Cohesion: Force of attraction between the molecules (or atoms) within a single phase. Contrast with adhesion.

Cold treatment: Cooling to a low temperature, often near – 100^0F, for the purpose of obtaining desired conditions or properties, such as dimensional or structural stability.

Cold working: Deforming metal plastically at a temperature lower than the recrystallization temperature.

Corrosion: The deterioration of a metal by chemical or electrochemical reaction with its environment.

Covalent bond: A bond between two or more atoms resulting from the completion of shells by the sharing of electrons.

Creep: Time-dependent strain occurring under stress. The creep strain occurring at a diminishing rate is called primary creep; that occurring at a minimum and almost constant rate, secondary creep; that occurring at an accelerating rate, tertiary creep.

Critical cooling rate: The minimum rate of continuous cooling just sufficient to prevent undesired transformations. For steel, the slowest rate at which it can be cooled from above the upper critical temperature to prevent the decomposition of austenite at any temperature above the martensite temperature (M_S).

Critical point: (1) The temperature or pressure at which a change in crystal structure, phase, or physical properties occurs. Some as transformation temperature. (2) In an equilibrium diagram, that specific value of composition, temperature and pressure, or combinations thereof, at which the phases of a heterogeneous system are in equilibrium.

Crystal: A solid composed of atoms, ions, or molecules arranged in a pattern which is repetitive in three dimensions.

Crystalline fracture: A fracture of a polycrystalline metal characterized by a grainy appearance.

Crystallization: The separation, usually from a liquid phase on cooling, of a solid crystalline phase.

Decarburization: The loss of carbon from the surface of a ferrous alloy as a result of heating in a medium that reacts with the carbon at the surface.

Defect: A condition that impairs the usefulness of an object or of a part.

Deflection temperature. The temperature at which a polymer will deform a given amount under a standard load (also called distortion temperature).

Degradation temperature: The temperature above which a polymer burns, chars, or decomposes.

Degree of polymerization: The average molecular weight of the polymer divided by the molecular weight of the monomer.

Delamination: Separation of individual plies of a fiber-reinforced composite.

Dendrite: A crystal that has a free-like branching pattern, being most evident in cast metals slowly cooled through the solidification range.

Die casting: (1) A casting made in a die. (2) A casting process where molten metal is forced under high pressure into the cavity of a metal mould.

Dislocation: A linear defect in the structure of a crystal. Two basic types are recognized but combinations and partial dislocations are most prevalent. An "edge dislocation" corresponds to the row of mismatched atoms along a straight edge formed by an extra, partial plane of atoms within the body of the crystal, that is, by a plane of smaller area than any other parallel section through the crystal. A "Screw dislocation" corresponds to the highly distorted lattice adjacent to the axis of a spiral structure in a crystal, the spiral structure being characterized by a distortion that has joined normally parallel planes together to form a continuous helical ramp winding about the dislocation as an axis with a pitch of one inter-planar distance.

Distortion: Any deviation from the desired shape or contour.

Drawing: (1) Forming recessed parts by forcing the plastic flow of metal in dies. (2) Reducing the cross section of wire or tubing by pulling it through a die. (3) A misnomer for tempering.

Ductility: The ability of a material to deform plastically without fracturing, being measured by elongation or reduction of area in a tensile test.

Duralumin: A term formerly applied to the class of age-hardenable aluminium-copper alloys containing manganese, magnesium, or silicon.

Elastic deformation: Change of dimensions accompanying stress in the elastic range, original dimensions being restored upon release of stress.

Elasticity: That property of a material by virtue of which it tends to recover its original size and shape after deformation.

Elastic limit: The maximum stress to which a material may be subjected without any permanent strain remaining upon complete release of stress.

Elongation: In tensile, the increase in the gage length, measured after fracture of the specimen within the page length, usually expressed as a percentage of the original gage length.

Enamel: A ceramic coating on metal.

Equilibrium diagram: A graphical representation of the temperature, pressure, and composition limits of phase fields in an alloy system as they exist under conditions of complete equilibrium. In metal system as they exist under conditions of complete equilibrium. In metal systems, pressure is usually considered constant (synonymous with phase diagram).

Etching: Subjecting the surface of a metal to preferential chemical or electrolytic attack in order to reveal structural details.

Eutectic: (1) An isothermal reversible reaction in which a liquid solution is converted into two or more intimately mixed solids on cooling, the number of solids formed being the same as the number of components in the system. (2) An alloy having the composition indicated by the eutectic point on an equilibrium diagram. (3) An alloy structure of intermixed solid constituents formed by a eutectic reaction.

Eutectoid: (1) An isothermal reversible reaction in which a solid phase (usually a solid solution) is converted into two or more intimately mixed solids on cooling, the number of solids formed being the same as the number of components in the system. (2) An alloy having the composition indicated by the eutectoid point on an equilibrium diagram. (3) An alloy structure of intermixed solid constituents formed by a eutectoid reaction.

Fatigue: The phenomenon leading to fracture under repeated or fluctuating stresses having a maximum value less than the tensile strength of the material. Fatigue fractures are progressive, beginning as minute cracks that grow under the action of the fluctuating stress.

Fatigue life: The number of cycles of stress that can be sustained prior to failure for a stated test condition.

Fatigue limit: The maximum stress below which a material can presumably endure an infinite number of stress cycles. If the stress is not completely reversed, the value of the mean stress, the minimum stress, or the stress ratio should be stated.

Fatigue strength: The maximum stress that can be sustained for a specified number of cycles without failure, the stress being completely reversed within each cycle unless otherwise stated.

Ferrite: (1) A solid solution of one or more elements in body-centered-cubic iron. Unless otherwise designated, the solute is generally assumed to be carbon. On some equilibrium diagrams there are two ferrite regions separated by an austenite area. The lower area is alpha ferrite; the upper, delta ferrite. If there is no designation, alpha ferrite is assumed.

Fibrous structure: (1) In forgings, a structure revealed as laminations, not necessarily detrimental, on an etched section or as a ropy appearance on a fracture. It is not to be confused with the "silky" or "ductile" fracture of a clean metal. (2) In wrought iron, a structure consisting of slag fibers embedded in ferrite.

Flame hardening: Quench hardening in which the heat is applied directly by a flame.

Flux: Additions to ceramic raw materials that reduce the melting temperature.

Fracture stress: (1) The maximum principal true stress at fracture. Usually refers to un-notched tensile specimens. (2) The (hypothetical) true stress which will cause fracture without further deformation at any given strain.

Fracture test: Breaking a specimen and examining the fractured surface with the unaided eye or with a low-power microscope to determine such things as composition, grain size, case depth, soundness, or presence of defects.

Free carbon: The part of the total carbon in steel or cast iron that is present in the elemental form as graphite or temper carbon.

Free ferrite: Ferrite that is structurally separate and distinct, as may be formed without the simultaneous formation of carbide when cooling hypoeutectoid austenite into the critical temperature range. Also proeutectoid ferrite.

Full annealing: Annealing a ferrous alloy by austenitizing and then cooling slowly through the transformation range. The austenitizing temperature for hypoeutectoid steel is usually above A_{c3}; and for hypereutectoid steel, usually between $A_{c1.3}$ and A_{cm}.

Galvanic corrosion: Corrosion associated with the current of a galvanic cell consisting of two dissimilar conductors in an electrolyte or two similar conductors in dissimilar electrolytes. Where the two dissimilar metals are in contact, the resulting reaction is referred to as couple action.

Galvanic series: A series of metals and alloys arranged according to their relative electrode potentials in a specified environment.

Gamma iron: The face-centered cubic form of pure iron, stable from 1670 to 2550^{0}F.

Glass: An amorphous material derived by cooling of a melt.

Glass-ceramics: Ceramic shapes formed in the glassy state and later allowed to crystallize during heat treatment to achieve improved strength and toughness.

Glass formers: Oxides with a high bond strength that easily produce a glass during processing.

Glass temperature (T_g): The temperature below which an undercooled liquid becomes a glass. This is not a fixed temperature.

Glaze: Ceramic coating applied to glass. The glaze contains glassy and crystalline ceramic phases.

Grain: An individual crystal in a polycrystalline metal or alloy.

Grain growth (coarsening): An increase in the size of grains in polycrystalline metal, usually effected during heating at elevated temperatures. The increase may be gradual or abrupt, resulting in either uniform or non-uniform grains after growth has ceased. A mixture of non-uniform grains is sometimes termed duplexed. Abnormal grain growth (exaggerated grain growth) implies the formation of excessively large grains, uniform or non-uniform.

Grain size: For metals, a measure of the areas or volumes of grains in a polycrystalline material, usually expressed as an average when the individual sizes are fairly uniform. Grain sizes are reported in terms of number of grains per unit area or volume, in terms of average diameter, or as a grain-size number derived from area measurements.

Graphitic carbon: Free carbon in steel or cast iron.

Graphite steel: Alloy steel made so that part of the carbon is present as graphite.

Graphitization: Formation of graphite in iron or steel. Where graphite is formed during solidification, the phenomenon is called primary graphitization; where formed later by heat treatment, secondary graphitization.

Gray cast iron: A cast iron that gives a gray fracture due to the presence of flake graphite. Often called gray iron.

Hardenability: In a ferrous alloy, the property that determines the depth and distribution of hardness induced by quenching.

Hardening: Increasing the hardness by suitable treatment, usually involving heating and cooling. When applicable, the following more specific terms should be used: age hardening, case hardening, flame hardening, induction hardening, precipitation hardening, and quench hardening.

Hardness: Resistance of metal to plastic deformation usually by indentation. However, the term may also refer to stiffness or temper, or to resistance to scratching, abrasion, or cutting. Indentation hardness may be measured by various hardness tests, such as Brinell, Rockwell, and micro-hardness.

Heat treatment: Heating and cooling a solid metal or alloy in such a way as to obtain desired conditions or properties. Heating for the sole purpose of hot working is excluded from the meaning of this definition.

Homogenizing: Holding at high temperature to eliminate or decrease chemical segregation by diffusion.

Honeycomb: A lightweight but stiff assembly of aluminium strip joined and expanded to form the core of a sandwich structure.

Hot working: Deforming metal plastically at such a temperature and rate that strain hardening does not occur. The low limit of temperature is the recrystallization temperature.

Impact energy (impact value): The amount of energy required to fracture a material, usually measured by means of an Izod or Charpy test. The type of specimen and testing conditions affect the values and therefore should be specified.

Impact test: A test to determine the behaviour of materials when subjected to high rates of loading, usually in bending, tension, or torsion. The quantity measured is the energy absorbed in breaking the specimen by a single blow, as in the Charpy or Izod tests.

Indentation hardness: The resistance of a material to indentation. This is the usually type of hardness test, in which a pointed or rounded indenter is pressed into a surface under a substantially static load.

Induction hardening: Quench hardening in which the heat is generated by electrical induction.

Intermediate annealing: Annealing wrought metals at one or more stages during manufacture and before final treatment.

Intermediate phase: In an alloy or a chemical system, a distinguishable homogeneous phase whose composition range does not extend to any of the pure components of the system.

Interstitial solid solution: A solid solution in which the solute atoms occupy positions within the lattice of the solvent. See also substitutional solid solution.

Ion: An atom, or group of atoms, that has gained or lost one or more outer electrons and thus carries an electric charge. Positive ions, or cations, are deficient in outer electrons. Negative ions, or anions, have an excess of outer electrons.

Izod test: A pendulum type of single-blow impact test in which the specimen, usually notched, is fixed at one end and broken by a falling pendulum. The energy absorbed, as measured by the subsequent rise of the pendulum, is a measure of impact strength or notch toughness.

Knoop hardness: Micro-hardness determined from the resistance of metal to indentation by a pyramidal diamond indenter, having edge angles of $172^0 30'$ and 130^0, making a rhombohedral impression with one long and one short diagonal. The long diagonal is measured microscopically to determine the KHN.

Laminated glass: Annealed glass with a polymer (e.g., polyviny1 butyral, PVB) sandwiched in between, used for car windshields.

Ledeburite: The eutectic of the iron-carbon system, the constituents being austenite and cementite. The austenite decomposes into ferrite and cementite on cooling below the Ar_1.

Linear polymer: Any polymer in which molecules are in the form of spaghetti-like chains.

Liquidus: In a constitution or equilibrium diagram, the locus of points representing the temperatures at which the various compositions in the system begin to freeze on cooling or to finish melting on heating.

Liquid-crystalline polymers: Exceptionally stiff polymer chains that act as rigid rods, even above their melting point.

Macrostructure: The structure of metals as revealed by examination of the etched surface of polished specimen at a magnification not exceeding ten diameters.

Malleability: The characteristic of metals which permits plastic deformation in compression without rupture.

Malleable cast iron: A cast iron made by a prolonged anneal of white cast iron in which decarburization or graphitization, or both, take place to eliminate some or all of the cementite. The graphite is in the form of temper carbon. If decarburization is the predominant reaction, the product will have a light fracture, hence, "white-heart malleable"; otherwise, the fracture will be dark, hence, "black-heart malleable," "Pearlitic malleable" is a black-heart variety having a pearlitic matrix along with perhaps some free ferrite.

Martempering: Quenching an austenitized ferrous alloy in a medium at a temperature in the upper part of the martensite range, or slightly above that range, and holding it in the medium until the temperature throughout the alloy is substantially uniform. The alloy is then allowed to cool in air through the martensite range.

Martensite: (1) In an alloy, a metastable transitional structure intermediate between two allotropic modifications whose abilities to dissolve a given solute differ considerably, the high-temperature phase having the greater solubility. The amount of the high temperature phase transformed to martensite depends to a large extent upon the temperature attained in cooling, there being a rather distinct beginning temperature. (2) A metastable phase of steel, formed by a transformation of austenite below the M_s (or Ar) temperature. It is an interstitial supersaturated solid solution of carbon in iron having a body-centered tetragonal lattice. Its microstructure is characterized by an acicular, or needle-like, pattern.

Martensitic transformation: A reaction which takes place in some metals on cooling, with the formation of an acicular structure called martensite.

Mechanical properties: The properties of a material that reveal its elastic and inelastic behaviour where force is applied, thereby indicating its suitability for mechanical applications; for example, modulus of elasticity, tensile strength, elongation, hardness, and fatigue limit.

Mechanical twin: A twin formed in a metal during plastic deformation by a simple shear of the lattice.

Melting point: The temperature at which a pure metal, compound, or eutectic changes from solid to liquid; the temperature at which the liquid and the solid are in equilibrium.

Metal: (1) An opaque lustrous elemental chemical substance that is a good conductor of heat and electricity and, when polished, a good reflector of light. Most elemental metals are malleable and ductile and are, in general, heavier than the other elemental substances. (2) As to structure, metals may be distinguished from non-metals by their atomic binding and electron availability. Metallic atoms tend to lose electrons from the outer shells, the positive ions thus formed being held together by the electron gas produced by the separation. The ability of these free electrons to carry an electric current, and the fact that the conducting power decreases as temperature increases, establish one of the prime distinctions of a metallic solid.

Metallograph: An optical instrument designed for both visual observation and photomicrography of prepared surfaces of opaque materials at magnifications ranging from about 25 to about 1,500 diameters. The instrument consists of a high-intensity illuminating source, a microscope, and a camera bellows. On some instruments provisions are made for examination of specimen surfaces with polarized light, phase contrast, oblique illumination, dark-field illumination, and customary bright-field illumination.

Metallurgy: The science and technology of metals. Process (chemical) metallurgy is concerned with the extraction of metals from their ores and with the refining of metals; physical metallurgy deals with the physical and mechanical properties of metals as affected by composition, mechanical working, and heat treatment.

Micro-hardness: Hardness determined by using a microscope to measure the diagonal of the impression left by a Knoop or Vickers indenter. The hardness is the load divided by the area of the impression.

Microstructure: The structure of polished and etched metals as revealed by a microscope at a magnification greater than ten diameters.

Miller indices, plane: Indices which identify a family of planes in crystal structure. The intercepts m, n, and ρ for any plane within a crystal give the reciprocals l/m, l/n, and l/p, which may be changed to a common denominator, resulting in the numerators h, k and l respectively. These numerators when written as (hkl) identify the family of planes to which the specific plane belongs.

Modulus of elasticity: A measure of the stiffness of metal. Ratio of stress, within proportional limit, to corresponding strain.

Monotectic: An isothermal reversible reaction in a binary system, in which a liquid on cooling decomposes into a second liquid of a different composition and a solid. It differs from a eutectic in that only one of the two products of the reaction is below its freezing range.

Modular cast iron: A cast iron that has been treated while molten with a master alloy containing an element such as magnesium or cerium to give primary graphite in the spherulitic form.

Nano-composite: A material in which the dispersed phase is nano-sized and is distributed in the continuous matrix.

Normalizing: Heating a ferrous alloy to a suitable temperature above the transformation range and then cooling in air to a temperature substantially below the transformation range.

Nucleation: The initiation of a phase transformation at discrete sites, the new phase growing on nuclei.

Nucleus: (1) The first structurally stable particle capable of initiating recrystallization of a phase or the growth of a new phase, and possessing an interface with the parent matrix. The term is also applied to a foreign particle which initiates such action. (2) The heavy central core of an atom, in which most of the mass and the total positive electric charge are concentrated.

Over-aging: Aging under conditions of time and temperature greater than those required to obtain maximum change in a certain property, so that the property is altered in the direction of the initial value. See aging.

Overheating: Heating a metal or alloy to such a high temperature that its properties are impaired. When the original properties cannot be restored by further heat treating, the overheating is known as burning.

Oxidation: A reaction in which there is an increase in valence resulting from a loss of electrons.

Parameter (lattice): In a crystal, the length, usually in angstrom units, of the unit cell along one of its axes or edges; also called lattice constant.

Pearlite: A lamellar aggregate of ferrite and cementite, often occurring in steel and cast iron.

Peritectic: An isothermal reversible reaction in which a liquid phase reacts with a solid phase to produce another solid phase on cooling.

Peritectoid: An isothermal reversible reaction in which a solid phase reacts with a second solid phase to produce yet a third solid phase on cooling.

Phase: A physically homogeneous and distinct portion of a material system.

Phase diagram: Same as constitution diagram or equilibrium diagram.

Plastic: A predominantly polymeric material comprising of other additives.

Plastic deformation: Deformation that does or will remain permanent after removal of the load which caused it.

Polymer: Polymers are materials made from giant (or macromolecular), chain-like molecules having average molecular weights from 10,000 to more than 1,000,000 g/mol. built by the joining of many units by chemical bonds. Polymers are usually, but not always, carbon based.

Polymorphism: See allotropy.

Powder metallurgy: Powder processing routes used for converting metal and alloy powders into useful shapes.

Precipitation hardening: Hardening caused by the precipitation of a constituent from a supersaturated solid solution. See also age hardening and aging.

Precursor: A starting chemical (e.g., a polymer fiber) that is carbonized to produce carbon fibers.

Process annealing: In the sheet and wire industries, heating a ferrous alloy to a temperature close to, but below, the lower limit of the transformation range and then cooling, in order to soften the alloy for further cold working.

Proportional limit: The maximum stress at which strain remains directly proportional to stress.

Pultrusion: A method for producing composites containing mats or continuous fibers.

Quenching: Rapid cooling. When applicable, the following more specific terms should be used: direct quenching, interrupted quenching, selective quenching, spray quenching, and time quenching.

Recovery: Reduction or removal of work-hardening effects, without motion of large-angle grain boundaries.

Recrystallization: (1) The change from one crystal structure to another, as occurs on heating or cooling through a critical temperature. (2) The formation of a new, strain-free grain structure from that existing in cold-worked metal, usually accomplished by heating.

Refractory metal: A metal having an extremely high melting point. In the broad sense, it refers to metals having melting points above the range of iron, cobalt, and nickel.

Residual stress: Stress present in a body that is free of external forces or thermal gradients.

Resilience: (1) The amount of energy per unit volume released upon unloading. (2) The capacity of a metal, by virtue of high yield strength and low elastic modulus, to exhibit considerable elastic recovery upon release of load.

Rockwell hardness test: A test for determining the hardness of a material based upon the depth of penetration of a specified penetrator into the specimen under certain arbitrarily fixed conditoins of test.

Rolling: Reducing the cross-sectional area of metal stock, or otherwise shaping metal products, through the use of rotating rolls.

Rust: A corrosion product consisting of hydrated oxides of iron. Applied only to ferrous alloys.

Scale: Formation of a layer of iron oxide on the surface of steel when heated at high temperature in a furnace.

Scratch hardness: The hardness of a metal determined by the width of a scratch made by a cutting point drawn across the surface under a given pressure.

Seam: On the surface of metal, an unwelded fold or lap which appears as a crack, usually resulting from a defect obtained in casting or in working.

Slip: Plastic deformation by the irreversible shear displacement (translation) of one part of a crystal relative to another in a definite crystallographic direction and usually on a specific crystallographic plane.

S-N diagram: A plot showing the relationship of stress *S* and the number of cycles *N* before failure in fatigue testing.

Solid solution: A single solid homogeneous crystalline phase containing two or more chemical species.

Solidius: In a constitution or equilibrium diagram, the locus of points representing the temperatures at which various compositions finish freezing on cooling or begin to melt on heating.

Solute: The component of either a liquid or solid solution that is present to a lesser or minor extent; the component that is dissolved in the *solvent*.

Solvent: The component of either a liquid or solid solution that is present to a greater or major extent; the component that dissolves the solute.

Sorbite: A fine mixture of ferrite and cementite produced either by regulating the rate of cooling of steel or tempering steel after hardening. The first type is very fine pearlite difficult to resolve under the microscope; the second type is tempered martensite.

Spheroidite: An aggregate of iron or alloy carbides of essentially spherical shape dispersed throughout a matrix of ferrite.

Spheroidizing: Heating and cooling to produce a spheroidal or globular form of carbide in steel. Spheroidizing methods freqently used are:

1. Prolonged holding at a temperature just below Ae_1.
2. Heating and cooling alternately between temperatures that are just above and just below Ae_1.

3. Heating to a temperature above Ae_1 or Ae_3 and then cooling very slowly in the furnace or holding at a temperature just below Ae_1.
4. Cooling at a suitable rate from the minimum temperature at which all carbide is dissolved, to prevent the reformation of a carbide network, and then reheating in accordance with methods 1 or 2 above. (Applicable to hypereutectoid steel containing a carbide network.)

Steel: An iron-base alloy, malleable in some temperature range as initially cast, containing manganese, usually carbon, and often other alloying elements. In carbon steel and low-alloy steel, the maximum carbon is about 2.0 percent; in high-alloy steel, about 2.5 percent. The dividing line between low-alloy and high-alloy steels is generally regarded as being at about 5 percent metallic alloying elements.

Steel is to be differentiated from two general classes of "irons"; the cast irons, on the high-carbon side, and the relatively pure irons such as ingot iron, carbonyl iron, and electrolytic iron, on the low-carbon side. In some steels containing extremely low carbon, the manganese content is the principal differentiating factor, steel usually containing at least 0.25 percent; ingot iron contains considerably less.

Stiffness: The ability of a metal or shape to resist elastic deflection. For identical shapes, the stiffness is proportional to the modulus of elasticity.

Strain: A measure of the change in the size or shape of a body, referred to its original size or shape. Linear strain is the change per unit length of a linear dimension. Conventional strain is the linear strain referred to the original gage length. When the term strain is used alone, it usually refers to the linear strain in the direction of the applied stress.

Strain hardening: An increase in hardness and strength caused by plastic deformation at temperatures lower than the recrystallization range.

Stress: Force per unit area, often through of as force acting through a small area within a plane. It can be divided into components, normal and parallel to the plane, called normal stress and shear stress, respectively.

Stress-induced crystallization: The process of forming crystals by the application of an external stress. Typically, a significant fraction of many amorphous plastics can be crystallized in this fashion, making them stronger.

Stress relieving: Heating to a suitable temperature, holding long enough to reduce residual stresses, and then cooling slowly enough to minimize the development of new residual stresses.

Stress-rupture test: A tension test performed at constant load and constant temperature the load being held at such a level as to cause rupture. Also known as creep-rupture test.

Super-cooling: Cooling below the temperature at which an equilibrium-phase transformation can take place without actually obtaining the transformation.

Synthesis: Steps conducted to make a ceramic powder.

Temper: (1) In heat treatment, to reheat hardened steel or hardened cast iron to some temperature below the eutectoid temperature for the purpose of decreasing the hardness steel. (2) In nonferrous alloys and in some ferrous alloys (steels that cannot be hardened by heat treatment), the hardness and strength produced by mechanical or thermal treatment, or both, and characterized by a certain structure, mechanical properties, or reduction in area during cold working.

Tempering: Re-heating a quench-hardened or normalized ferrous alloy to a temperature below the transformation range and then cooling at any rate desired.

Tempered glass: A high-strength glass that has a surface layer where the stress is compressive, induced thermally during cooling or by the chemical diffusion of ions.

Tensile strength: In tensile testing, the ratio of maximum load to original cross-sectional area. Also called ultimate strength.

Thermocouple: A device for measuring temperatures consisting of two dissimilar metals which produce an electromotive force roughly proportional to the temperature difference between their hot and cold junction ends.

Thermoplastics: Linear or branched polymers in which chains of molecules are not interconnected to one another.

Thermoplastic elastomers: Polymers that behave as thermoplastics at high temperatures, but as elastomers at lower temperatures.

Toughness: Ability of a metal to absorb energy and deform plastically before fracturing. It is usually measured by the energy absorbed in a notch-impact test, but the area under the stress-strain curve in tensile testing is also a measure of toughness.

Troostite: A previously unresolvable, rapidly etching fine aggregate of carbide and ferrite produced either by tempering martensite at low temperature or by quenching a steel at a rate slower than the critical cooling rate. Preferred terminology for the first product is tempered martensite; for the latter, fine pearlite.

Twin: Two portions of a crystal having a definite crystallographic relationship; one may be regarded as the parent, the other as the twin. The orientation of the twin is either a mirror image of the orientation of the parent about a “twinning plane” or an orientation that can be derived by rotating the twin portion about a “twinning axis”.

Twin band: On a polished and etched surface, the section through a twin and the parent crystal.

Ultimate strength: The maximum conventional stress-tensile, compressive, or shear-that a material can withstand.

Vacancy: A type of lattice imperfection in which an individual atom site is temporarily unoccupied. Diffusion (of other than interstitial solutes) is generally visualized as the shifting of vacancies.

Vickers hardness: Micro-hardness determined from the resistance of a metal to indentation by a 136^0 diamond-pyramid indenter making a square impression.

Vitrification: Melting or formation of a glass.

Whiskers: Very fine fibers grown in a manner that produces wingle crystals with no mobile dislocations, thus giving nearly theoretical strengths.

White cast iron: Cast iron that gives a white fracture because the carbon is in combined form.

Wrought iron: A commercial iron consisting of slab (iron silicate) fibers entrained in a ferrite matrix.

Yield point: The first stress in a material, usually less than the maximum attainable stress, at which an increase in strain occurs without an increase in stress. Only certain metals exhibit a yield point. If there is a decrease in stress after yielding, a distinction may be made between upper and lower yield points.

Yield strength: The stress at which a material exhibits a specified deviation from proportionality of stress and strain. An offset of 0.2 percent is used for many metals.

Appendix I

Hardness Conversion Table and Approximate Tensile Strength

S.N.	Various hardness number						
	Vickers hardness number	*Brinell number*		*Rockwell number*		*Sclero-scope*	*Tensile strength*
		Steel ball	*Carbide ball*	*C-Scale*	*B-Scale*	*Scope hardness*	
	HV	*BHN*	*BHN*	R_C	R_B	*Shore*	*Kg/mm²*
1	**940**	**-**	**-**	**68.0**	**-**	**97**	**-**
2	**920**	**-**	**-**	**67.5**	**-**	**96**	**-**
3	**900**	**-**	**-**	**67.0**	**-**	**95**	**-**
4	**880**	**-**	**-**	**66.4**	**-**	**93**	**-**
5	**860**	**-**	**-**	**65.9**	**-**	**92**	**-**
6	**840**	**-**	**745**	**65.3**	**-**	**91**	**-**
7	**820**	**-**	**733**	**64.7**	**-**	**90**	**-**
8	**800**	**-**	**722**	**64.0**	**-**	**88**	**-**
9	**780**	**-**	**710**	**63.3**	**-**	**87**	**-**
10	**760**	**-**	**698**	**62.5**	**-**	**86**	**-**
11	**740**	**-**	**684**	**61.8**	**-**	**84**	**-**
12	**720**	**-**	**670**	**61.0**	**-**	**83**	**-**
13	**700**	**-**	**656**	**61.1**	**-**	**81**	**-**
14	**680**	**-**	**638**	**59.2**	**-**	**80**	**250**
15	**660**	**-**	**620**	**58.3**	**-**	**79**	**243**
16	**640**	**-**	**601**	**57.3**	**-**	**77**	**234**
17	**620**	**-**	**582**	**56.3**	**-**	**75**	**225**
18	**600**	**-**	**564**	**55.2**	**-**	**74**	**216**
19	**580**	**-**	**545**	**54.1**	**-**	**72**	**208**
20	**560**	**-**	**525**	**53.0**	**-**	**71**	**200**
21	**540**	**496**	**507**	**51.7**	**-**	**69**	**190**
22	**520**	**480**	**488**	**50.5**	**-**	**67**	**185**
23	**500**	**465**	**471**	**49.1**	**-**	**66**	**175**
24	**480**	**448**	**452**	**47.7**	**-**	**64**	**168**
25	**460**	**433**	**433**	**46.1**	**-**	**62**	**156**
26	**440**	**415**	**415**	**44.5**	**-**	**59**	**150**
27	**420**	**397**	**397**	**42.7**	**-**	**57**	**141**
28	**400**	**379**	**379**	**40.8**	**-**	**55**	**133**
29	**380**	**360**	**-**	**38.8**	**-**	**52**	**127**
30	**360**	**341**	**-**	**36.6**	**-**	**50**	**116**

S.N.	Various hardness number						
	Vickers hardness number	Brinell number		Rockwell number		Sclero-scope	Tensile strength
		Steel ball	Carbide ball	C-Scale	B-Scale	Scope hardness	
	HV	BHN	BHN	R_C	R_B	Shore	Kg/mm^2
31	340	322	-	34.4	-	47	110
32	320	303	-	32.2	-	45	104
33	300	284	-	29.8	-	42	97
34	290	275	-	28.5	-	41	94
35	280	265	-	27.1	-	40	90
36	270	256	-	25.6	-	38	86
37	260	247	-	24.0	-	37	84
38	250	238	-	22.2	-	36	81
39	240	228	-	20.3	-	34	78
40	230	219	-	-	99.1	33	76
41	220	209	-	-	97.5	32	73
42	210	200	-	-	95.7	30	70
43	200	190	-	-	93.8	29	67
44	190	181	-	-	91.6	28	66
45	180	171	-	-	89.2	26	62
46	170	162	-	-	86.5	25	58
47	160	152	-	-	83.4	23	54
48	150	143	-	-	80.0	22	51
49	140	133	-	-	76.1	21	48
50	130	124	-	-	71.5	20	46
51	120	114	-	-	66.3	18	43
52	110	105	-	-	60.0		-
53	100	95	-	-	52.5		-
54	90	85.5	-	-	43.3		-
55	80	76.0	-	-	31.9		-
56	70	66.5	-	-	17.1		-
57	60	57.0	-	-	-		-
58	50	47.5	-	-	-		-
59	40	38.0	-	-	-		-
60	30	28.5	-	-	-		-

+ The hardness conversion table showing relationship between Vickers, Brinell and Rockwell hardness numbers apply to the following materials:

Appendix 2: Values of Physical Constants

S. N.	*Constant*	*Symbol*	*Value*
1	Atomic mass unit	a,m.u	1.660×10^{-27} kg
2	Avogadro's number	N	6.023×10^{23}/mol
3	Bohr Magneton (Magnetic moment)	μ_B	9.273×10^{-24} A.m^2
4	Boltzmann's constant	k	1.38×10^{-23} J/K $= 8.620 \times 10^{-5}$ eV/K
5	Electronic charge	e	1.620×10^{-19} C
6	Electronic mass	m	9.109×10^{-31} kg
7	Gas constant	R	8.314 J/mol. K
8	Permeability of free space	μ_0	$4\pi \times 10^{-7}$ H/m
9	Permittivity of free space	ε_0	8.854×10^{-12} F/m
10	Planck's constant	h	6.626×10^{-34} J.s

Appendix 3: Conversion Factors

S.N.	*Unit*	*Conversion factor*
1	1 angstrom $(\dot{A})$	10^{-10}m
2	1 atmosphere	0.1 MN/m^2
3	1 kgf/cm^2	98.1 kN/m^2
4	1 eV/entity	96.49 kJ/mol
5	1 kcal	4.18 KJ

The formula for converting Centigrade readings to equivalent Fahrenheit readings is

F = 1.8C + 32

$$\boldsymbol{C = \frac{F - 32}{1.8}}$$

Where

F = degrees Fahrenheit

C = degrees Centigrade

Appendix 4: Physical Properties of Selected Metals

S.N.	*Metal*	*Symbol*	*Atomic number*	*Melting point (^0C)*	*Density g/cm³*	*Crystal Structure (20^0C)*
1	**Aluminum**	Al	13	660	2.70	FCC
2	**Antimony**	Sb	51	630	6.70	RHOMBIC
3	**Arsenic**	As	33	814	5.78	RHOMBIC
4	**Beryllium**	Be	4	1280	1.85	HCP
5	**Bismuth**	Bi	83	271	9.80	RHOMBIC
6	**Boron**	B	5	2300	2.30	ORTHORHOMBIC
7	**Cadmium**	Cd	48	321	8.65	HCP
8	**Chromium**	Cr	24	1875	7.19	BCC
9	**Cobalt**	Co	27	1495	8.83	HCP
10	**Copper**	Cu	29	1083	8.90	FCC
11	**Gold**	Au	79	1063	19.25	FCC
12	**Iron**	Fe	26	1539	7.87	BCC
13	**Lead**	Pb	82	327	11.38	FCC
14	**Magnesium**	Mg	12	650	1.74	HCP
15	**Manganese**	Mn	25	1245	7.47	-
16	**Molybdenum**	Mo	42	2610	10.22	BCC
17	**Nickel**	Ni	28	1453	8.90	FCC
18	**Platinum**	Pt	78	1769	21.45	FCC
19	**Silver**	Ag	47	961	10.49	FCC
20	**Tantalum**	Ta	73	2996	16.60	-
21	**Tin**	Sn	50	232	7.30	BCT
22	**Tungsten**	W	74	3410	19.30	BCC
23	**Uranium**	U	92	1132	19.07	ORTHORHOMBIC
24	**Vanadium**	V	23	1900	6.10	-
25	**Zinc**	Zn	30	420	7.13	HCP

Appendix 5: The Atomic and Ionic Radii of Selected Elements

S.N.	Element	Atomic Radius $(\dot{A})$	Valence	Ionic Radius $(\dot{A})$
1	Aluminum	1.432	+3	0.51
2	Arsenic	1.15	+5	2.22
3	Barium	2.176	+2	1.34
4	Bismuth	1.60	+5	0.74
5	Boron	0.46	+3	0.23
6	Bromine	1.19	-1	1.96
7	Cadmium	1.49	+2	0.97
8	Calcium	1.976	+2	0.99
9	Carbon	0.77	+4	0.16
10	Chlorine	0.905	-1	1.81
11	Chromium	1.249	+3	0.63
12	Cobalt	1.253	+2	0.72
13	Copper	1.278	+1	0.96
14	Fluorine	0.6	-1	1.33
15	Gold	1.442	+1	1.37
16	Hydrogen	0.46	+1	1.54
17	Iodine	1.35	-1	2.20
18	Iron	1.241 (BCC)	+2	0.74
		1.269 (FCC)	+3	0.64
19	Lead	1.75	+4	0.84
20	Lithium	1.519	+1	0.68
21	Magnesium	1.604	+2	0.66
22	Manganese	1.12	+2	0.80
			+3	0.66
23	Mercury	1.55	+2	1.10
24	Molybdenum	1.363	+4	0.70
25	Nickel	1.243	+2	0.69
26	Nitrogen	0.71	+5	0.15
27	Oxygen	0.60	-2	1.32
28	Phosphorus	1.10	+5	0.35
29	Platinum	1.387	+2	0.80
30	Potassium	2.314	+1	1.33
31	Silicon	1.176	+4	0.42
32	Silver	1.445	+1	1.26
33	Sodium	1.858	+1	0.97
34	Sulfur	1.06	-2	1.84
35	Tin	1.405	+4	0.71
36	Titanium	1.475	+4	0.68
37	Tungsten	1.371	+4	0.70
38	Uranium	1.38	+4	0.97
39	Vanadium	1.311	+3	0.74
40	Zinc	1.332	+2	0.74

Note that 1 $\dot{A}=10^{-8}$ cm = 0.1 nanometer (nm)

hydrogen 1 **H** 1.0079																		helium 2 **He** 4.0026
lithium 3 **Li** 6.941	beryllium 4 **Be** 9.0122												boron 5 **B** 10.811	carbon 6 **C** 12.011	nitrogen 7 **N** 14.007	oxygen 8 **O** 15.999	fluorine 9 **F** 18.998	neon 10 **Ne** 20.180
sodium 11 **Na** 22.990	magnesium 12 **Mg** 24.305												aluminium 13 **Al** 26.982	silicon 14 **Si** 28.086	phosphorus 15 **P** 30.974	sulfur 16 **S** 32.065	chlorine 17 **Cl** 35.453	argon 18 **Ar** 39.948
potassium 19 **K** 39.098	calcium 20 **Ca** 40.078		scandium 21 **Sc** 44.956	titanium 22 **Ti** 47.867	vanadium 23 **V** 50.942	chromium 24 **Cr** 51.996	manganese 25 **Mn** 54.938	iron 26 **Fe** 55.845	cobalt 27 **Co** 58.933	nickel 28 **Ni** 58.693	copper 29 **Cu** 63.546	zinc 30 **Zn** 65.39	gallium 31 **Ga** 69.723	germanium 32 **Ge** 72.61	arsenic 33 **As** 74.922	selenium 34 **Se** 78.96	bromine 35 **Br** 79.904	krypton 36 **Kr** 83.80
rubidium 37 **Rb** 85.468	strontium 38 **Sr** 87.62		yttrium 39 **Y** 88.906	zirconium 40 **Zr** 91.224	niobium 41 **Nb** 92.906	molybdenum 42 **Mo** 95.94	technetium 43 **Tc** [98]	ruthenium 44 **Ru** 101.07	rhodium 45 **Rh** 102.91	palladium 46 **Pd** 106.42	silver 47 **Ag** 107.87	cadmium 48 **Cd** 112.41	indium 49 **In** 114.82	tin 50 **Sn** 118.71	antimony 51 **Sb** 121.76	tellurium 52 **Te** 127.60	iodine 53 **I** 126.90	xenon 54 **Xe** 131.29
caesium 55 **Cs** 132.91	barium 56 **Ba** 137.33	57-70 *	lutetium 71 **Lu** 174.97	hafnium 72 **Hf** 178.49	tantalum 73 **Ta** 180.95	tungsten 74 **W** 183.84	rhenium 75 **Re** 186.21	osmium 76 **Os** 190.23	iridium 77 **Ir** 192.22	platinum 78 **Pt** 195.08	gold 79 **Au** 196.97	mercury 80 **Hg** 200.59	thallium 81 **Tl** 204.38	lead 82 **Pb** 207.2	bismuth 83 **Bi** 208.98	polonium 84 **Po** [209]	astatine 85 **At** [210]	radon 86 **Rn** [222]
francium 87 **Fr** [223]	radium 88 **Ra** [226]	89-102 **	lawrencium 103 **Lr** [262]	rutherfordium 104 **Rf** [261]	dubnium 105 **Db** [262]	seaborgium 106 **Sg** [266]	bohrium 107 **Bh** [264]	hassium 108 **Hs** [269]	meitnerium 109 **Mt** [268]	ununnilium 110 **Uun** [271]	unununium 111 **Uuu** [272]	ununbium 112 **Uub** [277]		ununquadium 114 **Uuq** [289]				

*Lanthanide series	lanthanum 57 **La** 138.91	cerium 58 **Ce** 140.12	praseodymium 59 **Pr** 140.91	neodymium 60 **Nd** 144.24	promethium 61 **Pm** [145]	samarium 62 **Sm** 150.36	europium 63 **Eu** 151.96	gadolinium 64 **Gd** 157.25	terbium 65 **Tb** 158.93	dysprosium 66 **Dy** 162.50	holmium 67 **Ho** 164.93	erbium 68 **Er** 167.26	thulium 69 **Tm** 168.93	ytterbium 70 **Yb** 173.04
**Actinide series	actinium 89 **Ac** [227]	thorium 90 **Th** 232.04	protactinium 91 **Pa** 231.04	uranium 92 **U** 238.03	neptunium 93 **Np** [237]	plutonium 94 **Pu** [244]	americium 95 **Am** [243]	curium 96 **Cm** [247]	berkelium 97 **Bk** [247]	californium 98 **Cf** [251]	einsteinium 99 **Es** [252]	fermium 100 **Fm** [257]	mendelevium 101 **Md** [258]	nobelium 102 **No** [259]

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